SMART DASHBOARD: A WEB-BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED DATA PROCESSING

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In partial fulfillment of the requirements for the degree
Bachelor of Science in Information Technology
Specialized in *Web and Mobile Application Development*

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ABSTRACT

The Barangay Malinta Smart Dashboard is a web-based management platform developed to address the recurring problems in manual barangay processes such as slow certificate requests, disorganized records, and limited access to information. The project was created to modernize barangay operations and improve service delivery for both residents and officials by providing a centralized and accessible digital system. The platform integrates essential modules including resident registration, certificate request processing, appointment scheduling, event and announcement posting, blotter reporting, archives management, and real-time analytics for data-driven decision-making. The system is built with modern tools like Next.js, React, TypeScript, and Firebase, while Cloudinary handles image storage, Nodemailer manages email notifications, and PDF utilities generate official documents, ensuring a fast, reliable, and easily maintainable platform. A user-centered design approach guided development to ensure a simple and functional interface suited for community use. The project underwent system testing, compatibility testing, and user acceptance evaluation using the Technology Acceptance Model (TAM), all of which achieved positive and functional results. The ISO 25010 evaluation conducted by 2 IT experts yielded an overall compatibility rating of 5.0, indicating that the system strongly meets quality standards in reliability, usability, and performance. Overall, findings show that the Barangay Malinta Smart Dashboard effectively streamlines transactions, enhances transparency, improves communication within the community, and stands as a reliable and efficient tool for digital governance

Keywords: *Smart Dashboard, Barangay Management System, Digital Governance, Web-Based Application, Citizen Engagement, Data-Driven Decision-Making, Automated Services, Local Government Technology, Community Information System, Online Public Service Platform, User-Friendly Interface*

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Definition of Terms

Terms	Definition
Agile Scrum Methodology	An incremental project management method widely used in agile development, offering customizable Features and various benefits.
Automated Data Processing	Automates data handling-collecting, transforming, and analyzing large datasets with limited human intervention, streamlining processes for efficient and accurate management.
Centralized Platform	A single documentation system that stores all materials, tools, and resources in one place.
Citizen Engagement	The action that allows citizens and policymakers to exchange Information.
Data Analytics	The process of analyzing data to find trends and draw conclusions about the information they contain.
Data Driven Digital Governance	Use of data to improve public administration; utilizing policies and technologies to create more efficient transparent, and responsive government services.
Data Visualization	A graphical representation of Information and data using charts, graphs, maps, and other visual tools.
Flowchart	A process map that visually represents the steps of a process system, or workflow in sequential order.
Smart Dashboard	A centralized digital system that enables organizations to manage and track all visitor and internal mobility operations in real time.

Sustainable System	A system developed to adapt to future developments using technological advances to ensure long-term usability.
Technology Acceptance Model (TAM)	A framework used to assess user satisfaction, the perceived value of the system, and its overall usability.
User Acceptance Testing (UAT)	A testing stage that involves actual users ensure that the system satisfies their needs and requirements
User Interface (UI)	Serves as the vital connection between users and their computers, websites, or applications allowing for seamless interaction through input and output devices

CHAPTER I

INTRODUCTION

Project Context

According to the Philippine Statistics Authority (2023), there were over 42,000 barangays in the Philippines, serving more than 110 million people. These barangays played a crucial role in governance, dealing with everything from document requests to complaints and helping people access local services. But here's the catch: even with technology moving at lightning speed, many of these barangays were still stuck using manual systems. This old-school approach led to delays, lost files, inefficient tracking of services, and folks having to visit in person more than necessary, definitely not ideal for both the operations and the satisfaction of the citizens (Garcia, 2020; Velasco, 2022).

When things were processed manually, it was not just slow; it could also be less transparent and more prone to human errors. (Santos and Torres, 2021), pointed out that these outdated systems often created bottlenecks and led to mishandled data. We really saw how serious these issues were during the COVID-19 pandemic. With everyone's mobility limited, it became super clear that there was a pressing need for digital solutions that could allow people to access services remotely and conduct transactions without face-to-face interaction (Bringula et al., 2021).

In light of this, the Philippine government was now pushing for digital transformation at the barangay level. There had been some great teamwork between public institutions and private organizations, leading to systems like

BARS Philippines (Alberto, 2025; Thermozone, 2025), which helped automate resident information management, along with TECHSTATIC's Barangay eGovernance System (2025) that digitized document requests and issuances. These platforms were part of a broader trend towards using Information and Communication Technology (ICT) to enhance public service delivery (Lopez, 2021).

This study aimed to build on these exciting trends by developing a Smart Dashboard, a web-based system specifically designed for barangay operations. The idea was to make transactions easier, cut out unnecessary paperwork, and boost access to services. Residents were able to request certificates, file blotter reports, book appointments, and get announcements online. (Eliza and Orozco, 2025) Emphasized that local digital platforms were crucial for improving citizen access and fostering efficient governance.

The Smart Dashboard also promoted transparency by providing real-time analytics and visual data insights. (Magno, 2021) Highlighted that having timely access to data was key for decision-making in local government. With this dashboard, barangay officials could keep an eye on service volumes, understand resident demographics, and track response times, which helped them allocate resources better and improve service delivery (Sison, 2020).

The platform also strengthened internal management. As Lacasandile et al. (2020) mentioned, automated profiling could help identify community trends and recurring needs, which was super valuable for strategic governance. With features like dynamic resident directories, digital blotter logs, and reporting tools, the Smart Dashboard was set to modernize barangay operations. Plus, it

makes it easier for local leaders to communicate directly with their constituents by integrating announcements and event schedules.

To keep things improving,the system had feedback and monitoring tools built in. This meant citizens could track their requests online, which cut down on the need for follow-ups and boosted satisfaction. These features aligned well with international e-governance practices, striking a balance between tech convenience and responsive service (Williams, 2018).

Looking ahead,there's room for growth with mobile app integration, support for local languages, and automated alerts via SMS or email. (Rjnworks, 2024) Really emphasized that scalability and adaptability were crucial for effective governance platforms.

Given how digital literacy is on the rise, smartphones were everywhere, and internet infrastructure is getting better in the Philippines, that was a great time to roll out barangay-level digital solutions. Sure, there were still some connectivity gaps in rural areas, but strategies like localized hosting and offline-capable modules could help make sure everyone got access to the system (Eliza & Orozco, 2025).

To enhance accessibility and streamline initial transactions, the Smart Dashboard was designed as a Progressive Web App (PWA). This allowed residents to sign up or log in using any browser, whether on desktop or mobile, and conveniently access services such as viewing announcements, submitting document requests (e.g., barangay clearance), and filing blotter reports online. While official documents must still be claimed in person and, in some cases (like blotter follow-ups), require face-to-face meetings, the PWA setup significantly reduced the need for repeated visits. PWAs were capable of

functioning even with limited or intermittent connectivity by using service workers and caching strategies. This enabled users to access static content, like previously loaded announcements or form templates, even when offline. However, features that required data submission (like request forms or blotter entries) only worked when an internet connection was available. This hybrid online-offline behavior ensured that the platform remained usable and responsive, especially in barangays with unreliable internet, while also extending the system's reach to more citizens without requiring app store installation (Osmani, 2017).

The Smart Dashboard stepped in to tackle the inefficiencies of traditional barangay systems. It was all about creating a responsive, transparent, and accessible platform, marking a big leap forward in digital local governance. This empowered both citizens and officials to work towards a smarter, more inclusive public service framework.

Project Objectives

The main objective of this study is to develop a web-based Smart Dashboard platform that promotes citizen engagement and facilitates data-driven digital governance through the use of automated data processing. Specifically, it aims to:

1. To design and develop an interactive web-based Smart dashboard system with key features such as an announcements/events board, request submission forms, and automated notifications, tailored to enhance community engagement and governance transparency.

2. To provide an online platform that enables citizens to efficiently request and track barangay certifications, including Barangay Clearance, Residency, Indigency, Good Moral, Death Certificate, Business Closure, and Income or No Income Certificates, providing a faster, more transparent, and user-friendly experience for official purposes. Archive modules let barangay administrators temporarily save, restore, or permanently delete records to maintain secure data management.

3. This study aimed to evaluate the system's user acceptance using the Technology Acceptance Model (TAM) and software quality characteristics using the ISO 25010 standard.

Project Purpose

The purpose of this study was to develop a web-based dashboard system, Smart Dashboard, designed to enhance citizen engagement and promote data-driven digital governance at the barangay level. This platform served both barangay officials and residents by offering a centralized, user-friendly system that simplifies public service processes. Through the Smart Dashboard, citizens would be able to submit requests, manage documents, and schedule appointments conveniently online, while barangay administrators could track, manage, and respond to community needs in real-time.

This project directly supported several United Nations Sustainable Development Goals (SDGs). It contributed to Goal 16: Peace, Justice, and Strong Institutions by fostering transparent and responsive local governance. The system encouraged accountability and community participation by allowing residents to engage directly with barangay services. Furthermore, it supported

Goal 9: Industry, Innovation, and Infrastructure by integrating innovative ICT solutions into public administration, streamlining government processes, and promoting resilient digital infrastructure. Lastly, it aligned with Goal 10: Reduced Inequalities, as it ensured that all citizens, regardless of socioeconomic status or physical ability, could access government services equitably and efficiently through a digital platform.

The development of the Smart Dashboard aimed not only to strengthen the relationship between barangay officials and constituents but also to promote a more inclusive, efficient, and data-informed local governance system. By minimizing manual administrative processes, reducing service delivery delays, and improving information management, this project sought to enhance the overall quality of public service and encourage active community participation. The following were the key beneficiaries of the project:

Barangay Officials

The study's goal was to assist barangay officials by establishing an online platform for managing citizen requests, including legal and personal certifications, business-related requests, and various certificates. The system made permit creation easier, kept resident records, and tracked service responses in real time, improving speed and transparency while allowing barangay staff to focus on important governance tasks. The platform enhanced communication between residents and officials by providing timely updates on request statuses, reducing manual processing. This improvement allowed residents to stay informed about the progress of their requests without the need for repeated follow-ups or in-person inquiries became more efficient and transparent, helping build trust between the barangay.

Citizens

The system enhanced the experience of barangay residents by serving as a convenient and accessible platform to engage with their local government. Its main purpose was to provide residents with direct access to digital services that simplified transactions, improved service delivery, and reduced the barriers of time and distance. Through online access, residents were able to submit documents, monitor the progress of their requests, and schedule appointments without the need for physical visits. This not only minimized waiting times and travel inconveniences but also ensured that services were delivered more efficiently.

Community

The smart dashboard helped the larger community by making barangay operations more responsive, transparent, and well-organized, essentially improving operations' efficiency, transparency, and data-drivenness. It primarily aimed to support collective governance. Through the simplification of processes and the centralization of important data, the dashboard guaranteed improved communication, accountability, and service monitoring that promoted larger cooperation and trust between the barangay and its residents.

Future Researchers

Future researchers would benefit from this project through access to relevant data and system architecture that served as a foundation for further studies on digital governance and e-government solutions. The Smart Dashboard also contributed to the ongoing development of innovative digital

tools that addressed inefficiencies in public administration and service delivery at the grassroots level.

Scope and Limitation of the Study

The study focused on the design, development, and deployment of the Smart Dashboard, a web-based platform intended for barangay officials and residents. The system was accessible through desktop and mobile browsers and was enhanced with Progressive Web App (PWA) features, enabling faster load times, offline access to cached content, and smoother performance across devices. Its primary goal was to streamline citizen transactions, announcements, and barangay administrative processes by offering a centralized digital platform.

Login and Registration Module. The Login and Registration Module provided barangay residents with the opportunity to register by filling out a signup form with their personal details. For verification, only residents living within the barangay can register, as confirmed by the information provided regarding their barangay, municipality, province, and ZIP code. Once registered, residents could log in using their email and password. Barangay personnel would eventually handle pending registrations by checking, approving, or rejecting accounts. Ensuring that only authorized residents are granted access to the system.

Super Admin Dashboard Module. The Super Admin Dashboard Module served as the main control center to access the Smart Dashboard System. The super admin had complete access to all features and modules, which include managing users and roles, and monitoring important analytics

while generating reports. It enabled the super admin to manage all administrative accounts, control permissions, and make strategic decisions. The super admin dashboard provided huge visibility and total control over the entire system, facilitating efficient governance and secure administration.

Admin Dashboard Module. The Admin Dashboard Module acted as the central control panel for barangay staff. Through this dashboard, administrators could manage resident profiles, monitor document requests, post announcements, handle content updates, and access system reports. The dashboard was designed for efficient monitoring and management of all barangay system activities.

User Management Module. The User Management Module served as the main homepage for residents after login. It displayed the resident's profile information, announcement board, request forms, and notifications. The dashboard was designed for easy navigation, allowing users quick access to essential features such as updating their profiles, submitting requests, and viewing announcements. It also added a feature that allowed residents to submit feedback anonymously. This allowed residents to express their ideas or problems without revealing their identities, promoting honest and constructive feedback.

Profile Management Module. The Profile Management Module allowed residents to view and update their personal information, including their name, address, contact details, and profile picture. Administrators also managed and updated their own account information within the system. The Super Admin had the ability to edit key details of each administrator's profile, such as first name,

last name, middle name, suffix, phone number, address, and assigned barangay position. The module also includes a profile picture section where administrators can upload or change their display photo.

Announcement and Events Board Module. Serves as the official bulletin board where barangay officials post updates, emergency announcements, community events, and other important notices. Residents easily viewed categorized and timestamped announcements from their dashboard, keeping them informed about barangay activities.

Request Form Module. The Request Form Module allowed residents to submit requests for barangay documents such as Barangay Clearance, Certificate of Indigency, Business Permit or Renewal, and Blotter Reports through the Certificate Request page. Users tracked the status of their requests, which could be marked by the administrative staff as Processing, Approved, or Denied within the system. This module allowed residents to easily and directly upload their requirements, resulting in more efficient processing.

Notification Module. The Notification Module provided real-time system alerts regarding the status of resident requests, new announcements, and upcoming events. Notifications were accessible within the user dashboard and could also be delivered via email, depending on the user's preference.

Officials Management Module. The Officials Management Module allowed administrators to manage submitted requests and handle user accounts within the system. Administrators could add, edit, and delete accounts or requests; however, when an account is deleted, it is automatically moved to the archive instead of being permanently removed. This allowed deleted

records to be restored if they were removed by mistake. Archived accounts of former or inactive users were also kept to maintain complete and accurate records. Barangay officials were organized into categories such as Barangay Captain, Councilors, Secretary, Treasurer, and SK Chairperson. This organized setup makes account management easier and ensures that officials' accounts remain properly arranged. It also helped maintain an updated and consistent request processing flow, since the system could still reference past records when validating or reviewing submissions.

Content Management Module. The Content Management Module allowed barangay officials to manage and update all content displayed on the Smart Dashboard. Through this module, they could edit static information pages, publish announcements, and post upcoming events for the community. This ensured that the dashboard remained updated with the latest information relevant to residents.

Records Management Module. The Records Management Module allowed barangay staff to organize and manage important records, including resident databases, walk-in requests, pending requests, and blotter entries. It helps maintain a structured record system that could be referenced anytime necessary.

Monitoring Module. The Monitoring Module summarized critical barangay data, including the number of registered residents, pending and completed requests, and issued documents. It features visual aids such as charts and graphs and offers downloadable reports that assist barangay officials in policy-making, resource allocation, and future planning. It also

included an archives feature that allowed the administrator to view, restore, and permanently delete history records.

While the Smart Dashboard improved efficiency and accessibility, it had limitations. Offline functionality was restricted to viewing previously loaded content; new submissions or updates require an active internet connection. The system did not support real-time chat, reactions to announcements, or RSVP functionality for events. Furthermore, official documents must still be claimed in person, and certain services, such as blotter follow-ups, require face-to-face interaction. Full access was limited to registered and verified residents, and email notifications may experience delays depending on external providers.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presented the existing literature and research related to the creation of smart dashboards. It explored the components involved and the challenges faced when developing a smart dashboard that incorporated data analytics functionalities.

Smart Dashboard

A smart dashboard was an information management tool that gathered data from a connected database to create visualizations. It enabled end users to view summarized information in one place and find answers to specific questions. Dashboards were increasingly recognized as essential tools for presenting complex data in an accessible manner. They served as platforms for community engagement, enabling citizens to understand urban challenges and participate in decision-making processes (Pluto-Kossakowska, Fijałkowska, Denis, Jaroszewicz, & Krzysztofowicz, 2022) (Lock, Bednarz, Leao, & Pettit, 2019). Dashboards consolidated information into a single view, making it easier for governments and citizens to monitor urban systems such as energy, transport, and health (Lock et al., 2020). According to Matheus et al. (2020), Dashboards served various purposes in government agencies, such as supporting policy creation, assisting with planning, monitoring performance, and enhancing transparency.

The design of dashboards had to balance simplicity and complexity, depending on the user's needs. For instance, simple dashboards were used for basic monitoring, while complex dashboards incorporated advanced analytical

tools and geo-analytical features (Pluto-Kossakowska et al., 2022). However, challenges such as data publication, analytical tools, and cartographic presentation often limit the potential of dashboards in fostering a real dialogue with residents (Pluto-Kossakowska et al., 2022). The automation of dashboard development is feasible due to the common functional and non-functional requirements across cities. Frameworks like Cities-Board proposed the use of model-driven engineering to generate dashboards, reducing the need for significant human and financial resources (Rojas, Bastidas, & Cabrera, 2020). These frameworks allowed city authorities to create dashboard models that could be transformed into functional code artifacts through model-to-model (M2M) and model-to-text (M2T) transformations (Rojas et al., 2020).

According to Singh and Gill (2020), a smart dashboard transformed raw data into actionable insights by utilizing data visualization tools, APIs, and dynamic user interfaces. The primary goal was not only to monitor activities but also to provide predictive analytics and trends that could guide future decisions. For example, a barangay dashboard could show which services were most requested in real time, highlight backlogs, and flag potential issues like delayed certificate issuance or frequent complaints in specific zones.

Moreover, dashboards equipped with Geographic Information System (GIS) layers mapped community problems, such as illegal dumping or public safety concerns, which helped officials deploy resources more strategically (Kitchin, 2013). Smart dashboards were also designed with responsiveness and mobile optimization in mind, ensuring access for both barangay officials on the go and residents using various devices.

As said by Ganapati (2011), Dashboards served as a tool for presenting performance data. Their effectiveness relies on how organizations implement them, enhancing both external accountability and transparency, along with internal performance. This explained that they helped stakeholders make wise decisions and promoted continuous improvement by offering real-time insights and visual representations of important measurements.

In the study of Maheshwari and Janssen (2014), they also served as a practical resource for delivering information to policymakers in the government, additionally, providing clear and concise data that helped policymakers make informed decisions that effectively addressed public needs. This meant that when information was presented in an accessible manner, it allowed government officials to better understand the issues at hand and develop strategies that were more likely to result in positive outcomes for the community.

The development of modern web applications relied on a suite of tools and frameworks that enhanced productivity, scalability, and user experience. Visual Studio Code emerged as a preferred editor among developers due to its intelligent code completion, built-in debugging, and extensive extension ecosystem (Microsoft, n.d., 2025). For crafting the user interface, React.js offered a component-based architecture that promotes reusability and maintainability, making it ideal for dynamic and responsive web applications (Brien/CEO & Brien/CEO, 2023). To further enhance performance and SEO, Next.js, built on React, implements server-side rendering and static site generation for faster page loads and improved indexing (Pati & Zaki, 2025).

Styling was streamlined using Tailwind CSS, a utility-first framework that supported quick design implementation, responsive layouts, and consistent visual language (Ovcharov, 2025). Together, these technologies formed an efficient, scalable, and user-friendly development stack for building modern platforms.

Citizen Engagement

Citizen engagement was a critical component of smart city initiatives, enabling residents to contribute to decision-making processes and urban planning. Dashboards and digital platforms played a pivotal role in facilitating this participation by providing accessible and interpretable data (Pluto-Kossakowska, Fijałkowska, Denis, Jaroszewicz, & Krzysztofowicz, 2022) (Chokki et al., 2022). For example, participatory budgeting and collaborative decision-making platforms were successfully implemented in cities like Helsinki and Barcelona (Suter et al., 2024) (Cardullo & Kitchin, 2024).

Digital platforms significantly expanded these opportunities by enabling two-way communication between citizens and local authorities (Nam, 2012). Tools such as online request forms, complaint tracking, and public information boards enhanced transparency and civic involvement. Bussu and Galanti (2018) argued that effective citizen engagement was critical in cultivating trust and responsiveness within public institutions, particularly at the barangay level, where grassroots governance is essential. Bekkers et al. (2014) show that the increasing relevance of digital technologies in promoting citizen involvement and simplifying governance procedures. At the local level, features like interactive service requests, real-time notifications, and digital calendars enable citizens to engage actively in local issues without physical presence. These

online platforms reduced bureaucratic hurdles, promoted faster feedback systems, and fostered more responsive decision-making, acting as vital connectors between citizens and their local leaders.

Potter and Olaoye (2024) highlighted the importance of both digital and in-person strategies in fostering meaningful citizen participation in smart cities. Their study illustrated how smart tools, when designed inclusively, could bridge the gap between citizens and local decision-makers. Similarly, the World Bank's Digital Citizen Engagement (CE) Catalog provided a comprehensive resource for leveraging digital tools to enhance citizen participation in public administration, especially in low-capacity environments (World Bank, n.d.).

Co-creation techniques, such as those described in the SEMAT standard, involve citizens and stakeholders in the development of ICT services. These practices stimulated idea generation, document creation, and the development of design systems, fostering active citizen participation (Ricardo et al., 2024). Hybrid practices, which combine online and offline engagement, also emerged as effective strategies for fostering multi-actor networks and negotiating urban governance (Suter et al., 2024).

Despite the potential of digital platforms, challenges such as the quality of online discussions and the need for incentives to motivate citizen participation persisted. Design principles, such as AI-based feedback, can enhance the quality of contributions and interactions on urban participation platforms (Borchers et al., 2024). Additionally, the integration of open government data (OGD) and crowdsourcing initiatives has been shown to improve transparency and accountability (Rashdi, 2024). Through citizen participation in reporting concerns, submitting feedback and information.

Data Driven Governance

Data-driven governance relied on the collection, analysis, and visualization of data to inform decision-making processes. Dashboards and digital platforms were essential tools for presenting this data in a clear and digestible manner (Frick, 2023). For instance, the Data for All initiative provides municipalities with comprehensive and intuitive access to data, enabling evidence-based decision-making and performance monitoring (Frick, 2023). According to Stedman (2005), data governance encompasses the management of data accessibility, usability, confidentiality, and security within enterprise systems. It included implementing internal standards and policies that regulated the use of data.

To enhance decision-making, utilize resources more efficiently, and offer more responsive public services, governments worldwide were increasingly adopting strategies based on data. This change not only marked a significant advancement in technology but also represented a fundamental rethinking of how governments functioned and interacted with the public (Lee, 2025). However, governments have not yet fully utilized data to realize the power of government analytics. According to a recent report by the World Bank, "Data for Better Governance," governments needed to address issues of data quality and availability. Moreover, they had to be equipped with professionals who could analyze data and apply it to provide evidence-based support for decision-making. Above all, governments had to use their administrative data efficiently to maximize effectiveness and responsiveness (World Economic Forum, 2025) According to Witz et al. (2019), digital platforms that incorporate automated Analytics empowered governments to become more agile and citizen-centered.

Their study emphasized the potential of smart governance tools to improve transparency and performance through continual data monitoring. A system like the Smart Dashboard supported this concept by enabling barangay leaders to identify gaps in service, generate reports instantly, and respond to resident feedback more efficiently, making governance not only smarter but also more accountable.

Meijer et al. (2012) discussed how transparency enabled by digital tools led to increased trust in public institutions. They argued that access to structured, real-time information could reduce corruption risks and promote civic confidence in local leaders. The Smart Dashboard, by automating data processing and publicly displaying service statistics and community updates, directly aligned with this principle. When data was visible and auditable, both citizens and government actors were held to higher standards, resulting in more ethical and effective governance practices. Real-time dashboards that collect citizen inputs and service performance indicators are vital in this approach. These systems provided local administrators with live updates on community activity, which they can use to optimize responses and adjust programs accordingly (Islam & Sufian, 2022). Barangays equipped with such tools can track document requests, complaint resolutions, and response times, creating a more agile and accountable local government. Zuiderwijk et al. (2015) stated that open data initiatives and automated platforms had to be designed with usability in mind to truly empower both citizens and public servants. Systems that were difficult to navigate or lacked real-time functionality tended to fail in engaging users effectively. Data sovereignty and security were critical concerns in data-driven governance. People-centered data governance (PCDG) models

emphasize the inclusion of citizens as agents, beneficiaries, or enablers of data production and use (Nassar et al., 2025). These models employed mechanisms such as data donations, rights-based approaches, and civic deliberation to ensure that data was used responsibly and ethically (Nassar et al., 2025).

While data-driven governance offered significant opportunities for improving service delivery and transparency, challenges such as data quality, security, and privacy had to be addressed. These concerns required careful handling to ensure that information stored in the system remained accurate, protected, and accessible only to authorized users. Strategies such as the integration of advanced data visualization techniques and the use of customizable reports helped mitigate these challenges (Frick, 2023; Lee et al., 2024).

Automated Data Processing

Automated data processing was a key enabler of smart city initiatives, supporting the efficient analysis and visualization of large datasets. Tools such as the Cities-Board framework automate the development of dashboards, reducing the need for manual intervention and enabling real-time monitoring of urban systems (Rojas et al., 2020). Automation in governance transformed traditional administrative processes by reducing manual labor, enhancing accuracy, and expediting service delivery.

Automated data processing allowed for the seamless handling of repetitive tasks such as generating certificates, scheduling appointments, or processing blotter reports, freeing up personnel to focus on more strategic functions (Kim & Lee, 2019). Research by Subiyakto et al. (2021) highlighted

the successful implementation of automated public service dashboards in Indonesia, noting significant improvements in user satisfaction, system reliability, and processing time. For barangays, automation not only streamlined workflows but also ensured consistency and reduced human error. Automation was shown to reduce operational costs, improve service accuracy, and minimize turnaround time in administrative settings. A study by Bwalya and Mutula (2016) noted that e-government systems that incorporated automation features tended to score higher in transparency and public trust due to reduced human errors and corruption potential. For instance, the "iGovPhil Project" in the Philippines introduced a shared Government Cloud (GovCloud), which offers automated systems for HR, finance, and records management. This initiative demonstrated that automation was feasible and beneficial at various government levels, including LGUs (Department of ICT, 2020). Efficient data handling methods enabled automated data processing to offer precise and effective management of large data sets. However, Govnet Technology (2023) reported that manual data processing resulted in delays and inefficiency for government agencies. Mistakes and human error were common in manual reporting.

This process was time-consuming and was vulnerable to inconsistencies, as different personnel may interpret and enter data in different ways. Furthermore, the lack of real-time updates resulted in outdated information being used in important decision-making processes. Therefore, Dashthis (2025) stated in their study that automated dashboards reduced these errors by using defined rules to extract important data directly from the source. Through reducing the need for human intervention in the data extraction

process, these dashboards ensured that information was consistently pulled from reliable sources according to a defined specification. The automation of data processing and dashboard development offered several benefits, including reduced costs, improved efficiency, and enhanced decision-making. For example, the Data for All initiative leveraged automated tools to consolidate diverse data sources and present complex information in a clear and accessible manner (Frick, 2023). Despite the benefits of automation, challenges such as data quality and interoperability had to be addressed. The integration of advanced data visualization techniques and the use of model-driven engineering helped overcome these challenges, ensuring that automated systems were both functional and user-friendly (Rojas et al., 2020; Frick, 2023).

Synthesis and Relevance to the Study

This study explored the integration of smart dashboards in local governance, focusing on their potential to improve data transparency, public service delivery, and citizen engagement. The reviewed literature highlighted how dashboards consolidated data into visual, easily digestible formats that helped both government officials and community members make better decisions. By simplifying access to important information and automating administrative tasks, these platforms contributed to more efficient and accountable public service systems.

The literature also emphasized the evolving role of digital platforms in encouraging civic participation. Digital tools such as service request portals, feedback systems, and real-time notifications were shown to strengthen relationships between citizens and their local leaders. However, challenges persisted in sustaining participation, ensuring inclusivity, and maintaining the

quality of public interaction. These findings reinforced the importance of designing user-friendly, responsive platforms that supported both online and offline engagement.

Automation was identified as a key driver in improving governance efficiency. It helped reduce human error, minimized delays, and improved consistency in service delivery. At the same time, concerns around data quality, interoperability, and digital equity underlined the need for systems that were not only technically sound but also ethically designed and inclusive of all citizens.

This research contributed to the field by addressing these challenges through the development of a smart dashboard tailored to the needs of barangay-level governance. The system was designed to support real-time data processing, monitor key services, and enhance community participation. By offering practical solutions to known limitations in current governance systems, this study supported the broader goal of promoting inclusive, efficient, and transparent local governance in the digital age. Positive feedback from participants revealed that the system effectively promotes community engagement and simplifies barangay services. The findings revealed that the Smart Dashboard System can increase access to information and services, making it a useful tool for residents and officials. By centralizing data and services into one platform, the system allows users to easily view updates, requests, and important announcements. This reduces the need for face-to-face transactions and helps residents access services more conveniently. For barangay officials, the system supports faster decision-making by providing real-time and organized information.

Related Literature Matrix

Table 1. Matrix of Review of Related Literature

Year & Author	Title/Source	Findings	Relevance to the Study
Dashthis (2025)	Automated Dashboards	Automated dashboards extract data based on rules, reducing errors and inconsistencies.	Highlights how the Smart Dashboard reduces human error and ensures consistent data handling.
Nassar et al. (2025)	People-Centered Data Governance	Citizens should act as agents, beneficiaries, or enablers of data use.	Supports ethical data use and engagement practices within the Smart Dashboard framework.
Lee (2025)	Data-Driven Government	Governments must use data to improve public services and efficiency.	Connects to the use of real-time analytics in the Smart Dashboard to optimize barangay services.
World Economic Forum (2025)	Data for Better Governance	Governments need data professionals and infrastructure for analytics.	Underlines the need for proper data handling and analytics tools in local government systems.
Potter & Olaoye (2024)	Inclusive Citizen Participation	Smart tools can bridge citizens and local decision-makers when designed inclusively.	Emphasizes inclusivity in Smart Dashboard design for more effective citizen engagement.
Cardullo & Kitchin (2024)	Collaborative Governance	Digital platforms foster participatory decision-making in smart cities.	Strengthens the Smart Dashboard's role in promoting local co-creation and community input.

Ricardo et al. (2024)	SEMAT & Co-Creation	Co-creation frameworks promote active citizen participation in service design.	Aligns with citizen-led data reporting and service feedback features in the Smart Dashboard.
Suter et al. (2024)	Hybrid Participation	Hybrid digital/offline practices build multi-actor governance networks.	Encourages the integration of hybrid engagement models in dashboard interaction.
Borchers et al. (2024)	AI-Enhanced Urban Platforms	AI can improve online discussion quality in participation platforms.	Suggests improvements for Smart Dashboard's citizen feedback system using AI.
Lee et al. (2024)	Advanced Data Visualizations	Visualization tools improve communication of complex datasets.	Reinforces the dashboard's use of advanced charts and maps for community data.
Frick (2023)	Data for All	Data visualization tools support evidence-based decisions in municipalities.	Aligns with barangay-level decision-making through the Smart Dashboard.
Islam & Sufian (2022)	Real-Time Dashboards	Dashboards help local officials respond to community needs efficiently.	Reflects real-time response and monitoring capacities in the Smart Dashboard.
Pluto-Kossakowska et al. (2022)	Smart Urban Dashboards	Dashboards enable public participation and urban monitoring.	Supports the use of dashboards for community-driven barangay governance.

Subiyakto et al. (2021)	Automated Public Services	Automation improved system reliability and user satisfaction.	Validates automation in Smart Dashboard development for barangays.
Singh & Gill (2020)	Data Visualization in Dashboards	Dashboards transform raw data into predictive insights.	Shows how the dashboard helps predict community service needs.
Rojas et al. (2020)	Cities-Board Framework	Automated model-driven dashboards reduce development costs.	Provides the technical basis for Smart Dashboard automation.
Lock et al. (2020)	Urban Monitoring Systems	Dashboards consolidate urban system data for stakeholders.	Aligns with the policy and planning goals of barangay dashboards.
Matheus et al. (2020)	Government Dashboards	Dashboards support policy, planning, and performance monitoring.	Aligns with the policy and planning goals of barangay dashboards.
Kim & Lee (2019)	Automation in Governance	Automation reduces manual tasks and increases accuracy.	Highlights automation benefits in barangay administrative tasks.
Wirtz et al. (2019)	Smart Governance Tools	Analytics platforms improve transparency and responsiveness.	Supports the accountability objectives of the Smart Dashboard.
Bussu & Galanti (2018)	Trust in Local Governance	Engagement fosters trust and institutional responsiveness.	Highlights the Smart Dashboard’s community trust-building features.

Bekkers et al. (2014)	Digital Citizenship	Digital tools simplify governance and promote engagement.	Supports interactive features like requests and complaints in the dashboard.
Maheshwari & Janssen (2014)	Informing Policymakers	Dashboards present data clearly to help guide decisions.	Validates the use of dashboards for barangay-level policy support.
Kitchin (2013)	GIS and Smart Cities	GIS dashboards map community issues and improve response.	Encourages GIS integration in barangay-level problem-solving.
Ganapati (2011)	Dashboards and Performance	Dashboards support accountability and internal improvement.	Ties directly to barangay transparency through performance tracking.
Meijer et al. (2012)	Transparency through Digital Tools	Transparency via data fosters civic trust and reduces corruption.	Enhances the accountability and trustworthiness of the Smart Dashboard.
Nam (2012)	Digital Urban Platforms	Digital platforms enable two-way communication.	Aligns with Smart Dashboard's engagement and feedback system.
Stedman (2005)	Data Governance	Data must be accessible, secure, and well-governed.	Highlights the importance of data integrity and privacy in dashboard systems.

Conceptual Framework

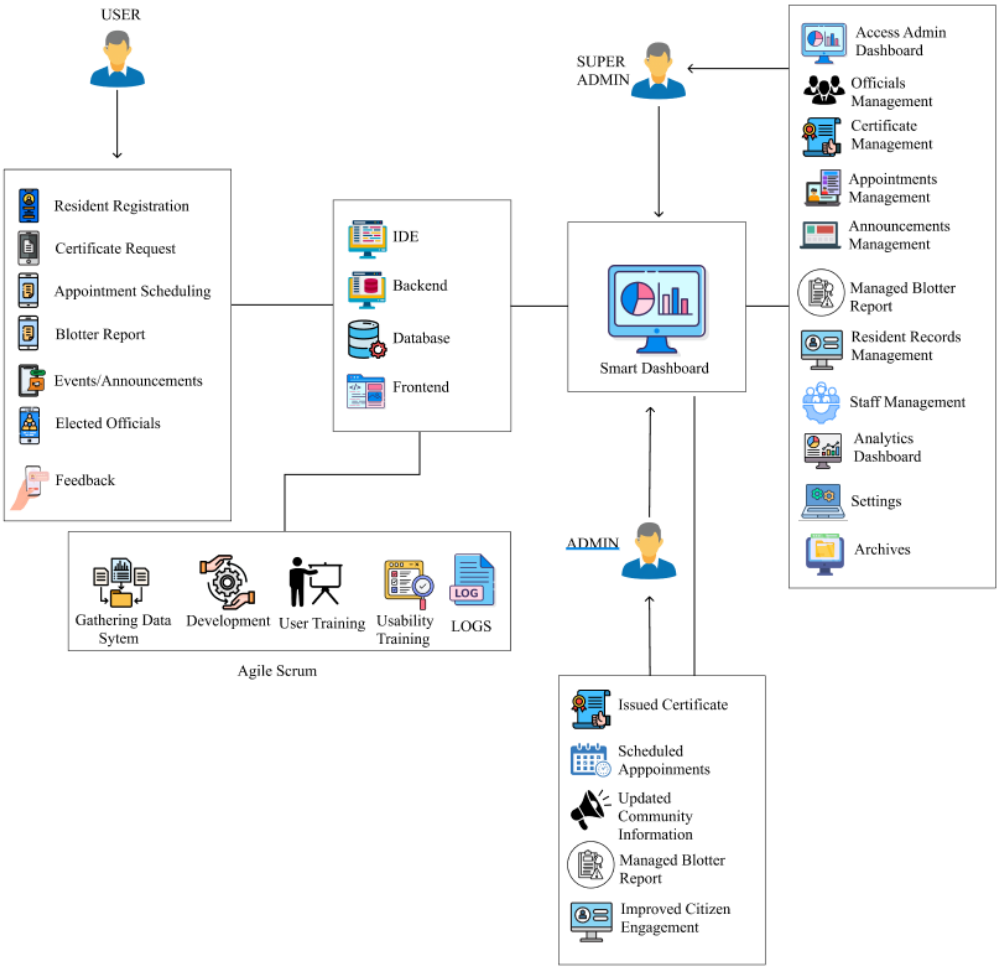


Figure 1. Conceptual Framework

Figure 1 displayed the conceptual framework; it showed how the barangay information system worked by connecting the users, the admins, and the system processes. On the left side, the User section represented ordinary residents who could use the system to register, request certificates, schedule appointments, submit blotter reports, and check events or elected officials. These were the basic features they could access to interact with the barangay online. At the center, the framework showed the development process, which followed the Agile Scrum approach. It includes gathering data from the community, developing the system, training the users, testing usability, and

recording system logs. This part explains how the system was created and continuously improved.

The framework also presented the system architecture, which consisted of the IDE, backend, database, and frontend. These components worked together to run the system smoothly, storing data, managing user inputs, and displaying the interface.

The Smart Dashboard in the middle acted as the main control panel for both the admin and the super admin. The admin managed issued certificates, scheduled appointments, community updates, blotter reports, and overall citizen engagement. Meanwhile, the Super Admin had more advanced controls, such as managing officials, certificates, appointments, announcements, resident records, and staff. They also had access to analytics, settings, and archives to oversee everything. The proposed smart dashboard showed how digital tools may help barangay administrators make better decisions by delivering organized, accurate, and easily available information. Data visualization and automated reporting enabled officials to quickly analyze changes, identify frequent issues, and respond more effectively to community needs. This reduced usage on manual processes and improved up service delivery. The accessibility of analytics and archival options for the super admin enabled data-driven decision-making and permanent record management.

CHAPTER 3

METHODOLOGY

This chapter presented an in-depth exploration of the processes and methodologies employed within the study. It encompassed various crucial elements such as software and hardware frameworks, data management protocols, characteristics of the study population, sampling methodologies, tools for data collection, project design, and conceptual modeling. A significant focus was placed on how automated data processing enhanced citizen engagement through the Smart Dashboard system. The evaluation of procedural efficacy was also discussed to ensure the system met the objectives of improving transparency and citizen participation in governance.

Project Design

The project design began with the Smart Dashboard, which was designed to facilitate the collection and processing of real-time data to enable dynamic interaction between citizens and the local government. Automated processing ensured that the system delivered accurate, up-to-date information, reducing manual intervention and improving system efficiency. This design ensured that the dashboard remained responsive, scalable, and secure, meeting the needs of modern governance. The system continuously captured, processed, and displayed relevant data, ensuring that citizens and government officials had access to real-time information. Information related to government events, announcements, and requests was gathered through forms and user interactions. This data was processed automatically, ensuring accuracy and relevance. Once processed, it was displayed in real-time on the dashboard,

allowing for quick and easy access. Citizens interacted by submitting requests or RSVPs, which were tracked and processed automatically, reducing manual handling. This approach enhanced the system's adaptability, supported decision-making, and improved service delivery while maintaining a user-friendly interface.

System Architecture

Figure 2 below shows how the system architecture of the Barangay Malinta Smart Dashboard works by connecting different users to one main database. When a user opened the system, they first went through the Login and Registration Module, where they could create an account or log in using their email and password. After logging in, residents and barangay officials were sent to their own dashboards. Residents could update their profile, request certificates, view announcements, and events, and get real-time notifications. Barangay officials, on the other hand, had access to tools for posting announcements, checking requests, managing resident records, and viewing statistics and reports. The Super Admin had the highest control and could manage all modules, including content updates, record management, monitoring, and adding or editing official accounts. All data coming from residents, officials, and the super admin was saved in one central database, making the system organized, secure, and easy to manage. This setup helped ensure smooth communication and faster processing of barangay services. With information readily available and updates provided in real time, both residents and officials experienced fewer delays and misunderstandings. This made it easier to track requests and respond to concerns promptly.

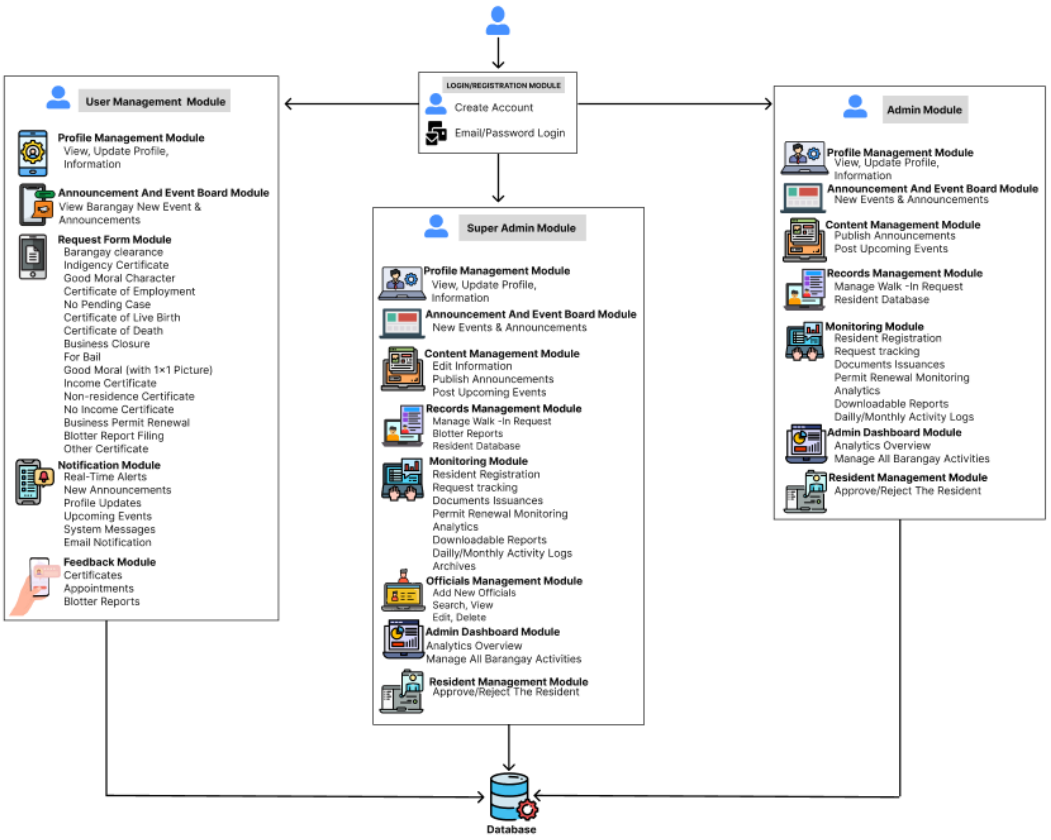


Figure 2. System Architecture

Smart Dashboard

The development of the Smart Dashboard followed a user-centered and system-driven process to ensure that the platform would function efficiently and meet the needs of both residents and barangay officials. The system collected data from digital forms such as announcements, events, and service requests and automatically processed it to keep information accurate and updated in real time. This automation reduced manual tasks and allowed users to instantly access important updates on the dashboard. Residents could easily submit and track requests, while barangay officials managed submissions through dedicated modules. The system architecture was composed of interconnected components like secure user authentication, request management, business permit tracking, announcements, and a barangay calendar. These modules

worked together to make navigation smooth, improve service delivery, and strengthen communication between the barangay and its residents. The Smart Dashboard provided an organized, reliable, and real-time platform that enhanced transparency and streamlined barangay operations.

Project Development

Employing the Agile Software Development Cycle led to effective design and development of the smart dashboard web-based system built as a Progressive Web App (PWA) to enhance user experience. The Agile Scrum Methodology improved the smart dashboard by encouraging continuous improvement and adaptability to changing requirements. It emphasized active user interaction, regular feedback, simplified functionality, and ongoing interface refinement. The system ensured compliance with modern regulations through the use of the most recent valid instructions via repeated sprint cycles. Agile Scrum helped a responsive system by guaranteeing consistent value delivery and breaking down complex legal requirements into manageable tasks. The use of Agile Scrum increased transparency and collaboration among developers, stakeholders, and users. Regular sprint reviews and assessments gave chances to assess progress, address issues, and match system functionality with user expectations. This process ensured that the smart dashboard web-based system developed in an organized yet flexible manner, resulting in a dependable, and scalable Program. By following a structured development approach, the researchers were able to manage tasks efficiently while still allowing adjustments when needed. This helped address issues early and improve system features based on feedback. the system can adapt to growing needs while maintaining reliable performance and functionality.

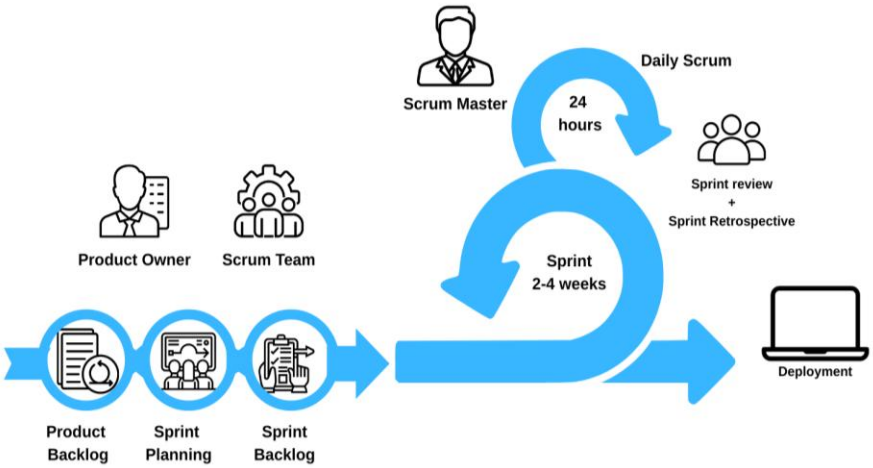


Figure 3. Agile Scrum Diagram

Product Backlog

The product backlog for the project was the initial source for the framework's first phase which included a list of tasks to be completed. It had a list of procedures for creating the system. Table 2 displayed the features in the user's product backlog and specifications for the entire system from the perspective of the user. The most important items were listed first in the backlog, which was arranged according to priority. It also included estimated hours per day, which showed how much work each task required. The product backlog supported the development team's systematic planning, tracking, and management of the project's progress. Through clearly outlining each task and its accompanying requirements, the team was able to efficiently manage resources while also ensuring that important features were developed within the period specified. This organized strategy reduced misunderstandings while also ensuring that project goals and expectations. By establishing structured communication and systematic planning, the development process.

Table 2. Product Backlog for Super Admin

Priority	Super Admin Story	Estimated Time
1	As a Super Admin, I want to manage all admin activities and oversee their actions.	5 hrs
2	As a Super Admin, I want to generate and view blotter reports.	4 hrs
3	As a Super Admin, I want to manage user roles and permissions effectively.	3 hrs
4	As a Super Admin, I want to manage resident profiles and information.	4 hrs
5	As a Super Admin, I want to configure system settings and preferences.	3 hrs
6	As a Super Admin, I want to monitor system performance and user activities.	4 hrs
7	As a Super Admin, I want to manage officials and staff records.	4 hrs
8	As a Super Admin, I want to generate reports on user activities and statistics.	5 hrs
9	As a Super Admin, I want to ensure data integrity and backup processes.	4 hrs
10	As a Super Admin, I want to gather feedback from users regarding their experience with the system to improve it.	2 hrs

Table 3 shows the list of features developed for the admin side of the system, arranged according to priority. Each admin story represents a task or function that the administrator needs to perform in order to manage the system properly, such as logging in securely, handling requests, and monitoring important data.

Table 3. Product Backlog for Admin

Priority	Admin Story	Estimated Time
1	As an admin, I want to log in and access the smart dashboard securely.	2hrs
2	As an admin, I want to view and manage pending certificate requests.	3hrs
3	As an admin, I want to filter and track requests by type, date, and status.	3hrs
4	As an admin, I want to manage Announcements and publish them on the dashboard.	2hrs
5	As an admin, I want to monitor demographic data (e.g., male/female count, total residents.	3hrs
6	As an admin, I want to view and edit the barangay calendar with important dates and events.	2hrs
7	As an admin, I want to generate reports based on request statistics and resident data.	4hrs
8	As an admin, I want to manage officials and assign system roles.	4hrs

Sprint Planning

Table 4 shows the sprint planning tasks for the Super Admin, showing the prioritized features needed to effectively monitor and manage the entire system. Each item describes a specific responsibility of the Super Admin, securing system access, and monitoring overall system performance.

Table 4. Sprint Planning for Super Admin

Priority	Item	Task	Estimated Time
1	As a Super Admin, I want to view all admin activities immediately.	Develop a comprehensive dashboard to list and manage all admin activities.	12 hrs
2	As a Super Admin, I want secure access to the admin panel.	Implement login functionality with role-based access for the admin panel.	10 hrs
3	As a Super Admin, I want to see quick stats about user activities and system performance.	Design and develop stat widgets for user activities, system performance, and key metrics.	16 hrs
4	As a Super Admin, I want to view all system alerts and notifications.	Create a notification section with recent alerts and important messages.	8 hrs
5	As a Super Admin, I want to manage user roles and permissions easily.	Integrate a user management interface for role assignments and permission settings.	12 hrs
6	As a Super Admin, I want to navigate easily to all modules from the admin panel.	Implement navigation links to Reports, Users, System Settings, and other modules.	6 hrs
7	As a Super Admin, I want to generate comprehensive reports from the system data.	Add report generation logic and export features (PDF/Excel) for user activities and system metrics.	10 hrs

Table 5 shows the admin role, arranged based on priority to guide the development process. Each item represents a specific function that the admin needs to perform, such as viewing pending certificate requests, accessing the dashboard securely, and monitoring important barangay data.

Table 5. Sprint Planning for Admin

Priority	Item	Task	Estimated Time
1	As an admin, I want to view pending certificate request immediately.	Develop a table view on the dashboard to list and manage pending certificate requests.	12hrs
2	As an admin, I want secure access to the dashboard.	Implement login functionality with role-based access for the dashboard	10hrs
3	As an admin, I want to see quick stats like total residents and voters.	Design and develop stat widgets for residents, voters, and requests.	16hrs
4	As an admin, I want to view barangay announcements and alerts	Create a new news/announcement section with recent posts.	8hrs
5	As an admin, I want to view and manage the barangay calendar.	Integrate an interactive calendar with event management capabilities.	12hrs
6	As an admin, I want to navigate easily to all modules from the dashboard.	Implement navigation links to Records, Officials, Requests, and other modules.	6hrs
7	As an admin, I want to generate report from the dashboard data.	Add report generation logic and export features (PDF/Excel).	10hrs

System Flowchart

This displayed the process and flow of the process visually. Figure 4 shows the flowchart overview for the smart dashboard system. This displayed the user's access to the smart dashboard features.

A Smart Dashboard that simplified services for administrators, officials, and residents was shown in the flowchart. Users could access modules according to their roles once they had logged in and been authenticated. Managing announcements, submitting blotter reports, requesting certificates, and viewing resident statistics. Permit renewals, digital records, and content were managed by supporting modules. The system provided access through PWA and reports and analytics automation.

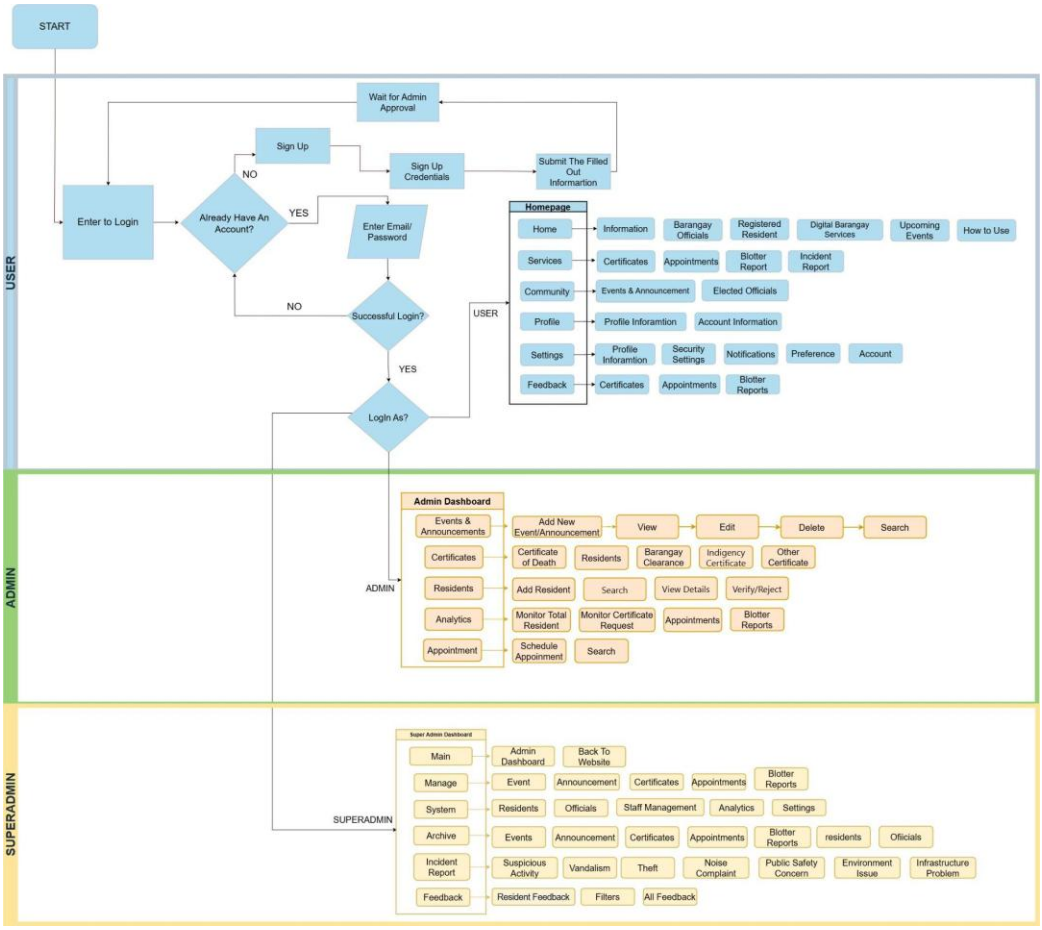


Figure 4. System Flowchart

Use Case Diagram

Figure 5 shows the capabilities of a web-based system. The administrator and the user were the two primary actors in a barangay management system, and this use case diagram showed their roles and interactions. In addition to managing resident records, supervising business permit renewals, posting community announcements, viewing dashboard statistics, creating reports, and managing other officials, the administrator had full access to the system. The User, who was usually a resident, had more restricted access that was centered on public services; they could request certificates (for example, barangay clearances) and monitor the progress of those requests. In order to ensure effective administration and user-friendly service delivery, the diagram clearly delineated the range of functionalities accessible to each user type, aiding in the differentiation of responsibilities and system access levels.

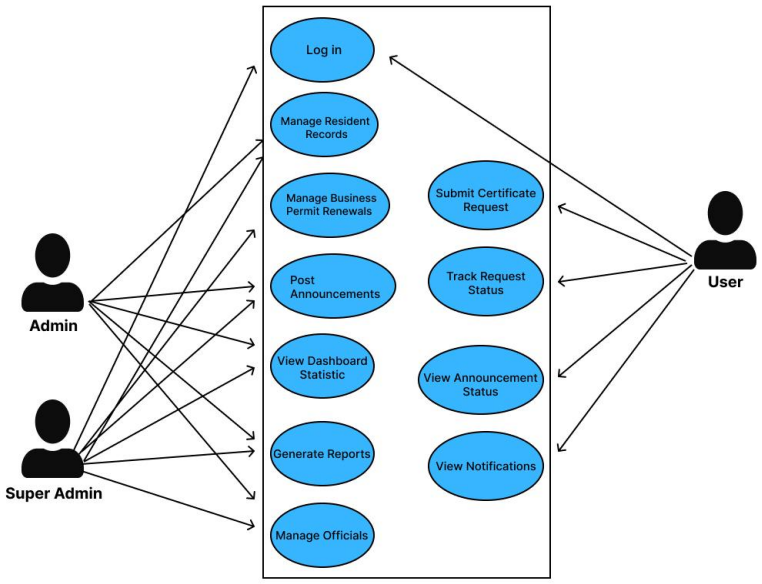


Figure 5. Use Case Diagram

Context Diagram

The context diagram of the smart dashboard system provided a high-level representation of how the system interacted with its two main external entities: the resident/user and the barangay official/admin. Residents or users interacted with the system by creating accounts, logging in, requesting certificates, submitting blotter reports, and viewing announcements and events. In return, the system provided them with access to certificate statuses, an event calendar, and public announcements. Meanwhile, the barangay official or admin uses the system to manage content, post announcements, handle blotter reports, approve certificates, update resident information, monitor service requests, add community events, and create user accounts. The system, in turn, provides the admin with access to dashboards, analytics, request overviews, and notifications to aid in monitoring and decision-making.

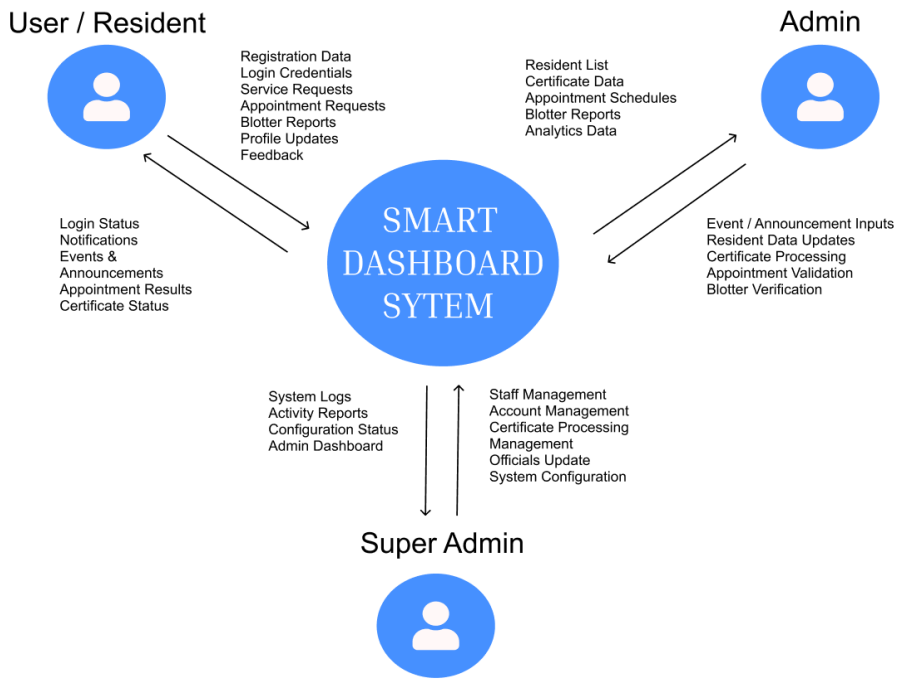


Figure 6. Context Diagram

Data Flow Diagram

The Data Flow Diagram (DFD) illustrated how data moved through the Smart Dashboard System by showing the main processes, data stores, and interactions among the Super Admin, Admin, and Resident/User.

Residents interacted with the system through the Authentication process (P1), where their login or signup information was verified using the Users Database (D1). After successful authentication, residents accessed the User Services process (P2), which retrieved user profiles, announcements, events, and appointments from the Events & Appointments Database (D3) and returned the necessary service data to the user.

Admins used the Admin Dashboard (P3) to manage certificates, announcements, and events. Their actions updated the Certificates Database (D2) and the Events & Appointments Database (D3), ensuring that residents received accurate information and that certificate records were properly maintained.

The Super Admin interacted with the system through the Super Admin Panel (P4), which managed system access and administrative controls. All Super Admin activities were recorded in the Logs / Audit Database (D4), enabling secure tracking and monitoring of system operations. Ensuring that only authorized users could modify information. The logging and monitoring mechanism increased transparency and accountability by keeping full records of all super admin operations. These records may be examined for system reviews, troubleshooting, and compliance purposes, making it easier to discover errors and ensuring that administrative activities followed established standards and procedures.

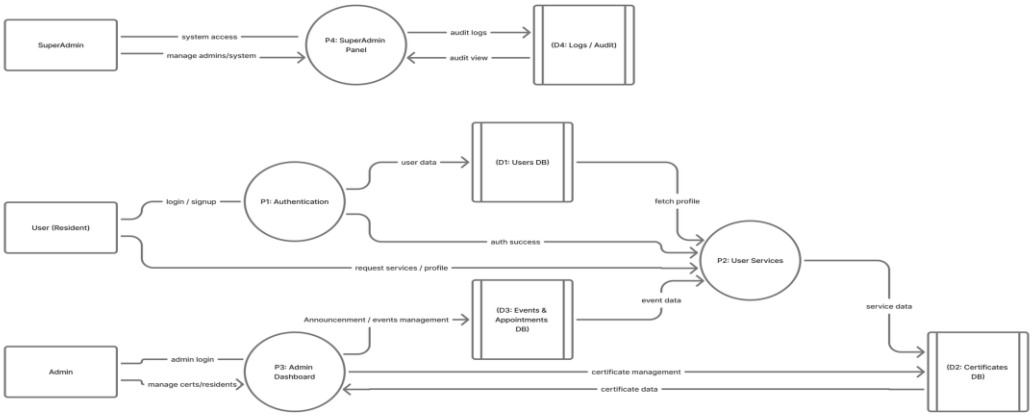


Figure 7. Data Flow Diagram

Materials

This section presented the hardware and software resources utilized in the development of the Smart Dashboard System. These tools were essential in ensuring that the system met the required functionality, performance, and user accessibility. Development tools, programming languages, and database management systems were carefully chosen to provide scalability, security, and ease of use.

Software

The software tools listed below were used during the development of the web based Smart Dashboard System. Each tool played a significant role in building the system's frontend and backend components. Firebase was used not only for real-time data synchronization and user authentication but also for hosting the application, ensuring that the system could scale efficiently and securely in a production environment.

Frontend Development

Visual Studio Code. The development team had opted for Visual Studio Code as the primary code editor due to its versatility, powerful integrated development environment (IDE) features, and robust support for a wide range of programming languages and frameworks. Visual Studio Code offered a comprehensive set of features such as intelligent code completion, debugging, and a wide variety of extensions, making it an ideal tool for both frontend and backend development. Its lightweight nature, combined with these advanced features, allowed the developers to efficiently manage the project and ensure smooth code execution throughout the development lifecycle.

ReactJS. For the front-end development, ReactJS was selected as the JavaScript library for building interactive user interfaces. ReactJS enabled developers to create highly dynamic and responsive web pages by building reusable UI components. This component-based structure enhanced development efficiency, as developers could easily manage and maintain different parts of the user interface independently. React's ability to update and render the right components when data changed made it ideal for creating interactive and user-friendly interfaces that responded to user inputs in real time.

NextJS. To further improve the performance and user experience, NextJS was integrated into the system. NextJS was a powerful React framework that facilitated server-side rendering (SSR), which enhanced the page load times by generating static pages at build time and improved search engine optimization (SEO) through better indexing by search engines. With NextJS, the application

benefited from faster initial page loads and improved SEO performance, which was crucial for increasing visibility and accessibility, especially for users who might have relied on search engines to find the system.

Tailwind CSS. For styling, Tailwind CSS was applied to design the web pages using a utility-first approach. This approach allows developers to apply pre-defined utility classes directly in the HTML structure, making it easier to create responsive and modern designs without the need for writing custom CSS from scratch. Tailwind CSS empowers developers to quickly build aesthetically pleasing and functional user interfaces while maintaining high levels of flexibility and customization. By using utility classes, the team was able to maintain a consistent design language throughout the application while also ensuring it was fully responsive on various devices.

ShadCN. was chosen for managing the UI components, which helped ensure a polished and consistent look throughout the application. As a component library, ShadCN provides pre-built components that follow modern design principles, making it easier for developers to implement polished UI elements without having to build them from scratch. This streamlined the development process, allowing the team to focus more on functionality and user experience, knowing that the UI components would be consistently styled and aligned with best design practices. Development became more efficient, collaboration improved, and the overall product quality was enhanced through a unified and maintainable design system. Having a consistent design structure allowed the team to work more smoothly and avoid confusion during development. It also made it easier to update features without affecting other parts of the system.

Backend Development

Firebase. For the backend infrastructure, Firebase was chosen as the cloud platform to manage real-time data synchronization, user authentication, and hosting the system. Firebase is a fully managed platform that provides a scalable and secure environment for building and deploying web applications. Its real-time database capabilities ensure that changes made by one user are immediately reflected across all connected clients, providing a seamless and dynamic user experience. Additionally, Firebase's built-in user authentication system allows for easy implementation of secure login and registration features, ensuring that only authorized users can access certain parts of the application.

TypeScript. The backend logic was written in TypeScript, a statically typed language that offered significant advantages over traditional JavaScript. TypeScript helped prevent runtime errors by providing compile-time checks, which ensured that potential issues could be detected early in the development process. This feature was particularly useful in large-scale projects were maintaining code quality and preventing bugs could be challenging. With TypeScript, developers were able to write more maintainable and less error-prone code, which improved long-term project stability and reduced the time spent debugging.

Firebase Real-time Database. The system’s data was managed using Firebase’s real-time database, a cloud-based NoSQL database designed for seamless data synchronization. The database stored and organized user data, event records, announcements, and other critical information. By storing data in JSON format, Firebase eliminated the need for manual updates or refreshes.

Any changes made to the database were instantly reflected across all connected clients, making it ideal for applications that required real-time data updates. This feature was particularly important in the Smart Dashboard system, as it ensured that users received the most up-to-date information in real time, without delays.

Firebase Hosting. was utilized to deploy the system, providing a fast, secure, and scalable platform for hosting the web application. Firebase Hosting delivered content through a global content delivery network (CDN), ensuring that users across different geographic locations experienced minimal latency and fast load times. The hosting platform seamlessly integrated with the rest of Firebase's services, such as the real-time database and authentication system, making it an ideal choice for the Smart Dashboard system.

With Firebase Hosting, the system efficiently handled growth in the user base and traffic, scaling automatically without the need for manual intervention. Firebase Hosting's ease of deployment and management eased web application maintenance. Developers were able to quickly update and deploy new versions of the system, enabling upgrades and fixes to be incorporated without affecting service. This flexibility allowed for constant system improvement and guaranteed that the platform remained responsive to changing user needs. It also supported faster updates and smoother adaptations, ensuring long-term reliability and user satisfaction. With a well-structured system in place, improvements and fixes could be applied without disrupting existing features. This made it easier to maintain the system and respond to changing user needs.

Hardware

The system was developed and tested on a workstation capable of running all the required development tools and frameworks smoothly. The table below outlines the hardware specifications used during development.

Table 6. Hardware Specification

Hardware	Specification
RAM	8GB
CPU	Intel core i7
Operating System (OS)	Windows 10

Table 6 shows that the system was developed using a computer that was operated. An intel core i7 gen cpu running windows 10 and 8gb ram. Enough resources are guaranteed for development and software devices with this setup.

Table 7. Minimum Requirements for Laptop/PC

Hardware	Specification
RAM	4GB
Storage (HDD/SSD)	128GB
CPU	Intel core i3 / AMD Ryzen 3

Table 7 lists the minimal hardware requirements for the developed system to function efficiently. For a large number of users, these minimal requirements guarantee that the system operates without crashing. Those who need to multitask or have heavy workloads can benefit greatly from upgrading components, especially RAM and storage. Complying with the recommended

hardware specifications helps ensure stable and reliable system performance. As the system grows and more features are implemented, improved hardware capacity. This ensures that the system remains dependable and continues to meet the operational demands of the barangay over time.

Table 8. Requirements for Smartphone or Tablet

Hardware	Specification
Android OS	Android 8.0(Oreo) or latest
RAM	2GB
BROWSER	At least 2GB free space
PROCESSOR	Quad - core

The minimal requirements for gaining access to the smart dashboard are listed in Table 8. management system through Android tablets and smartphones' web browsers. Providing a seamless user experience on the mobile device is essential to meeting these requirements.

Sprint Backlog

The sprint backlog for administrators and users is displayed in Table 9 presents the sprint backlog for the Super Admin, outlining the prioritized tasks needed to manage and oversee the entire system. Each Super Admin story focuses on high-level responsibilities such as monitoring admin activities, managing user roles, handling resident records, and configuring system settings. This table highlights the Super Admin’s role in maintaining system control, performance, and data integrity, ensuring the platform runs smoothly.

Table 9. Sprint Backlog for Super Admin

Priority	Super Admin Story	Task	Estimated Time
1	As a Super Admin, I want to manage all admin activities and oversee their actions.	Implement a comprehensive dashboard for monitoring admin activities.	5hrs
2	As a Super Admin, I want to generate and view blotter reports.	Create a module for generating and viewing blotter reports.	4hrs
3	As a Super Admin, I want to manage user roles and permissions effectively.	Develop a user management interface for role assignments and permissions.	3hrs
4	As a Super Admin, I want to manage resident profiles and information.	Create a module for adding, editing, and viewing resident profiles and information.	4hrs
5	As a Super Admin, I want to configure system settings and preferences.	Create an interface for managing system settings.	3hrs
6	As a Super Admin, I want to monitor system performance and user activities.	Develop analytics tools for tracking system performance metrics.	4hrs
7	As a Super Admin, I want to manage officials and staff records.	Create a module for adding, editing, and removing officials and staff members.	4hrs
8	As a Super Admin, I want to generate reports on user activities and statistics.	Implement reporting functionality for tracking user activities and statistics.	5hrs

Table 10 shows the sprint backlog for the admin role, emphasizing daily barangay operations and management tasks. The admin stories focus on secure system access, dashboard monitoring, posting announcements, handling certificate requests, and managing users and events.

Table 10. Sprint Backlog for Admin

Priority	Admin Story	Task	Estimated Time
1	As an Admin, I want to log in securely to manage barangay operations.	Implement login authentication with admin role verification.	3hrs
2	As an Admin, I want to view a dashboard summarizing key barangay statistics.	Display widgets for total residents, certificate requests, voters, etc.	4hrs
3	As an Admin, I want to post and update barangay announcements.	Build an interface to create, edit, and delete announcements.	3hrs
4	As an Admin, I want to view and monitor all certificate requests.	Create an admin view for tracking all incoming certificate requests and their statuses.	4hrs
5	As an Admin, I want to add and assign roles to other users.	Develop a user management panel with role assignment options.	3hrs
6	As an Admin, I want to monitor system activity and requests.	Provide a dashboard section for tracking system logs, request frequency, and trends.	3hrs
7	As an Admin, I want to schedule and manage barangay events.	Add functionality to create and manage events in the barangay calendar.	4hrs

Table 11 presents the sprint backlog for the User, detailing the features that support user interaction with the system. The user stories focus on account access, viewing dashboard information, managing certificate requests, checking announcements and events, and navigating the system easily.

Table 11. Sprint Backlog for User

Priority	User Story	Task	Specification
1	As a User, I want to create an account and log in.	Develop a login and registration page with authentication.	3hrs
2	As a User, I want to view dashboard statistics and glance.	View quick summary cards showing total residents, voters, gender breakdown etc.	4hrs
3	As a User, I want to manage my certificate requests easily.	Access and track my pending certificate requests from a table.	4hrs
4	As a User, I want to view important barangay announcements.	Display latest announcements in a dedicated section or pop-up panel.	2hrs
5	As a User, I want to check barangay events and calendar view.	Open and navigate a monthly calendar that shows upcoming barangay events.	3hrs
6	As a User, I want to easy access to all barangay modules.	Use the sidebar to quickly navigate to announcements, officials,and other sections.	3hrs
7	As a User, I want to see my Profile information on the dashboard	View my name, email and role,details displayed in the interface.	3hrs
8	As a User, I want the dashboard to work well on any device.	Experience smooth, responsive layout on mobile, table, desktop.	3hrs
9	As a User, I want to provide feedback on my experience.	Implement a feedback form for users to submit their experiences and suggestions.	2hrs

Project Testing and Debugging

The developers employed different techniques for testing to guarantee the level of quality and reliability of the system. This stage of software testing involved testing and examining every module component of the web-based system to ensure that the developed system operated as planned. Lastly, acceptance testing allowed users to evaluate the developed system to determine if it satisfied the specifications and standards of the intended system, in which case researchers took user input into account to optimize the system completely.

Unit testing

Unit testing was important in the development and maintenance of the Smart Dashboard platform. This approach involved isolating individual components or features of the system and thoroughly testing them to ensure they functioned as intended. In the context of the Smart Dashboard, unit testing verified core functionalities such as data input, resident request handling, automated processing workflows, and announcement management. By identifying and resolving issues at the unit level, developers ensured that each module, whether it was the event notification system or form validation process, performed reliably. This proactive testing method enhanced the stability and performance of the platform and supported continuous integration practices, allowing for seamless updates without disrupting existing features. Ultimately, unit testing helped maintain the quality and trustworthiness of the dashboard, reinforcing its role in promoting resident engagement and efficient, data-driven governance.

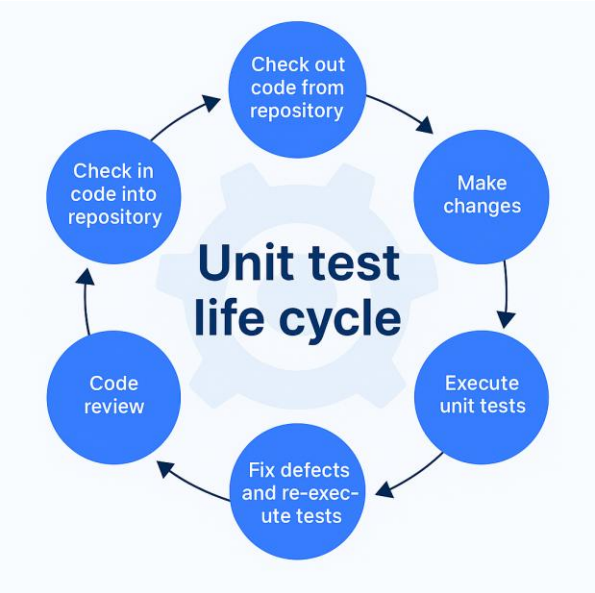


Figure 8. Unit Testing

Integration testing

Integration testing involved combining and testing multiple modules of a system to ensure they worked together as expected. In the development of the Smart Dashboard platform, this phase focused on verifying the interactions between various integrated components, such as the resident request submission form, the automated data processing module, and the announcement/event board. Integration testing ensured that data flowed smoothly between modules, such as when a resident's Request was submitted, processed, and the status was reflected accurately in the notification system. This process helped uncover interface-related issues and communication gaps between subsystems, contributing to the reliability and consistency of the platform. By validating these interactions, integration testing strengthened the overall functionality of the Smart Dashboard, supporting its mission of efficient, data-driven governance and enhanced resident engagement.

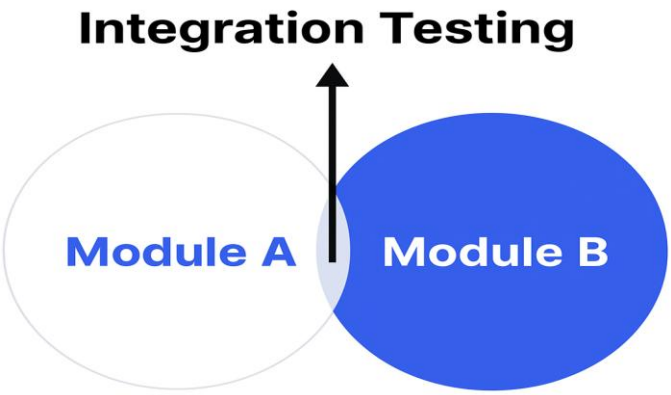


Figure 9. Integration Testing

System Testing

System testing is a critical phase in the development lifecycle of the smart governance platform, as it ensures the overall functionality, stability, and reliability of the system. This phase involves comprehensive validation of the platform's modules, ensuring that key features such as resident request submissions, announcement visibility, data analytics, and notification services operate as expected across web and mobile platforms.

Through user acceptance testing (UAT) and structured feedback mechanisms, testers evaluate whether the platform meets user needs, supports citizen engagement, and aligns with the governance objectives of local barangay units. UAT results provide insights into usability, responsiveness, and user satisfaction, contributing to system refinement and enhanced adoption. Furthermore, performance testing is conducted to measure the system's responsiveness and scalability under various load conditions. This includes assessing how the dashboard handles concurrent users, simultaneous data

requests, and peak usage scenarios, which is vital for ensuring uninterrupted access and functionality in real-time governance operations.

Compliance and security testing are also integral to this stage, ensuring that the platform complies with relevant data protection regulations, privacy standards, and information security protocols. These tests mitigate risks related to unauthorized access, data breaches, and legal non-compliance. System testing plays a vital role in verifying the efficiency, performance, and security of the smart governance platform, supporting its goal of fostering transparent, participatory, and data-driven local governance.

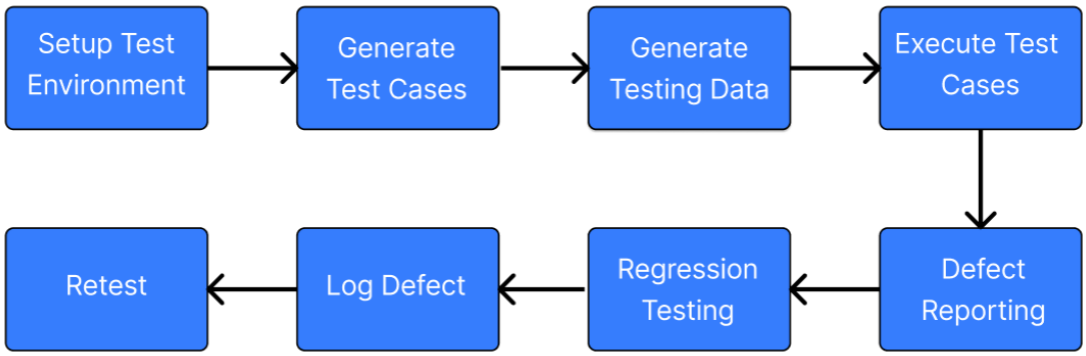


Figure 10. System Testing

Load Testing

Load Testing was conducted using Apache JMeter to analyze the system's performance with 100 users. The system achieved a maximum response with 100 users and performed even better with 200, maintaining a 0% error rate and proper response time. Among 500 users, the system-maintained accuracy and remained error-free. When the load reached 800 users, response time increased; however, the system remained stable and error-free. Overall,

the results showed that the system could handle larger user loads while maintaining functionality.

Table 12. Load Testing Performance

Users	Max Response Time (ms)	Error (%)	Performance Level	Remarks
100	312ms	0.00%	Excellent	The system responds quickly and operates efficiently under low user load
200	1350ms	0.00%	Very Good	Performance remains stable with a slight increase in response time
500	1350ms	0.00%	Good	The system maintains reliable performance under moderate user load
800	6178ms	0.00%	Fair	Response time increases however, the system continues to function correctly without errors

Stress Testing

Stress Testing was performed using JMeter to evaluate the Smart Dashboard System's performance, stability, and reliability under high-usage situations. The purpose of this test was to find out how the system responded when tested with workloads that exceeded its usual functional capacity, as well as to identify potential issues that might have impacted system availability and user experience. The results of the JMeter stress tests gave useful information for system optimization and performance adjustment. By fixing the issues

discovered during testing, the developers were able to improve resource handling, increase system durability using Jmeter.

Table 13. Stress Testing Performance

Users	Max Response Time (ms)	Error (%)	Performance Level	Remarks
100	390ms	0.00%	Excellent	The system operates quickly and maintains stability under light user loads
200	498ms	0.00%	Excellent	Performance remains fast with no errors as users increase
800	498ms	0.00%	Very Good	System maintains quick response time under moderate load
1200	2606ms	0.00%	Fair	Response time increases but the system remains stable with no errors
2000	5752ms	0.00%	Acceptable	System can handle high load, though response time is slower

Compatibility Testing

This procedure evaluated the system across many platforms; in this case, a limited number of browsers were used to evaluate the web system. The system had to run accordingly on the most used browsers, including Google Chrome, Microsoft Edge, and Opera Browser, which were audited using Lighthouse. In this situation, when one browser became inappropriate, the developers were more aware of the system’s viability. Multi-browser testing guaranteed that the system's interface, functionality, and performance were

consistent regardless of which browser the end user used. By finding errors in display, loading speed, and responsiveness, the developers were able solve compatibility issues early in the testing process. In addition, Lighthouse gave measurable insights into performance, accessibility, best practices, and SEO compliance. These evaluations assisted developers in optimizing the system to fulfill quality standards and enhance the overall user experience. As a result, the system became more dependable and available to an increased number of users on multiple platforms.

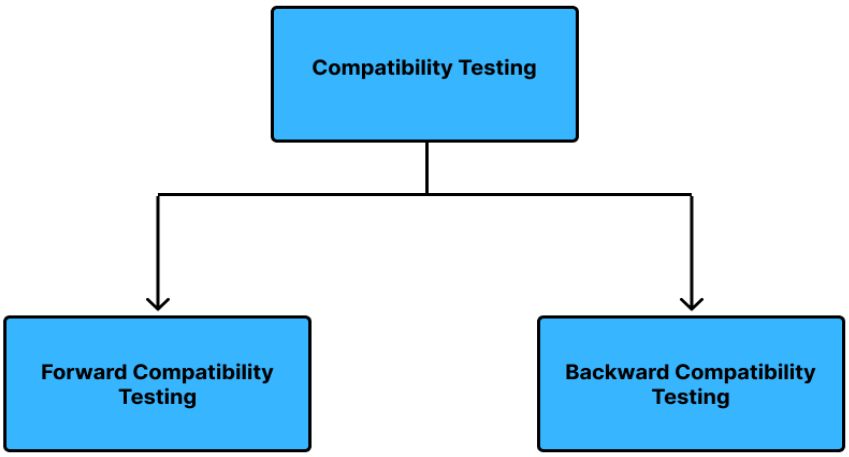


Figure 11. Compatibility Testing Model Diagram

Acceptance Testing

Acceptance testing was the final stage, during which barangay officials and residents tested the developed system based on real-world usage and requirements. This phase assessed whether the system met the predefined acceptance criteria and addressed the actual needs of the users.

Participants evaluated all system features, including creating accounts, submitting blotter reports, requesting certificates, and viewing announcements and event schedules. Based on their feedback, developers verified that the

system was fully functional, user-friendly, and ready for deployment within the community. The findings revealed that the Smart Dashboard System can increase access to information and services, making it a useful tool for both citizens and barangay officials in day-to-day operations.

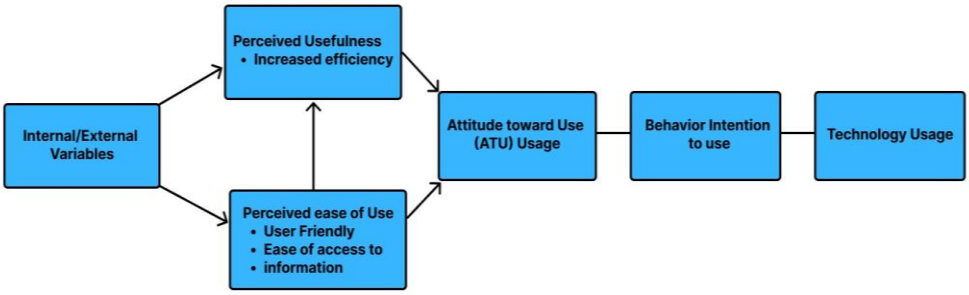


Figure 12. Acceptance Testing Model

Sampling Design

Purposive sampling was employed in this study, wherein participants were selected based on their relevance to the objectives of the research. This non-probability sampling method was chosen because it enabled the selection of individuals who were most accessible and possessed the necessary background to provide meaningful insights within a limited timeframe.

A total of 50 respondents were involved in the study. This included fifty (50) barangay personnel and residents, as well as two (2) IT experts or professionals with knowledge and experience in web-based systems. The barangay participants were selected based on their familiarity with the community's processes and services, while the IT experts were chosen for their expertise in evaluating system usability, performance, and technical functionality. The participation of IT professionals improved the system's

technical evaluation by providing professional assessments of its design, structure, and general performance. Their inputs helped validate the system's compliance with web-based standards and best practices.

Population of the Study

The population of this study consisted of barangay staff, residents, and IT professionals who participated in the evaluation and testing of the developed Smart Dashboard system. These individuals were selected as they represented the primary users of the system and were directly involved in the processes the platform aimed to improve. Their feedback was essential in evaluating the system's performance, identifying strengths, and pinpointing areas for improvement.

Barangay staff were chosen to assess the system using the Technology Acceptance Model (TAM) questionnaire. This model evaluated user perception regarding the system's ease of use and usefulness. Since the barangay staff interacted with the system regularly for administrative tasks, their experiences were crucial in determining the system's practicality and effectiveness.

The table below shows the two (2) IT expert respondents who assessed, analyzed, and evaluated the feasibility of creating a web-based system, as well as the overall performance and features of the developed Smart Dashboard system. These experts provided valuable insights into the system's functionality and offered recommendations for future improvements and optimizations. Their feedback helped identify areas that could be enhanced to make the system more efficient and user-friendly. This input was important in refining the system and guiding possible upgrades for future development.

Table 14. Respondents of the Research Project (ISO 25010)

CATEGORY OF RESPONDENTS	SAMPLE
IT Experts	2
Total	2

The table below shows the respondents (3) Barangay Staff and (47) Residents by assessing, analyzing, and evaluating the web-based system using the TAM Questionnaire for the evaluation of the overall performance and features of the developed web-based system.

Table 15. Respondents of the Research Project (TAM)

CATEGORY OF RESPONDENTS	SAMPLE
Barangay Staff	3
Residents	47
Total	50

Data Collection Instrument

To collect data, the study used both structured questionnaires and semi-structured interviews. The questionnaires measured the users' perceptions based on two main TAM factors: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). These helped identify how useful the employees found the system in their daily work and how easy it was for them to use its features. This feedback also shows their willingness to adopt the system and their interest in using it regularly. Alongside the questionnaires, in-depth interviews

were conducted to gather more detailed opinions from the users. The interview guide contained 20 questions that focused on different parts of the system, such as design, accessibility, security, and accuracy of data. These face-to-face interviews provided deeper insights that supported the results of the questionnaires.

To measure user satisfaction, the study used a Likert Scale ranging from 1 to 4, where 1 meant “Strongly Disagree” and 4 meant “Strongly Agree.” These ratings were analyzed to understand how well the system’s features met user expectations and to identify areas that might have needed improvement.

Table 16. Likert Scale for User Feedback

RATING	RANGE	INTERPRETATION
4	3.40 – 4.19	Strongly Agree
3	2.60 – 3.39	Agree
2	1.80 – 2.59	Disagree
1	1.00 – 1.79	Strongly Disagree

Statistical Treatment

The IT Expert Evaluation was conducted to assess the technical quality of the Smart Dashboard system based on ISO 25010 software quality standards. A total of IT experts participated in the study and provided their feedback through structured questionnaires.

The IT experts rated each system feature using a 5-point scale, as shown below:

Table 17. Interpretation of Rating Scale for Evaluating System

RATING	INTERPRETATION
5	Excellent
4	Good
3	Fair
2	Needs Improvement
1	Poor

This statistical treatment allowed the study to summarize and quantify the IT experts' feedback. It clearly identified which parts of the system had met the expected standards and which areas had needed improvement. By using both descriptive statistics and expert evaluation, the study provided reliable technical insights that supported the overall assessment of the Smart Dashboard system.

Ethical consideration

Before proceeding with the study, the researchers obtained informed consent from all participants. Participation in the survey questionnaires and in-depth interviews was entirely voluntary, and respondents had the right to withdraw from the study at any point without facing any consequences. The researchers clearly explained the study's objectives, scope, and potential benefits to ensure transparency and promote participant understanding. All personal data provided by respondents was treated with strict confidentiality and was used exclusively for the purpose of evaluating the Smart Dashboard system. The researchers were committed to treating every participant with

respect, fairness, and impartiality throughout the data collection process. All data was collected and analyzed with integrity, ensuring that no information was misrepresented or falsified. The results were reported accurately, reflecting the genuine feedback and insights obtained during the research. To prevent unauthorized use or disclosure, only the researchers had access to the obtained data. To secure sensitive information, appropriate data handling and storage methods were followed. Through following recognized ethical research guidance resulting the validity of the Smart Dashboard.

CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter presented the results and discussions of the Smart Dashboard Web-Based platform. It highlighted the platform's key features, interfaces, and functionalities, explaining how each module aligned with the specific objectives of the study. The results emphasized the performance of various modules such as certificates management, appointment scheduling, feedback handling, analytics, and administrative tools, demonstrating how the platform enhanced efficiency, transparency, and accessibility in barangay operations. Each module offered a detailed discussion of outcomes, providing insights into how the Smart Dashboard met the needs of both residents and administrators.

Development of the Smart Dashboard Web-Based Platform for enhanced community engagement and governance transparency

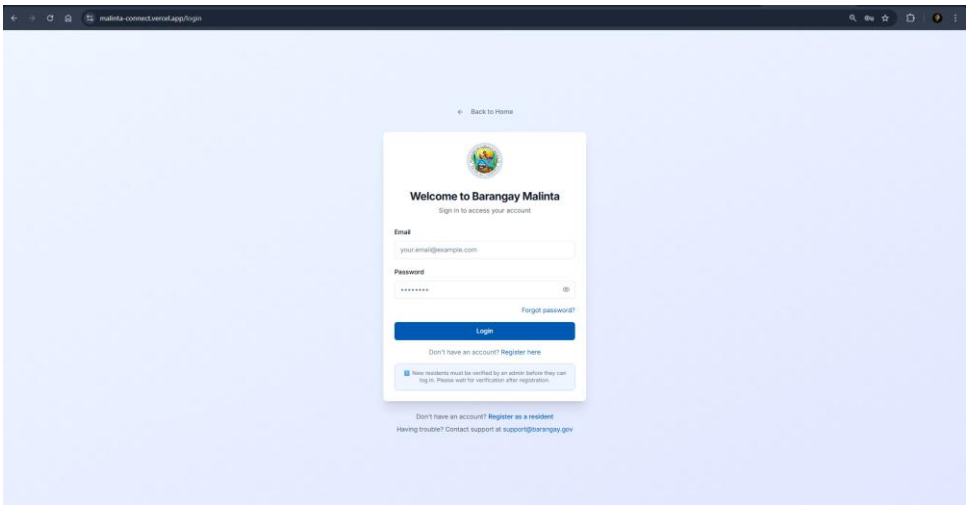


Figure 13. Smart Dashboard Log in for Users & Administrators

The development of the Smart Dashboard was first managed with the goal of creating a secure, intuitive, and accessible system for both residents and barangay officials. The login interface featured a clean and user-friendly authentication process that allowed citizens to easily register, sign in, and access the services provided by the barangay. The login module ensured data privacy while streamlining access to various features, such as certificate requests, announcements, and community updates.

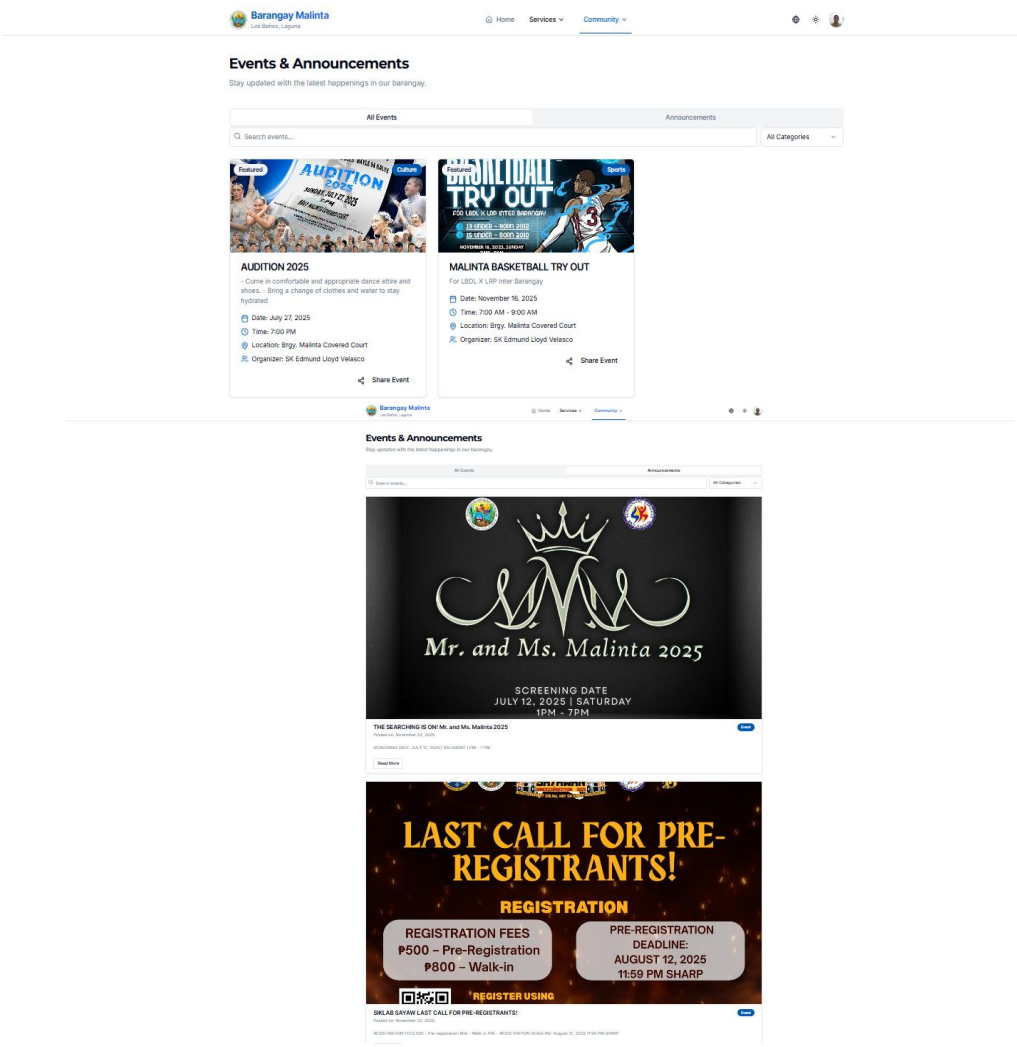


Figure 15. Events & Announcements

In this figure, the events and announcements served as the interface to keep the residents updated on the barangay activities, including community meetings, health programs, and other local events. This platform not only

spread important information but also encouraged resident participation and strengthened a sense of community engagement within the barangay.

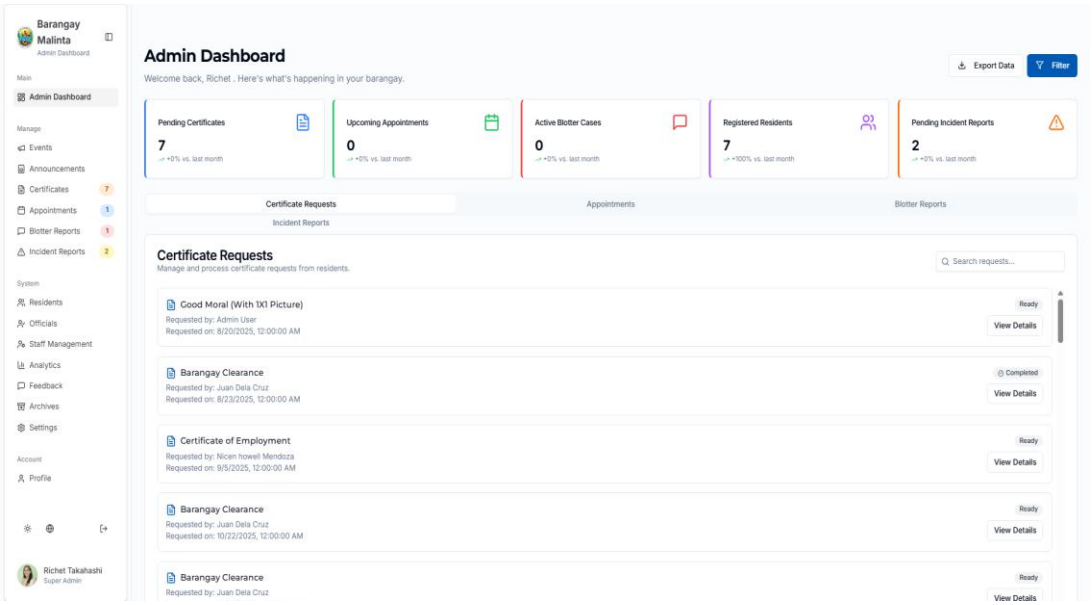


Figure 16. Administrator Dashboard

Figure 16 shows the Administrator Dashboard. It was essential for achieving the Smart Dashboard goal of improving data management, monitoring, and decision-making within the barangay. This module allowed administrators to access real-time analytics, service summaries, totals of certificate requests, user activities, and appointment updates, all presented in a clear and organized format. This centralized control panel enabled administrators to efficiently view system operations, respond to requests, and identify trends that facilitated data-driven governance. By offering a comprehensive overview of community interactions and service performance, the Administrator Dashboard enhanced operational efficiency and ensured smoother service across all platform modules. It centralizes essential management tools, allowing administrators to monitor transactions, manage records, track user activities, and review system reports in real time.

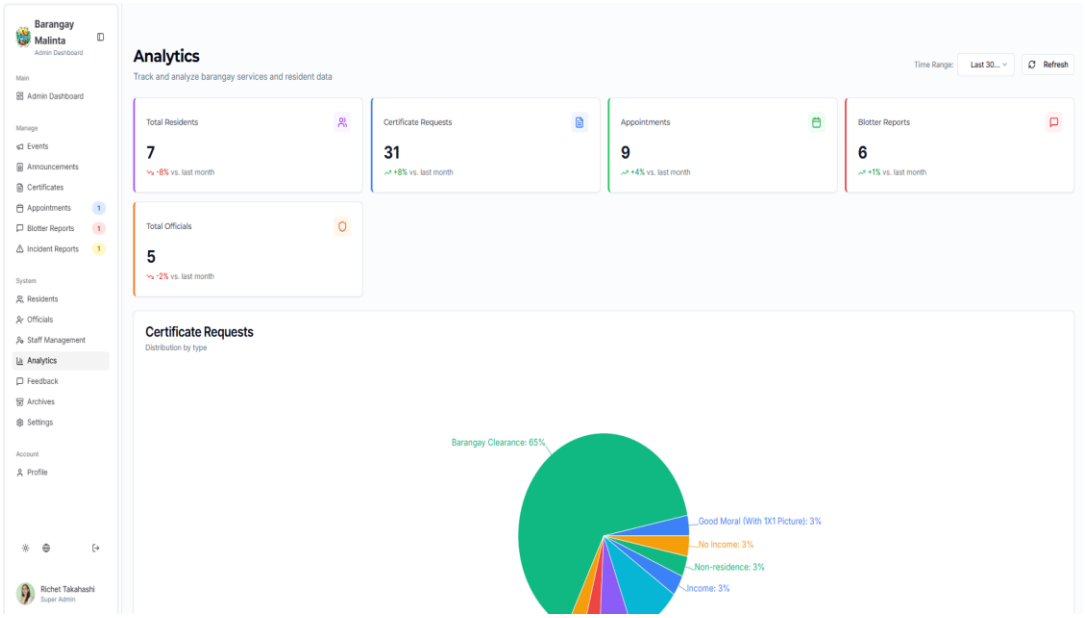


Figure 17. Analytics Overview Administrator

Figure 17 shows the Analytics Overview Administrator, which allowed administrators to view a clear visual summary of key barangay operations. This section displayed organized charts and statistics that reflected metrics such as certificate request volumes, appointments, and other important system activities. By offering real-time insights, the analytics tool assisted officials in identifying patterns, monitoring community needs, and evaluating the effectiveness of services. This enabled administrators to make informed decisions, allocate resources efficiently, and improve overall service delivery. Through this feature, the Smart Dashboard effectively transformed raw data into practical information, thereby enhancing transparency and operational performance. The vision not only improved operational efficiency, but it also increased community trust in local governance by proving that citizens' needs were planned for and fulfilled on time. Overall, the Smart Dashboard's variety of editable reports and predictive analytics validated it as a critical tool for promoting data-driven decision-making in barangay administration.

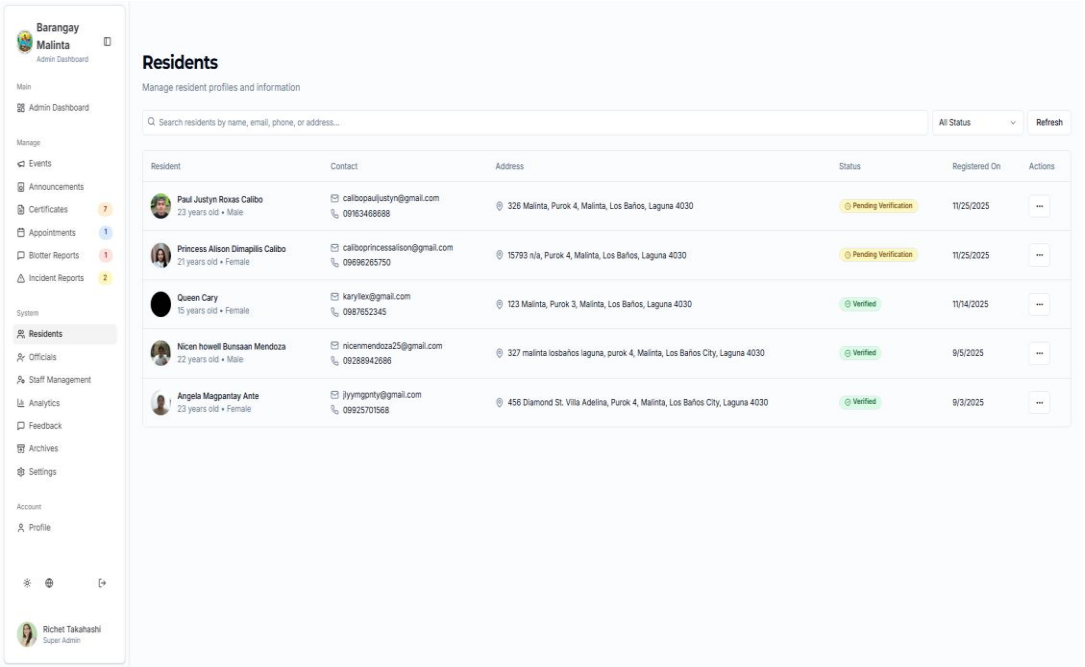


Figure 18. Residents Management Administrator

Figure 18 shows the Residents Management Administrator. This module was designed to support the Smart Dashboard's goal of maintaining an organized and accurate database of community members, which enhanced the efficiency of barangay operations. This feature allowed administrators to view, update, and manage resident profiles, ensuring that all vital information was complete and up-to-date. By streamlining the management of resident data, this module improved workflow efficiency, reduced errors from manual record-keeping, and enhanced the overall reliability of barangay services. It aligned with the platform's objective of providing a modern, well-structured, and responsive digital governance system. In addition, the module provided advanced search and filtering tools that allowed administrators to quickly view resident records, reducing time during community transactions and decision-making. This improvement allowed residents to complete requests more quickly and with less effort.

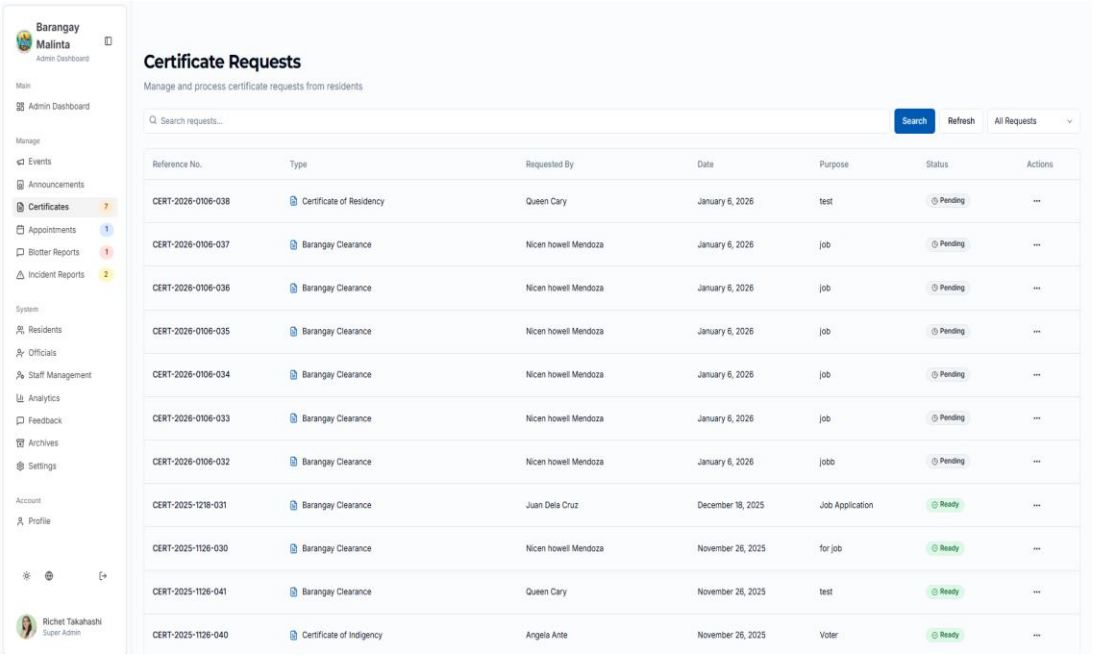


Figure 19. Certificate Management Administrator

The Certificate Management Administrator enabled administrators to support the Smart Dashboard's purpose of streamlining service delivery by centralizing and organizing all transactions related to certificates. With this feature, administrators reviewed, approved, rejected, and updated certificate requests submitted by residents. The system offered a clear overview of pending, ongoing, and completed applications, allowing administrators to manage workflows with increased accuracy and speed. By automating the tracking and processing of certificates, this module reduced the need for manual paperwork, minimized delays, and ensured that residents received their documents on time. Overall, Certificate Management strengthened the platform's goal of promoting efficient, transparent, and systematic governance. Moreover, this feature allowed administrators full monitoring capabilities to track the status of each Request in real time, maintaining security and reducing the risk of losing or misplacing applications. It also provided audit records, verifying

any action taken on a certificate request, leading to trust and transparency in the system.

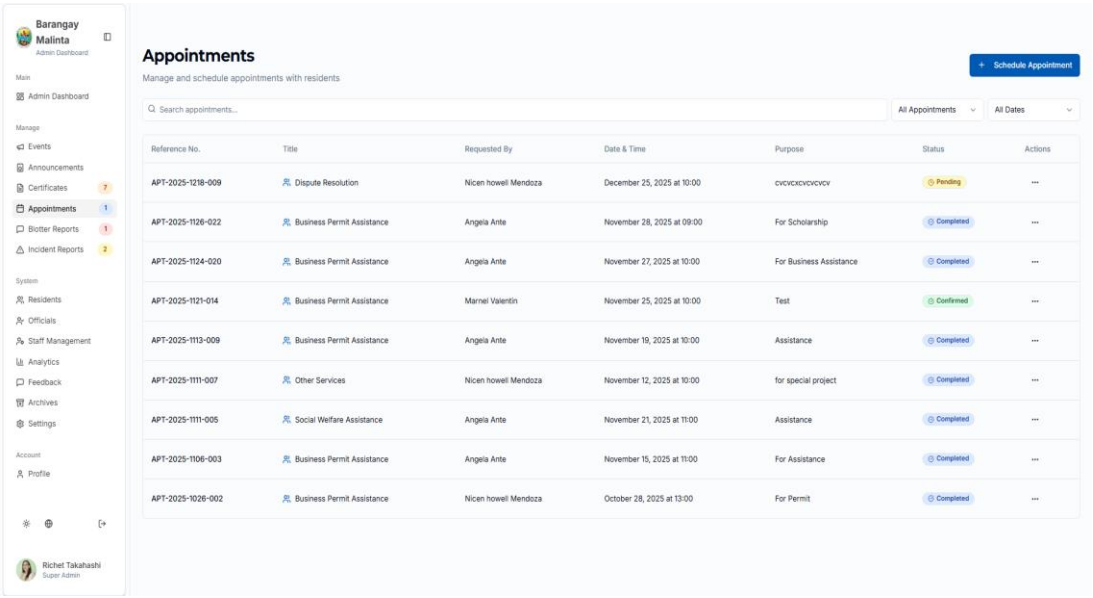


Figure 20. Appointments Management Administrator

Figure 20 displayed the Appointments Management Administrator, which enabled the Smart Dashboard to enhance service coordination and improve operational efficiency within the barangay. This feature allowed administrators to view, organize, and monitor all scheduled appointments submitted by residents. Administrators were also able to update the status of appointments, ensuring that residents received timely notifications and accurate information. By centralizing all appointment activities in one accessible interface, this module improved workflow organization, reduced manual handling, and supported a smoother and more responsive service process in the barangay. The module provided administrators with a full picture of appointment schedules, allowing users to effectively organize services and avoid conflicts or redundancies. It also provided real time-tracking and updates, ensuring that administrators and residents remained aware of any changes or corrections. Each appointment change was recorded in a documentation

system, which improved accountability and transparency in the management process.

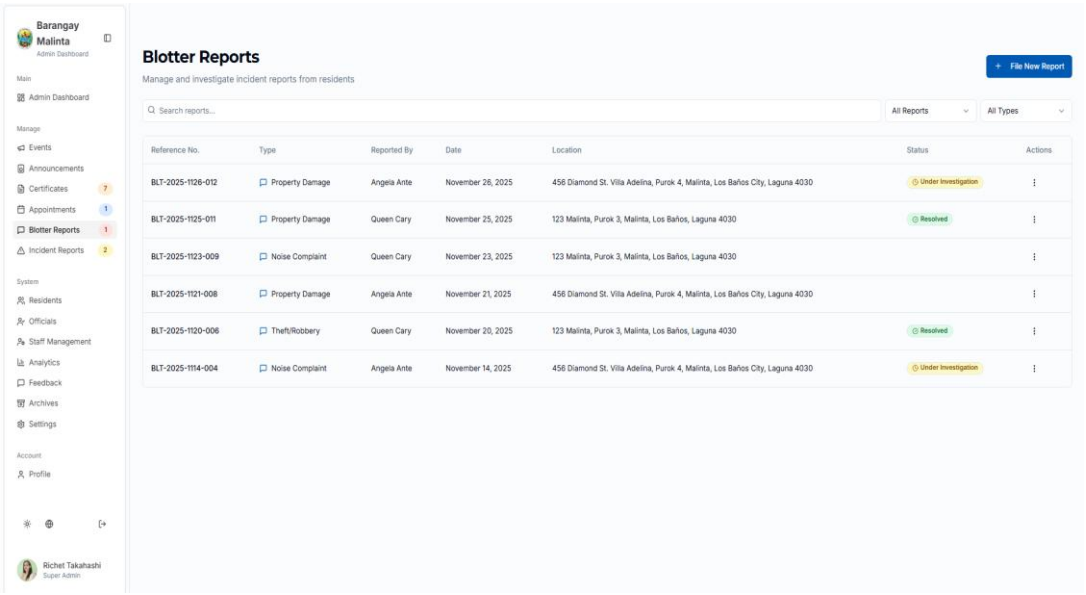


Figure 21. Blotter Reports Management Administrator

Figure 21 shows the Blotter Reports feature, which allowed administrators to see and manage incident reports submitted by residents. The table displayed important information such as the reference number, type of incident, reporter, date, location, and case status. The features provided to administrators allowed them to search, filter reports, and create new reports. Every report also had an action button that enabled the administrator to view or update the case, helping in the organization and efficiency of incident management. The feature also supported systematic record-keeping by ensuring that all incidents were properly documented and easily accessible when needed. It also helped administrators monitor the progress of each case through updated statuses, which improved follow up and actions and resolution tracking. Overall, this feature enhanced transparency and accountability within the system, encouraging an environment where barangay incident reports

could be managed effectively. It authorized both the administrators and residents to ensure better community safety.

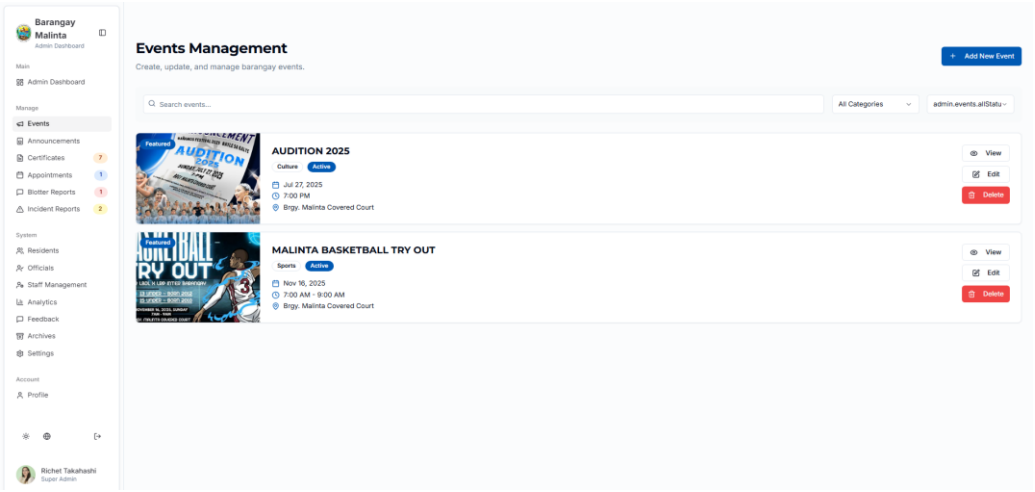


Figure 22. Events Management Administrator

The Events Management Administrator enhanced the Smart Dashboard's goal of boosting community engagement by providing administrators with an efficient system for organizing, updating, and announcing barangay events. This feature allowed officials to easily create event posts, upload images, and share essential details such as schedules, venues, and descriptions.

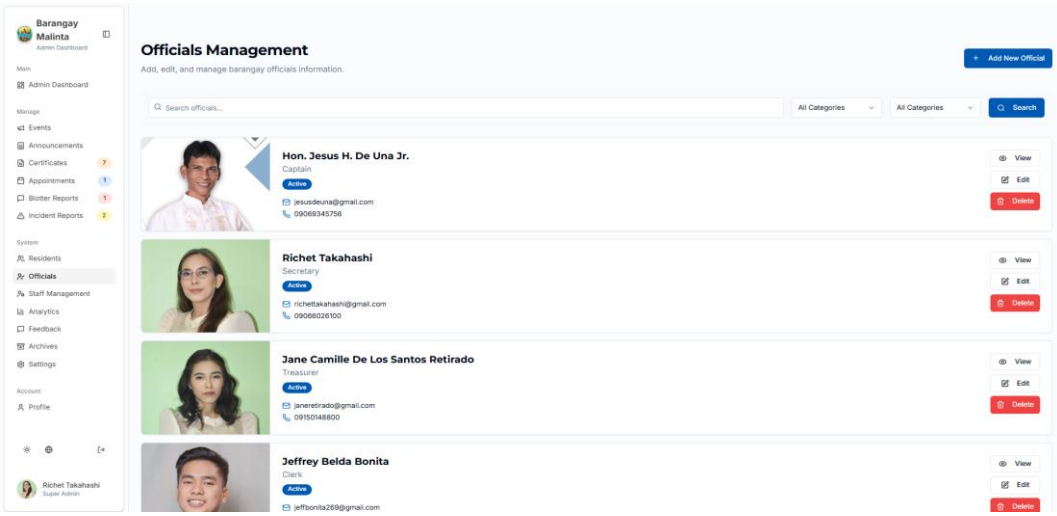


Figure 23. Officials Management Administrator

Figure 23 shows the Officials Management page. This section displayed the full list of barangay officials, including their profile photos, contact information, and email addresses. Administrators were able to view, edit, or delete official records using the control buttons. Overall, this feature ensured that the barangay's organization remained neat, organized, accurate, and updated, allowing for effective cooperation and transparency among barangay officials.

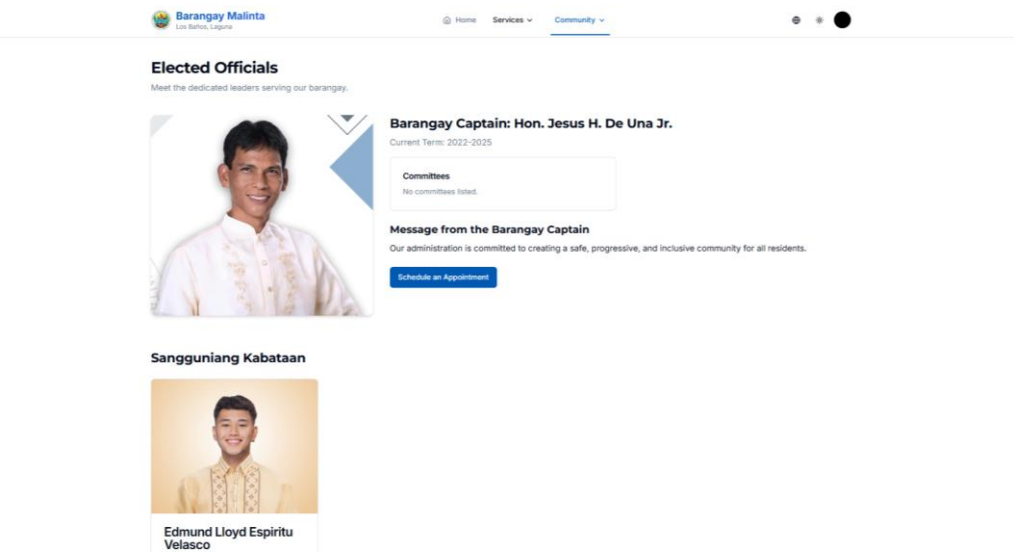


Figure 24. Elected Officials Page

The Elected Officials Page supports the Smart Dashboard's goal of improving transparency and accessibility by providing residents with clear and organized information about their barangay leaders. This module displays the profiles, roles, and responsibilities of elected officials, making it easy for users to identify the individuals responsible for governance and community services. The page helped strengthen community awareness by allowing residents to become more familiar with their local leaders and their respective functions. It also served as a reliable reference for official information.

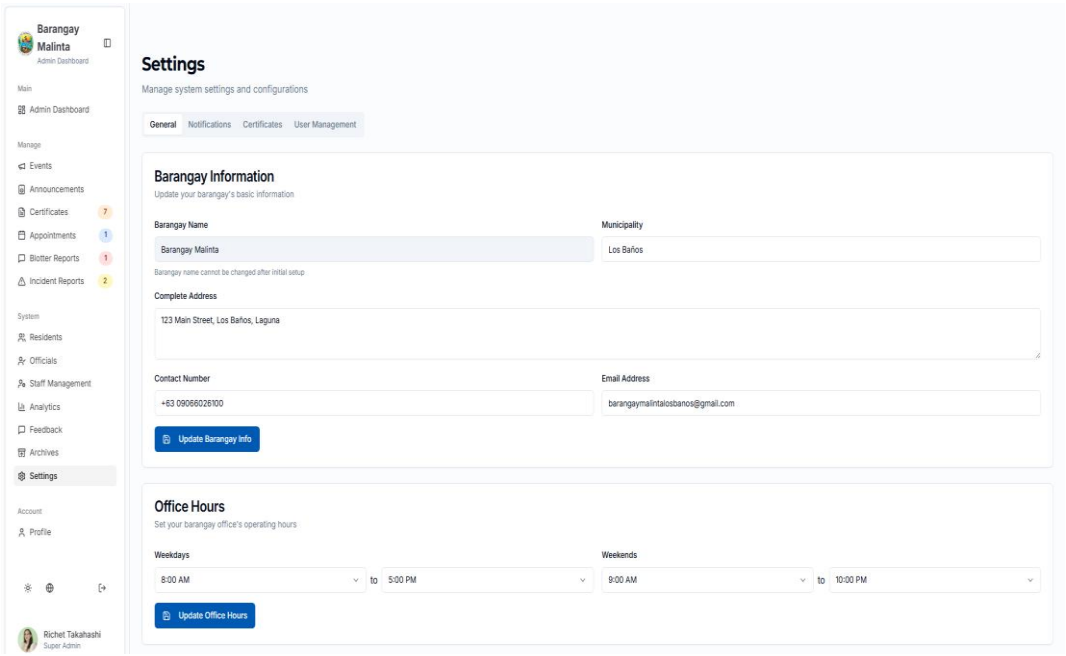


Figure 25. Settings Page Administrator

The section settings of the Smart Dashboard allowed administrators to manage and update essential system configurations. It included the Barangay Information panel, where authorized users could modify key details such as the barangay name, municipality, complete address, contact number, and official email address. This ensured that all displayed information remained accurate and up to date. In addition, restricting access to these settings to authorized users helped maintain data integrity and security. Proper configuration management supported smoother system operations and ensured that changes were traceable and controlled. Overall, the section settings feature contributed to the reliability, professionalism, and effective administration of the Smart Dashboard System. Through maintaining accurate barangay information, the system enhanced credibility and improved communication between the barangay and its residents. Additionally, the section settings feature allowed administrators to implement timely updates and modifications.

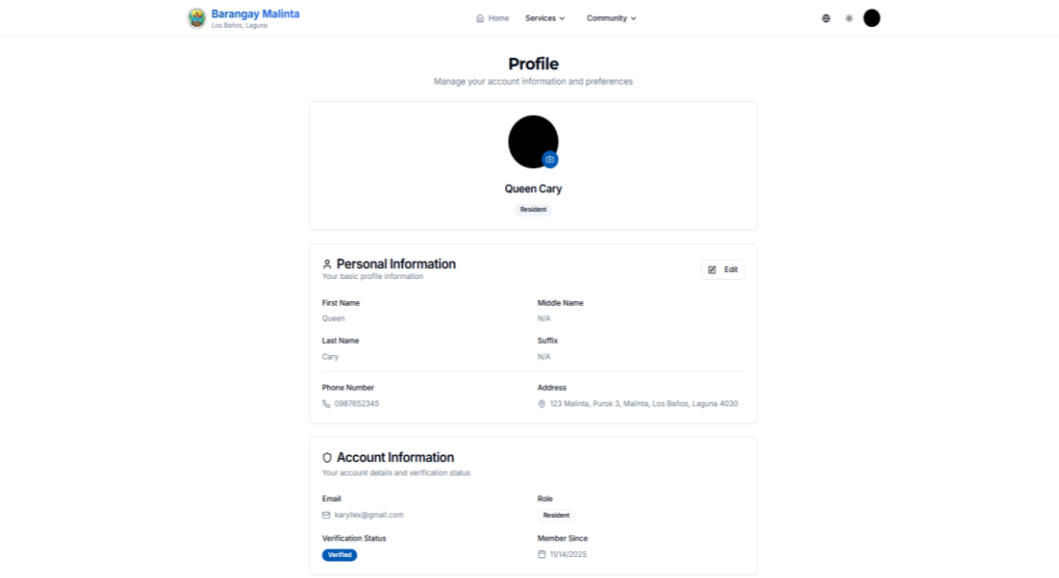


Figure 26. User's Profile Page

The user's profile page supported the Smart Dashboard's objective of providing personalized and streamlined access to barangay services by allowing residents to manage their own information with ease. This module enabled users to view and update their basic details, ensuring that their records remained accurate and up to date for certificate requests, appointments, and other transactions.

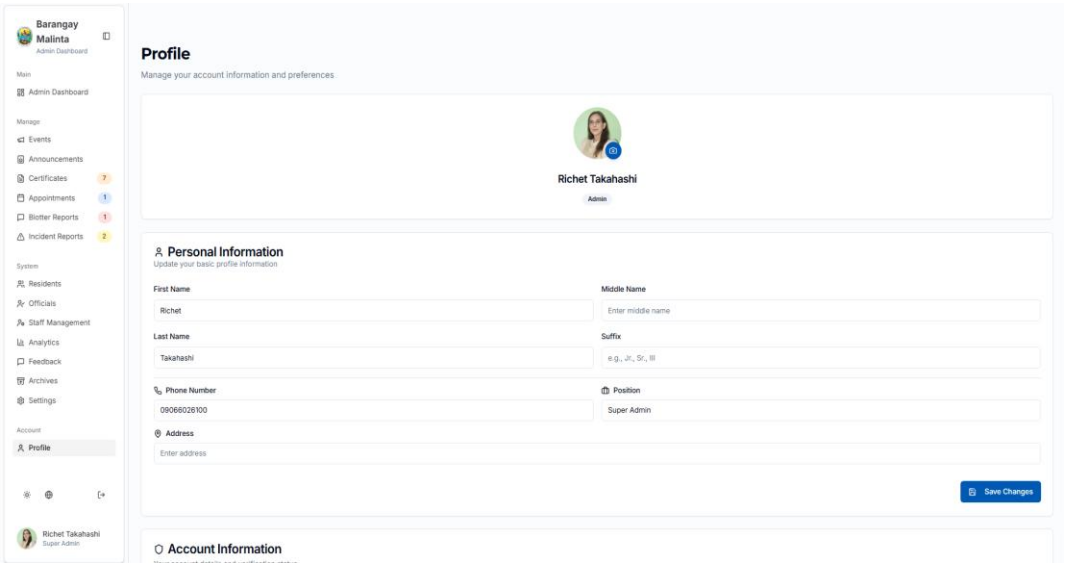


Figure 27. Administrator's Profile Page

The administrator's profile page was essential for maintaining secure and efficient management within the Smart Dashboard system. This section enabled administrators to view and update their personal account details, ensuring that their information remained accurate and consistent throughout all system operations. By keeping their profiles up to date, administrators successfully managed barangay records, viewed user activities, and handled various administrative tasks. The profile page enhanced account security by enabling administrators to manage their credentials and monitor their account usage effectively.

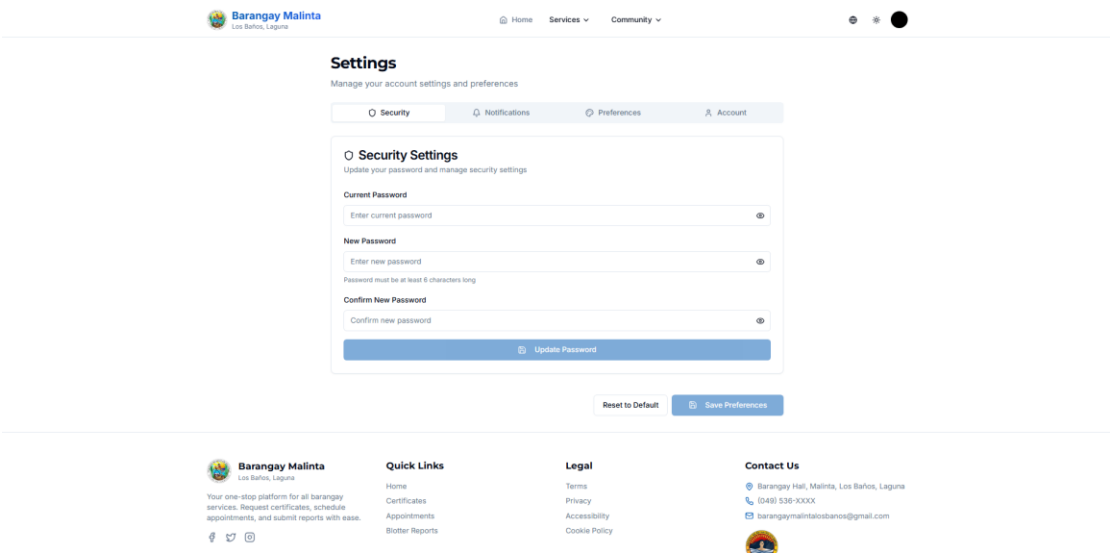


Figure 28. User's Settings Page

Figure 28 displayed the User's Settings Page. In this module, residents managed essential preferences such as updating their login credentials, account details, and configuring notification settings. It allowed users to directly modify their account information. The system ensured accuracy, strengthened data privacy, and minimized the assistance of administrators. Furthermore, the User's Settings Page contributed to a more personalized user experience by allowing residents to tailor system notifications based on their preferences,

ensuring timely and relevant updates. Security features such as password confirmation and input verification were integrated to protect user accounts from unauthorized changes. This module also reduced administrative workload by empowering users to independently manage routine account updates. Overall, the User's Settings Page supported efficient self-service, improved system reliability, and reinforced trust between users and the system through secure and transparent account management.

Development of the Smart Dashboard Web-Based Platform for generating barangay certificate requests and blotter reports

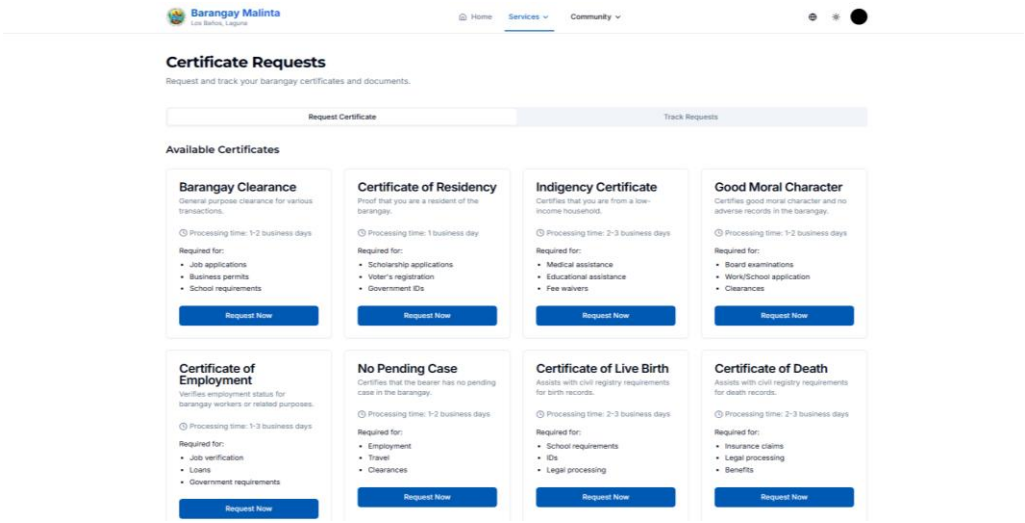


Figure 29. User's Certificate Requests

Figure 29 shows the certificate request. The certificate requests page displayed all accessible certificates along with their descriptions, processing times, and requirements, making it easy for residents to understand the details of each document. The Request Now button allowed users to quickly submit applications for certificates, achieving the goal of clarifying the request. Furthermore, the certificate requests interface was designed to provide a clear and user-friendly experience by presenting all relevant information in an

organized manner. By displaying descriptions, processing times, and requirements upfront, the system reduced confusion and minimized the need for residents to make follow-up inquiries at the barangay office. This transparency helped users make informed decisions before submitting their requests. In addition, the streamlined request process improved efficiency for both residents and barangay personnel. Once a request was submitted through the system, it could be tracked and processed more systematically, reducing manual paperwork and processing delays. As a result, the certificate request feature enhanced overall service delivery and supported faster, more convenient access to essential barangay documents.

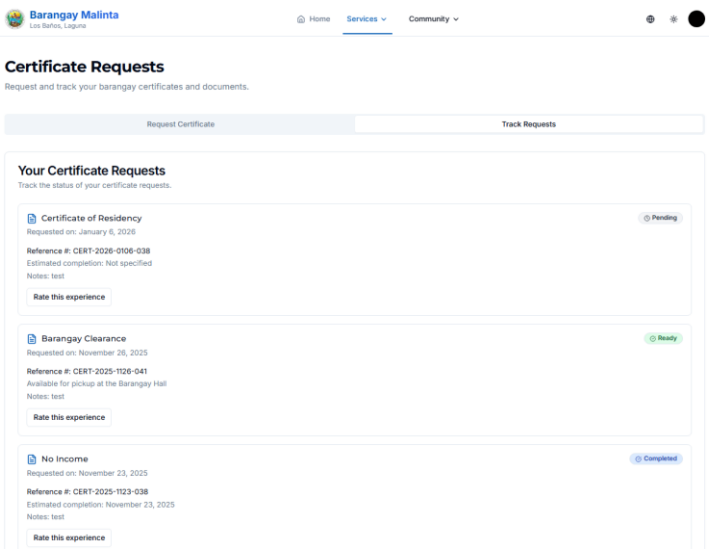


Figure 30. User's Track Requests

Figure 30 shows the track request. This interface enabled users to monitor the status of their applications in real time, which enhanced transparency and improved the user experience. Overall, this module demonstrated how the Smart Dashboard effectively provided a more efficient, organized, and user-friendly platform for the community to request certificates.

The track request feature improves clarity by keeping users informed about each stage of their application process, saving time for both residents and administrators while ensuring smoother and more efficient service delivery.

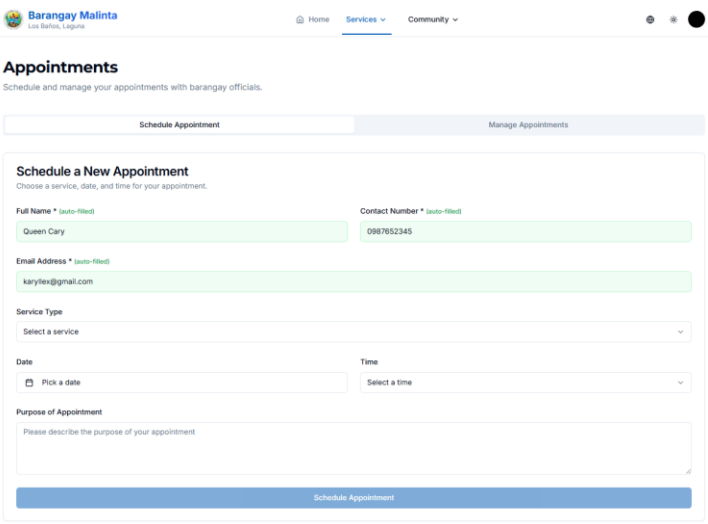


Figure 31. User's Appointments Page

Figure 31 shows the User's Appointment Page. It was designed to help residents schedule appointments with barangay officials in a streamlined and organized way. The interface allowed users to enter their important information, including their name, contact number, email, type of service needed, preferred date and time, and the purpose of their visit. This structured process reduced the need for manual coordination and minimized waiting times, ensuring that residents could easily secure appointments without having to visit the barangay. The module also kept residents updated about any changes or confirmations to their appointments. The program provided barangay staff useful input that could be utilized to assess performance and enhance service delivery, resulting in a more responsive and user-centered appointment system. Furthermore, the Manage Appointments module improved time management by allowing users to monitor both upcoming and previous appointments in one convenient spot.

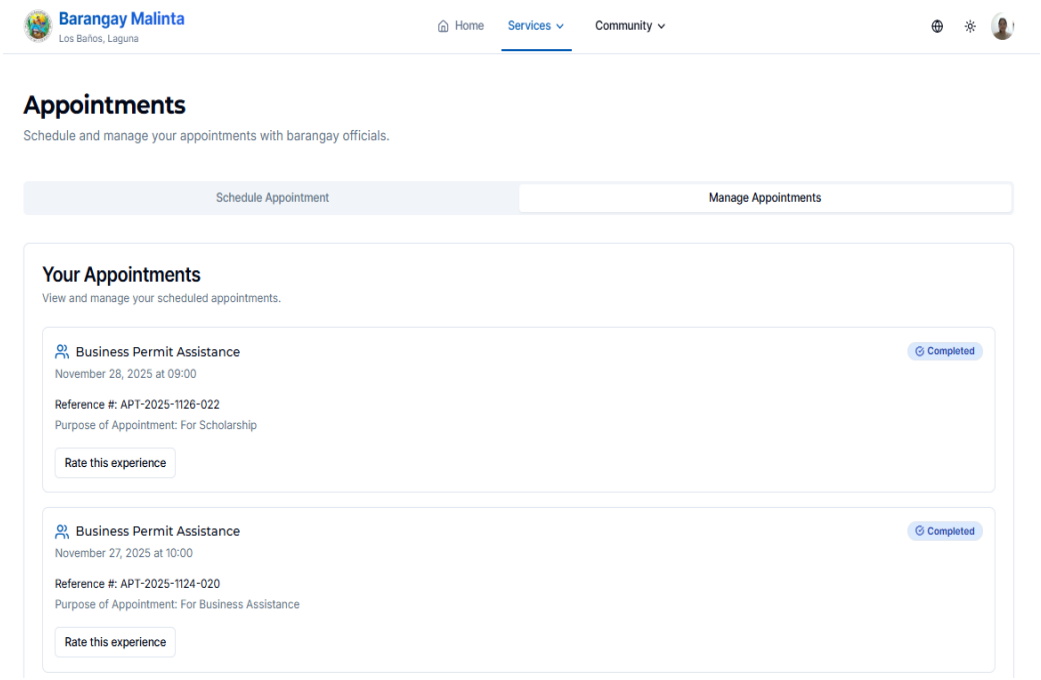


Figure 32. User's Manage Appointments

Figure 32 shows the User's Manage Appointments module, which allows users to view, track, and manage their scheduled appointments easily. Residents can monitor request status, make adjustments, and stay informed without visiting the barangay office. The module also includes a rating feature where users can rate their appointment experience, helping improve service quality. This feature minimizes schedule conflicts, increases transparency, and supports more efficient appointment management for both users and barangay staff. Once the documents have been recovered, administrators have two primary options they can restore the records to the system or choose to delete them permanently. This functionality is especially significant since it protects against the accidental loss of information. If a record is removed, administrators can still recover it if necessary, providing an additional degree of protection to data management methods. This measure helped ensure that sensitive

information, such as resident records and official documents, remained secure from unauthorized access.

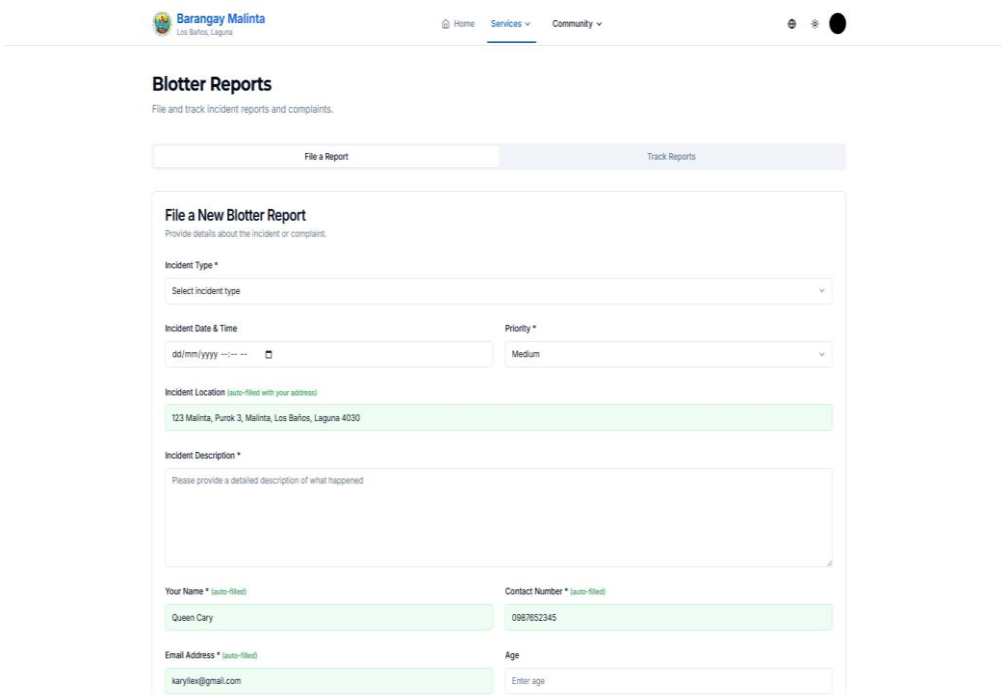


Figure 33. User's Blotter Reports

Figure 33 shows the blotter reports, allowing residents to use the system to directly file and submit official complaints or incident reports. By providing an organized form where users identified the incident type, location, description, and people involved, the system enabled users to upload an image as proof for validation, assisting administrators in accurately assessing the incident. Furthermore, the User's Settings Page contributed to a more personalized user experience by allowing residents to tailor system notifications based on their preferences, ensuring timely and relevant updates. Security features such as password confirmation and input verification were integrated to protect user accounts from unauthorized changes. This module also reduced administrative workload by empowering users to independently manage routine account updates. Overall, the User's Settings Page supported efficient self-service,

improved system reliability, and reinforced trust between users and the system through secure and transparent account management.

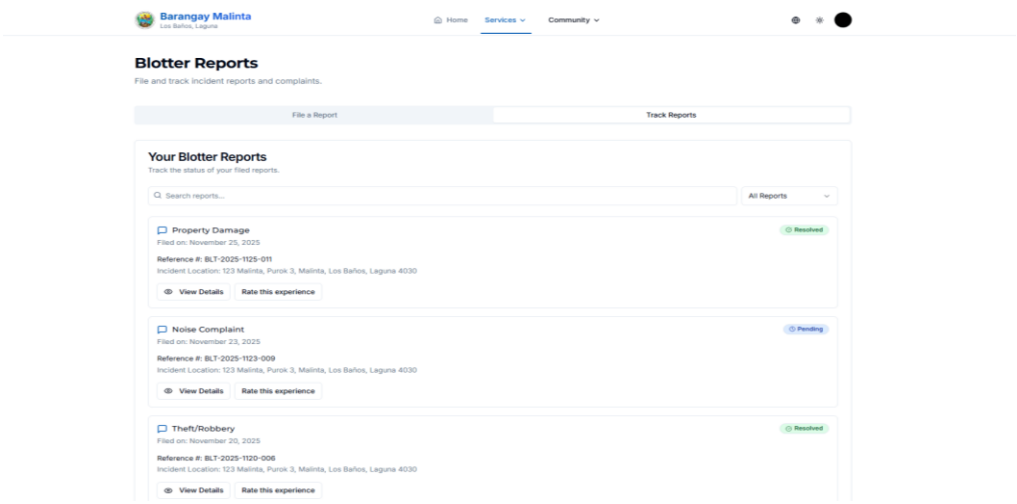


Figure 34. User's Blotter Track Reports

Figure 34 shows the User's Blotter Track Reports module, which allows residents to monitor the status and progress of their blotter reports in real time. It improves transparency and communication by providing timely updates without requiring office visits. The module also includes a rating feature, enabling users to give feedback to help improve barangay services and accountability.

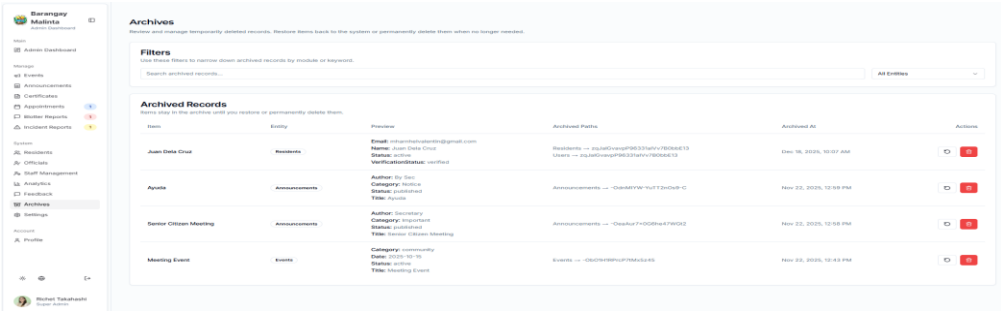


Figure 35. Archives Administrator

Figure 35 displayed the archives, where administrators managed or viewed the records that had been temporarily removed from the system. It

included a search bar and filter options that allowed administrators to filter the results by word or item type. Administrators were able to restore records to the system or permanently delete them if they were no longer needed. This feature helped with data management and guaranteed that deleted records could still be recovered if necessary. Overall, this feature not only enhances data management but also ensures compliance with organizational policies regarding data maintenance and deletion. By offering the flexibility to recover previously deleted records.

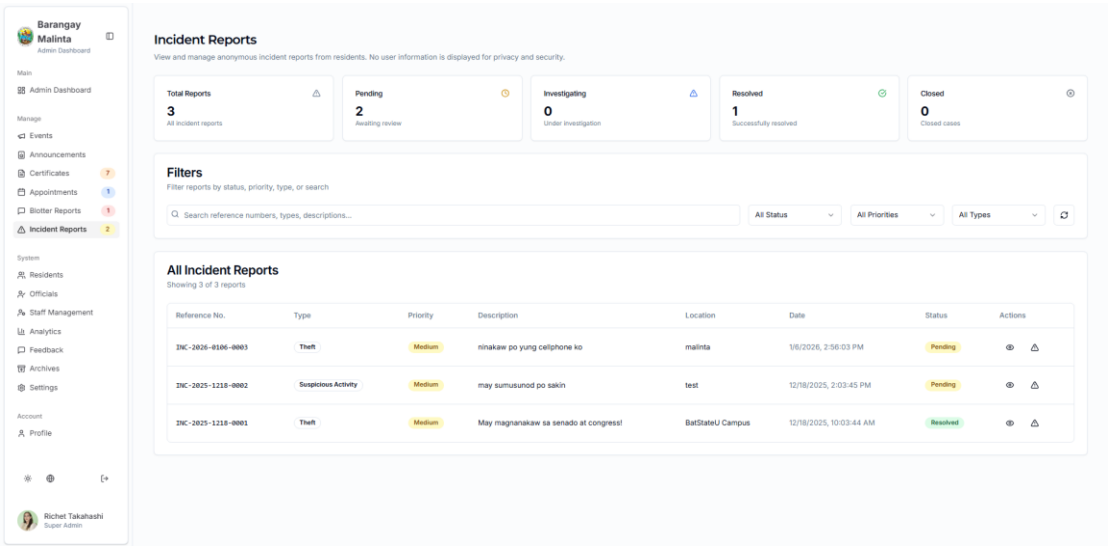


Figure 36. Incident Reports Administrator

Figure 36 shows the Incident Reports Administrator, which allowed administrators to view and manage anonymous incident reports. This module featured report summaries, search and filter options, and a list displaying essential details, such as type, priority, location, date and status. Also, monitoring anonymous incident reports using this module promoted community engagement by providing a secure and confidential reporting channel. Administrators could evaluate reports fairly while protecting residents' identity, promoting transparency and confidence throughout the community. Overall, the

issue Reports Administrator improves issue management, decision-making, and reactivity in the Smart Dashboard System.

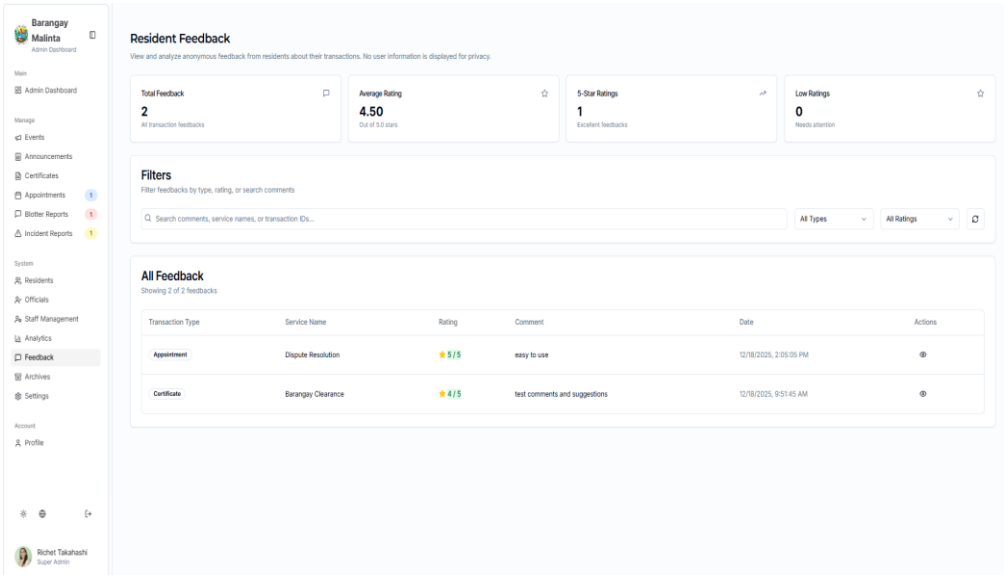


Figure 37. Resident Feedback

Figure 37 presents the Resident Feedback of the system. This feature allows administrators to view and analyze anonymous feedback submitted by residents after completing transactions. It displays key summaries such as the total number of feedback entries, average rating, five-star ratings, and low ratings. The module also includes filters and a feedback list showing the service type, rating, comments, and date, helping the barangay monitor service quality and identify areas for improvement while maintaining resident privacy.

Evaluation of the system's software quality characteristics using the ISO 25010 standard.

Based on the ISO 25010 Software Acceptance Evaluation in Table 18, the system showed strong performance in most quality attributes. Respondents rated Functional Stability, Performance Efficiency, and Security with a mean score of 3.33, which demonstrated considerable agreement on the system’s

effectiveness in these items. Compatibility, Usability, Reliability, and Portability received a mean score of 3.50, reflecting the users' strong approval of the system's consistency and user-friendliness. Maintainability scored a mean of 3.00, which reflected a positive evaluation and was interpreted as Agree. It suggested a potential area for improvement compared to the other items. Overall, the results indicated that users strongly believed the system performed well and met essential quality standards. The results showed that the system efficiently supported daily operations while providing secure and reliable performance. High usability and compatibility ratings indicated that users found the system simple to learn and adaptable to a variety of settings. Although the maintainability rating was slightly lower, it still displayed user confidence in the system's structure and long-term support, highlighting areas for future improvement.

Table 18. ISO 2510 Software Acceptance Evaluation

ITEMS	MEAN	INTERPRETATION
Functional Suitability	3.33	Strongly Agree
Performance Efficiency	3.33	Strongly Agree
Compatibility	3.50	Strongly Agree
Usability	3.50	Strongly Agree
Reliability	3.50	Strongly Agree
Security	3.33	Strongly Agree
Maintainability	3.00	Agree
Portability	3.50	Strongly Agree

Based on the findings of the questionnaire using the Likert Scale approach, Table 18 revealed that the system operated well in all categories assessed by the respondents. The total mean of every criterion resulted in a mean of 3.53, and researchers interpreted the data as users agreeing with the system. The standard deviation values were low, showing consistent responses across the evaluated items. Furthermore, the results revealed that consumers had a generally positive view of the system's performance and quality. The consistency of answers indicated that the system provided a stable and satisfying experience across multiple characteristics. Overall, the data confirmed that the system met user expectations and was well received by its intended users. Also, the high overall mean demonstrated the system's success in meeting user needs and performing its intended functions. The minimal range of responses suggested that satisfaction was not limited to a single group of users, but was clear among all respondents. These findings supported the system's architecture, functionality, and overall capacity for acceptance and regular usage. The results indicate that the system was able to meet the basic needs and expectations of its users in terms of performance and usability. The positive feedback suggests that the design and features were practical and easy to understand, which may encourage consistent use of the system. This shows that the system has the potential to improve efficiency and contribute positively to the overall workflow of the system

Evaluation of the system's user acceptance software quality characteristics using the Technology Acceptance Model

Table 19 shows that users strongly agree the system is useful, easy to use, and improves efficiency in barangay operations. They also have a positive attitude toward it and are willing to use it regularly, indicating high acceptance and readiness for implementation.

Table 19. TAM User Acceptance Evaluation

ITEMS	MEAN	INTERPRETATION
Perceived Usefulness		
The system improves the efficiency of managing barangay operations and resident services.	3.50	Strongly Agree
The system provides real-time and accurate information that enhances decision -making for barangay officials.	3.60	Strongly Agree
Using this system increases productivity and improves service delivery within the barangay.	3.52	Strongly Agree
The system helps prevent potential issues in barangay management before they become critical.	3.56	Strongly Agree
I find the system valuable in managing barangay activities effectively.	3.42	Strongly Agree
Perceived Ease of Use		
The system interface is user-friendly and easy to navigate for residents and officials.	3.48	Strongly Agree
I find it easy to learn how to operate the system and access the information of services I need.	3.58	Strongly Agree
Using the system requires minimal effort or technical knowledge.	3.50	Strongly Agree
The online platform and dashboard make it convenient to access barangay services anytime and anywhere	3.44	Strongly Agree
I am comfortable with the overall setup and functionality of the system.	3.54	Strongly Agree

Attitude Towards Using the System		
I feel positive about using the system for barangay transactions and services	3.44	Strongly Agree
The system fits well with the current processes and practices of barangay management.	3.46	Strongly Agree
I believe this technology can greatly benefit barangay operations and public service delivery.	3.38	Strongly Agree
Using this system is an enjoyable and beneficial experience.	3.56	Strongly Agree
I would recommend this system to other barangays or local government units.	3.60	Strongly Agree
Behavior (Intention to Use the System)		
I intend to use this system regularly for barangay-related activities.	3.28	Strongly Agree

Legend: 3.25 – 4.00 = Strongly Agree, 2.50 – 3.24 = Agree, 1.75 – 2.49 = Disagree, 1.0 – 1.74 = Strongly Disagree

Table 20 presents the overall mean and standard deviation for each TAM construct, providing a clear overview of user acceptance. All factors received mean scores interpreted as Strongly Agree, showing that users have a positive perception of the system. The low standard deviation values indicate consistent responses among users, confirming that the system is useful, easy to use, and likely to be used continuously.

Table 20. TAM User Acceptance Evaluation

ITEMS	MEAN	SD	INTERPRETATION
Perceived Usefulness	3.49	0.26	Strongly Agree
Perceived ease of use	3.56	0.25	Strongly Agree
Attitude Toward Using the System	3.54	0.27	Strongly Agree
Behavior Intention to use the system	3.59	0.32	Strongly Agree

Unit Testing Results

This was the method developers used during the testing phase after the system’s development. The primary modules worked as designed and fulfilled the client’s expectations. The other features were also developed properly and tested before the user test, wherein researchers found that the actual output had met the expected results, thus meeting the client’s other expectations on the developed Smart Dashboard System.

Table 21 shows the unit testing of the login page, which focused on user credentials validation, error management, and redirection processes. Researchers evaluated the system’s capacity to authenticate accounts, provide error warnings for invalid attempts, and direct users to their appropriate dashboards. The results showed that all functions operated as expected, and the module produced successful outputs with passing remarks. The login page was successfully unit tested, confirming that the system's authentication process was secure and dependable. Proper processing of valid and invalid credentials meant that only authorized users could get access to the system, while clear error messages helped users understand and resolve login issues without misunderstanding. This improved both system security and the user experience. Furthermore, accurate redirection to relevant dashboards based on user roles proved that access control and role management were properly implemented. These findings suggest that the login module efficiently enables safe system entrance and smooth navigation, adding to the Smart Dashboard System's overall stability and trustworthiness.

Table 21. Unit Testing Log in Page

Test Case	Expectation	Actual Output	Response Time	REMARKS
Validate User Credentials	System accepts correct email/passwords and rejects invalid ones	Credentials validated correctly	1sec	Passed
Redirect After Login	User is redirected to their dashboard	Redirection successful	1-2sec	Passed
Display Login errors	System shows error messages for incorrect credentials	Error messages displayed properly	1sec	Passed

Table 22 shows the unit testing of the Certificate Request Module. Researchers evaluated the submission of certificate requests, understanding of provided details, and cancelling pending requests. The result matches the system's intended behavior, demonstrating proper request record saving, retrieval, and removal. All test cases demonstrated expected functioning and received passing remarks. Moreover, the successful unit testing of the Certificate Request Module demonstrated that users could interact with the system efficiently and intuitively, with clear guidance provided for each step of the certificate request process. The ability to cancel pending requests enhanced user control and flexibility, reducing errors and unnecessary administrative workload. Residents could correct mistakes or change their minds without needing to contact officials directly, making the system more convenient and user-friendly. This feature empowered users to manage their own requests efficiently.

Table 22. Unit Testing Certificate Request

Test Case	Expectation	Actual Output	Response Time	REMARKS
Submit Certificate Request	Request is submitted and saved to database	Request successfully saved	1-2 sec	Passed
View submitted Request	User can view request details	Details displayed accurately	1 sec	Passed
Cancel Request	Admin can remove pending request	Request removed from records	1-2 sec	Passed

Table 23 shows the track request unit testing, which included testing request status loading, history retrieval, and system updates in response to backend changes. Researchers confirmed that the module accurately tracked request progress and presented comprehensive request histories. All outputs matched the expected results, resulting in passing scores for all test scenarios.

Table 23. Unit Testing Track Request

Test Case	Expectation	Actual Output	Response Time	REMARKS
Display Request status	Status should load based on backend data	Status displayed correctly	1 sec	Passed
Update Request status	System should reflect changes instantly	Status updates appear immediately	1-2 sec	Passed
Retrieve full request history	Users can view request history	History loads with no missing data	1 sec	Passed

Table 24 shows the unit testing of events and announcements to ensure that events were posted, displayed, and updated correctly. The tests involved adding new announcements, modifying previous posts, and confirming proper presentation on the user interface. The actual results were consistent with expectations, demonstrating proper data handling and display, and all test cases were marked as passed.

Table 24. Unit Testing Events & Announcements

Test Case	Expectation	Actual Output	Response Time	REMARKS
Display announcements	All announcements appear on page	Announcements load correctly	1-2 sec	Passed
Create announcement (Admin)	Admin can add an event/announcement.	Event saved and displayed	2 sec	Passed
Edit/delete announcement	Admin can update or removed posted items	Changes reflected immediately	1-3 sec	Passed

Table 25 shows the blotter reports unit testing, which emphasized report submission, data retrieval, and administrative adjustments to blotter records. Researchers confirmed that users could effectively submit reports, and administrators could review and change records. Because all processes performed as predicted, the module achieved passing grades on all test scenarios. This demonstrates that the system functions reliably according to its design and requirements. Each feature, from data input to output, worked without errors, confirming that the module is stable and dependable. The successful test results also indicate that users can confidently rely on the module for their daily tasks.

Table 25. Unit Testing Blotter Reports

Test Case	Expectation	Actual Output	Response Time	REMARKS
Submit blotter entry	User submits complete report	Report saved successfully	2 sec	Passed
View blotter details (Admin)	Admin can view full blotter info	Details displayed correctly	1-2 sec	Passed
Update blotter status	Admin can mark reports as processed	Status updated correctly	1 sec	Passed

Table 26 shows the elected officials unit testing, which tested the system's capacity to display official information, update details, and submit profile photographs. Researchers discovered that the module consistently provided accurate data and matched administrative changes. The actual results matched the expected results; hence, all test scenarios passed.

Table 26. Unit Testing Elected Officials

Test Case	Expectation	Actual Output	Response Time	REMARKS
Display official list	Page shows all elected officials	List loads with correct data	1 sec	Passed
Update officials info	Admin can edit official details	Updated data displays correctly	1-2 sec	Passed
Change official photo	Admin uploads image and displays it	Image upload successful	2 sec	Passed

Table 27 shows the unit testing of the dashboard. Researchers tested the dashboard module's loading of summary elements, analytics displays, and automatic data refreshing. All dashboard elements successfully obtained live system data and were updated accordingly. Because the outputs were

consistent with the expected behaviors, all test cases received passing evaluations.

Table 27. Unit Testing Dashboard

Test Case	Expectation	Actual Output	Response Time	REMARKS
Load dashboard	Dashboard displays all summaries	Widgets load correctly	1-2 sec	Passed
Generate statistics	System displays updated and analytics	Analytics render successfully	2 sec	Passed
Refresh data	Dashboard updates when data changes	Data refresh works properly	1 sec	Passed

Table 28 shows the unit testing of resident modules, which included adding resident data, changing information, and confirming search capabilities. Researchers discovered that the module appropriately maintained and updated resident data while returning accurate search results. All functions achieved the intended results, obtaining passing high marks on all test scenarios.

Table 28. Unit Testing Residents

Test Case	Expectation	Actual Output	Response Time	REMARKS
Add resident record	Admin can save new resident data	Record saved properly	2 sec	Passed
Edit resident	System updates saved information	Changes appear instantly	1-2 sec	Passed
Search resident	Search bar returns correct matches	Search works as expected	1 sec	Passed

Table 29 shows the unit testing of officials management, which included checking the adding, editing, and deletion of official records. Researchers

confirmed that all CRUD actions were correctly performed and quickly reflected in the system. Each test case obtained passing remarks because its results met expectations.

Table 29. Unit Testing Officials Management

Test Case	Expectation	Actual Output	Response Time	REMARKS
Add official	Admin can add official with complete details	Record saved properly	2 sec	Passed
Edit official	Changes to info should update instantly	Changes appear instantly	1-2 sec	Passed
Delete official	Official is removed from system	Search works as expected	1-3 sec	Passed

Table 30 shows the unit testing of staff management for account creation, role adjustment, and account deletion. Each function performed as expected, with accounts being successfully registered and updated. All of the goals were achieved, resulting in passing remarks for the module. The account creation function was tested to ensure that new staff members could be added to the system without errors. During testing, the module successfully saved user information, including names, roles, and login credentials, confirming that the registration process works properly. Role adjustment was also tested to verify that administrators could change the roles of staff members when needed. The system correctly updated roles and permissions, allowing users to access only the areas and functions appropriate for their responsibilities. This feature ensures that sensitive information is protected and that each user can perform tasks relevant to their role.

Table 30. Unit Testing Staff Management

Test Case	Expectation	Actual Output	Response Time	REMARKS
Create staff account	Admin can register new staff	Account created successfully	2 sec	Passed
Update roles permissions	System updates staff roles	Roles updated correctly	1-2 sec	Passed
Disable staff account	Disabled cannot log in	Restriction enforced properly	1 sec	Passed

Table 31 shows the unit testing of analytics, which evaluated chart display, filter responsiveness, and data refresh accuracy. Researchers confirmed that the module accurately presented data and updated chart outputs based on criteria. All actions performed as expected; therefore, the test cases obtained passing remarks. The system functions correctly based on the defined requirements and test scenarios. The successful execution of all test cases indicates that the features are working properly and no critical errors were encountered during testing. This also suggests that the system is stable and ready for further evaluation or actual use. the testing process helped validate the overall quality of the system. By identifying and addressing minor issues during development, the researchers were able to improve system performance and usability. Improvements in usability focused on making the interface more user-friendly and easier to navigate. Clear menus, organized dashboards, and simple instructions allowed users to complete tasks without confusion or difficulty. Users could perform their tasks efficiently and focus on important activities rather than struggling with the system interface.

Table 31. Unit Testing Analytics

Test Case	Expectation	Actual Output	Response Time	REMARKS
Displays charts	System loads graphs based on dataset	Charts render accurately	2 sec	Passed
Filter analytics	Filters adjust chart output	Filters working properly	1-2 sec	Passed
Refresh analytics	Data updates reflect in visuals	Updated visuals displayed	2 sec	Passed

Table 32 shows the unit testing of archives, which focused on the loading of archived records, search capability, and previewing of stored records. All information stored was accurately retrieved, and search queries returned correct results. The actual results met the expected results, obtaining satisfactory outcomes for all test activities. The effective search and preview features improved usability by allowing administrators to quickly locate and review stored records without unnecessary delays. These results indicate that the archives module is stable and dependable, contributing to the overall robustness of the Smart Dashboard System and supporting efficient record management in real-world usage. The results also suggest that the module can handle regular system operations without affecting overall performance. Smooth loading and accurate data display indicate that the system can support daily administrative tasks. This helps reduce delays and improves overall work efficiency. By streamlining processes and providing timely access to information, tasks can be completed faster and with fewer errors. As a result, both citizens and staff benefit from a more efficient workflow, leading to better overall system performance and a more satisfactory user experience.

Table 32. Unit Testing Archives

Test Case	Expectation	Actual Output	Response Time	REMARKS
Load archived records	System retrieves archived data	Records displayed successfully	1-2 sec	Passed
Search archive	User can search past entries	Search returns correct results	1 sec	Passed
Restore/preview records	System opens or restores archived record	Record preview successful	1-2 sec	Passed

Integration Testing Results

Throughout integration testing, the system module passed the requirements and tasks that had been coded into the system. This meant that the system's main modules and other features were effectively incorporated, fulfilling the expectations regarding user involvement and system functionality. Therefore, researchers determined that the system was well-established and successfully deployed. The successful integration testing results revealed that the system was stable and fit for usage in a real-world scenario. Through validating component interactions, the testing method reduced the risk of system failure after deployment and increased trust in the Smart Dashboard System's overall quality, dependability, and usability. As a result, the integration testing phase provided complete verification that the Smart Dashboard System is resilient, and capable of fulfilling real time operation. The successful results show that the system can handle several user activities without losing speed.

This proves that the Smart Dashboard System can withstand operating demands in practical settings. Overall, the integration testing phase

strengthened confidence in the system's architectural design and operational readiness, demonstrating that it can support making decision. Furthermore, the system's ability to execute dependably under real-world operating settings suggests that it can handle typical workloads, user interactions, and data processing requirements without degrading performance. This dependability is critical for maintaining continuous service availability and minimizing disruptions in real-world deployment. Furthermore, the favorable integration testing results demonstrated that the system's components function together to ensure accurate and timely information flow. This seamless coordination allows administrators and decision-makers to rely on the system to make informed and successful decisions.

The Smart Dashboard System is well positioned to serve as a reliable tool for assisting with operational and strategic decisions in the barangay environment. providing a centralized platform for monitoring services, incident reports, and community interactions, the Smart Dashboard System enhances transparency and accountability within the barangay. By providing clear and organized access to information, the system allows both officials and staff to easily monitor ongoing activities and records. This reduces confusion and helps ensure that actions taken by the barangay are properly documented. As a result, important information becomes more accessible and easier to verify. The system also supports accountability by keeping track of transactions, requests, and updates in a centralized platform. Allowing officials to monitor actions, the system promotes transparency and builds trust among users. It provides a clear record of transactions and activities within the platform, making it easier to verify processes and ensure that services are delivered properly.

Table 33. Integration Testing of Smart Dashboard System

Test Case	Expectation	ACTUAL OUTPUT	REMARKS
User Log in	Users can register, and log in	The system successfully validates user credentials and redirects to the dashboard	Passed
Certificate & Track Request	Administrators can view and manage data	The system successfully submits certificate requests and stores them in the database and displays updated status information fetched from backend without errors	Passed
Events & Announcements	Administrators can post and users can view it	The system load correctly, and display on the page without missing data	Passed
Blotter Reports	Users can submit blotter reports and admin can view reports	The system saves blotter report submissions and displays them correctly to authorized administrator	Passed
Elected Officials Page	The page shows the complete list of elected officials with proper images and details	The system displays officials information accurately with proper formatting	Passed
Dashboard	The dashboard shows analytics and summary data for admin or staff depending on role	The system loads the the charts, and activity logs correctly	Passed
Residents Module	Administrator can view to add, edit. And manage resident records	The system loads resident data, allows profile and updates database successfully	Passed
Officials Management	Administrator can add, edit, or remove officials profiles with images and details	The system functions perform correctly, changes reflect immediately	Passed
Staff Management	Administrator can manage staff accounts and roles, including creating disabling accounts	The system process staff updates and reflects changes accurately in the system	Passed
Analytics Module	The system display charts and graphs about certificates, blotters, etc	The system load stored records correctly and retrieval is successful	Passed
Archives Module	Admin should be able to access historical records.		Passed

Compatibility Testing Results

The researchers used the Google Lighthouse tool to evaluate the system. Overall, Chrome and Microsoft Edge displayed similar patterns in terms of performance, accessibility, best practices, and SEO. However, Chrome consistently outperformed in these areas. Microsoft Edge had poorer performance and accessibility ratings than the others. Meanwhile, Opera delivered the most balanced results, consistently receiving the highest marks across all criteria, mainly accessibility and SEO. Overall, the results showed that Opera provided the most optimized and reliable user experience among the tested browsers. Opera's better findings show that the system is in line with modern web standards and efficient rendering processes. The disparities noticed between browsers also gave useful insights for future improvements, notably in terms of performance and accessibility on less optimized platforms. Overall, the test confirmed the system's preparedness for deployment by exhibiting consistent performance and user experience across all major web browsers. Furthermore, the evaluation using Google Lighthouse highlighted the importance of cross-browser testing in ensuring that all users, regardless of their preferred browser, can access the Smart Dashboard System effectively. By identifying differences in performance, accessibility, and best practices among browsers, the researchers were able to pinpoint areas where optimization might be necessary, particularly for browsers that scored lower. In addition, the consistent results in Chrome and Opera reinforced that the system’s design and code structure adhere to modern web development standards, ensuring fast loading times and responsive interactions.

Table 34. System Compatibility Testing for Browsers in Desktop

Browser	Web Page	Performance	Accessibility	Best Practice	SEO
Google Chrome	Log in Page	96	88	96	90
	Certificate Requests	99	96	100	100
	Track Requests	100	96	100	100
	Events & Announcements	86	90	96	100
	Blotter Reports	100	95	100	100
	Elected Officials	80	100	96	100
	Dashboard	89	90	100	100
	Residents	94	91	100	100
	Officials Management	78	89	100	100
	Staff Management	98	89	100	100
	Analytics	93	89	100	100
	Archives	89	91	100	100
	Feedback	97	91	100	100
	Incident Reports	98	91	96	100

Microsoft Edge	Log In Page	97	90	96	91
	Certificate Requests	100	96	100	100
	Track Requests	99	96	100	100
	Events & Announcements	89	90	96	100
	Blotter Reports	99	95	100	100
	Elected Officials	73	100	96	100
	Dashboard	72	90	96	100
	Residents	91	91	96	100
	Officials Management	90	89	96	100
	Staff Management	98	89	96	100
	Analytics	95	89	96	100
	Archives	99	91	96	100
	Feedback	94	91	96	100
	Incident	98	91	96	100
Opera	Log In Page	91	90	96	91
	Certificate Requests	99	96	100	100

Track Requests	97	95	96	100
Events & Announcements	94	90	96	100
Blotter Reports	97	95	100	100
Elected Officials	100	100	96	100
Dashboard	88	90	96	100
Residents	92	91	96	100
Officials Management	91	89	96	100
Staff Management	92	89	96	100
Analytics	95	89	96	100
Archives	97	91	96	100
Feedback	94	91	96	100
Incident	94	91	96	100

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATIONS

This chapter contained a summary of the capstone project, which included a discussion of the conclusions found during the study and the recommendations that contributed to the project's success.

Summary

The Smart Dashboard system was developed to improve barangay communication, citizen participation, and transparency in governance through an online platform.

The Smart Dashboard system was developed to improve barangay communication, citizen participation, and transparency in governance through an online platform. The study focused on the development and evaluation of a Smart Dashboard designed to solve frequent barangay issues such as slow manual certificate processing, paperwork delays, and inefficient service delivery. To meet the increasing need for digital solutions, the Smart Dashboard was developed as a user-friendly online platform that allows citizens to request certifications, report concerns, view announcements, and access real-time information, while barangay officials can use digital record-keeping and real-time data for planning and decision-making. Based on e-governance and digital transformation concepts, the system prioritized automated workflows, user-friendly design, and secure features to improve efficiency, decrease errors, and increase transparency. The system was developed following the Software Development Life Cycle (SDLC) and went through thorough design,

implementation, and testing to assure usability, dependability, and security. The evaluation results showed high user satisfaction, increased productivity, fewer human errors, and better communication, demonstrating that the Smart Dashboard improved digital governance and community participation in barangay services.

Conclusion

In conclusion, the development of the Smart Dashboard showed that it was a practical tool in solving the common problems faced by many barangays. The outcomes of the project clearly showed that the first objective to design and develop an interactive web-based Smart Dashboard equipped with announcements, event boards, request forms, and automated notifications was fully achieved. The system was able to turn slow, manual tasks such as certificate requests, blotter documentation, resident profiling, and posting announcements into quicker and more organized digital processes. It also improved communication within the community, made daily administrative work easier for barangay staff, and gave residents a more convenient way to access information and services without always needing to go to the barangay hall. Many users shared that the interface was simple to use and that the notifications and organized layout helped them stay updated on barangay matters more easily.

The second objective, to provide citizens with a reliable online platform where they could request and track different barangay certifications, was also accomplished. The Smart Dashboard offered various certificate services such as Barangay Clearance, Certificate of Residency, Indigency Certificate, Good

Moral Certificate, No Income Certificate, and many others needed for school, work, legal matters, and other official purposes. With the online request feature, unnecessary walk-ins and long waiting times were reduced. Residents shared that the online process was more convenient and clearer than the manual method. Barangay staff also noted that the system reduced paperwork and made certificate processing more organized, showing that the project improved service delivery for both sides.

The last objective to evaluate the system's user acceptance using TAM and assess its quality based on the ISO/IEC 25010 standard was also achieved. Feedback from user testing showed that both residents and barangay officials found the system useful, dependable, and easy to use. The system performed well in terms of usability, reliability, and functionality, meeting the expected standards for a digital governance platform. Users appreciated how the dashboard provided clear and organized information that helped with planning and decision-making. Overall, the evaluation results confirmed that the Smart Dashboard had met the expected quality standards and proved to be effective in encouraging community participation.

Recommendation

These recommendations are based on the findings of the project and the suggestions of IT experts and some instructors. The study identified ways to improve the system’s performance, security, and usability, making it more convenient for users.

1. Developing a Progressive Web App (PWA) was recommended to allow users to access the platform anytime and anywhere using various

devices. This approach enabled easier submission of requests, real-time access to updates, and viewing of information without the need to install a traditional mobile application.

2. Implementing an online payment feature was suggested to make transactions faster and safer. Users were able to pay fees digitally, which reduced manual processing, errors, and waiting times.
3. Adding SMS notification features was recommended, allowing residents to still receive updates even without internet access. Important alerts such as request status, announcements, and reminders were sent directly to their mobile phones.
4. Integrating offline access capabilities is suggested so the system can still work even without an internet connection. This allows users and barangay staff to view important information and continue basic tasks during network interruptions, making the system more reliable for areas with weak or unstable internet
5. Using blockchain technology was suggested to make certificates secure and trustworthy. This ensured documents could not be tampered with and could be verified easily.
6. A digital signature QR code verified documents as authentic, confirmed the barangay captain's approval, and improved transparency.
7. Adding an API Scanner for ID Verification was also recommended. This feature allowed the system to scan and validate resident IDs automatically, making the verification process faster, more accurate, and more secure.

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APPENDICES

APPENDIX A

PLANNING & REQUIREMENT ANALYSIS PHASE

LAGUNA STATE POLYTECHNIC UNIVERSITY- LOS BAÑOS CAMPUS

COLLEGE OF COMPUTER STUDIES

PROJECT TITLE: **SMART DASHBOARD: A WEB-BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED DATA PROCESSING**

THE RESEARCHERS: Angela Ante
Nicen Howell Mendoza
Karylle Vidal

Dear IT Expert,

Good Day! We are 4th year students, currently pursuing a Bachelor of Science in Information Technology, specializing in Web and Mobile Application Development. As part of our academic endeavors, we have developed a web-based system adhering to the ISO 25010 standard.

To assess the effectiveness and user satisfaction of our system, we have devised a set of questions. Your responses, rated on a scale from 1 to 4, will help us evaluate various aspects of the system, from usability, functionality, security, and overall user experience.

1 = Poor 2= Needs Improvement 3 = Fair 4 = Good 5 = Excellent

RUBRIC FOR 5-Point Likert Scale to evaluate product/system quality using ISO 25010

Characteristics	Excellent (5)	Good (4)	Fair (3)	Needs Improvement (2)	Poor (1)
Functional Suitability	Fully meets all functional requirements; performs all barangay services accurately and reliably (e.g., certificate requests, appointments, blotter reports).	Meets most functional requirements; minor issues that don't affect core barangay functions.	Meets basic functions but with some limitations in accuracy or process completion.	Partially meets requirements; several functions are incomplete or inconsistent..	Fails to meet key barangay functions; major errors or missing features.
Reliability	Consistently stable; no crashes or data errors during use.	Generally stable; rare minor issues that don't affect performance.	Occasional lags or brief downtimes but still usable.	Frequent system slowdowns or temporary crashes.	Very unstable; crashes often or loses data.
Performance efficiency	Very fast and efficient; pages load quickly and respond instantly.	Fast performance with minor delays during heavy use.	Acceptable speed; some noticeable delays.	Slow response time and occasional freezing.	Very slow; often lags or times out.
Usability	Very easy to use; clear interface and smooth navigation for residents and staff.	User-friendly with minor areas that could be improved.	Some confusing parts; users need time to adjust.	Difficult to navigate without guidance.	Hard to use; confusing layout and poor design.
Security	Strong security; login, encryption, and role permissions protect all data.	Secure overall; minor vulnerabilities.	Adequate security but with a few risks.	Weak protection; potential exposure of user data.	Very poor security; data easily compromised.
Compatibility	Works perfectly on all browsers and devices.	Works on most platforms; minor display issues.	Runs on main browsers only; limited support for others.	Works on few devices; many compatibility issues.	Not compatible with most systems.
Maintainability	Code is clean and modular; easy to fix, update, and enhance.	Easy to maintain with minor complex areas.	Can be maintained but takes more time to update.	Difficult to debug or modify.	Hard to maintain; major revisions needed for changes.
Portability	Runs smoothly on any environment; easy to deploy or transfer.	Works on most environments with minor adjustments.	Needs configuration to run properly in new environments.	Hard to set up in new environments.	Not portable; fails to run outside original setup.

Signature of Respondent: _____ Date: _____

Results of Post - Evaluation for IT Specialist

Characteristic				Overall Mean
Functional Suitability	5	4	5	4.5
Reliability	5	5	5	5.0
Performance Efficiency	5	4	5	4.5
Usability	5	4	5	4.5
Security	4	5	5	4.5
Compatibility	5	5	5	5.0
Maintanability	4	4	4	4.0
Portability	5	5	5	5.0

LAGUNA STATE POLYTECHNIC UNIVERSITY- LOS BAÑOS CAMPUS
COLLEGE OF COMPUTER STUDIES

TECHNOLOGY ACCEPTANCE MODEL QUESTIONNAIRE

Dear respondent,

We are students of the Laguna State Polytechnic University Los Baños Campus under the College of Computer Studies conducting development on the project entitled **"SMART DASHBOARD: A WEB-BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED DATA PROCESSING"** We would like to seek your opinion on the acceptability of the said project by scoring the statements below by checking the appropriate numbers, wherein 1 = Strongly Agree; 2 = Agree; 3 = Disagree; and 4 = Strongly Disagree. Rest assured that any information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for research purposes only. Thank you for your cooperation.

Angela Ante
Nicen Howell Mendonza
Karylle Vidal

Perceived Usefulness	**4**	**3**	**2**	**1**
1. Mas napapadali ng sistema ang pag-aasikaso ng mga gawain at serbisyo sa barangay. *The system improves the efficiency of managing barangay operations and resident services.*				
2. Nagbibigay ang sistema ng mabilis at tamang impormasyon na nakakatulong sa mga opisyal sa paggawa ng desisyon. *The system provides real-time and accurate information that enhances decision-making for barangay officials.*				
3. Nagbibigay ang sistema ng mabilis at tamang impormasyon na nakakatulong sa mga opisyal sa paggawa ng desisyon. *Using this system increases productivity and improves service delivery within the barangay.*				
4. Nakakatulong ang sistema na maiwasan ang mga problema sa barangay bago pa ito lumala. *The system helps prevent potential issues in barangay management before they become critical.*				
5. Malaking tulong sa akin ang sistemang ito sa maayos na pamamahala ng mga gawain sa barangay. *I find the system valuable in managing barangay activities effectively.*				

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Perceived Ease of Use				
1. Madaling gamitin at intindihin ang itsura ng sistema para sa mga residente at opisyal. *The system interface is user-friendly and easy to navigate for residents and officials.*				
2. Madali kong natutunan kung paano gamitin ang sistema at hanapin ang mga kailangan ko. *I find it easy to learn how to operate the system and access the information or services I need.*				
3. Hindi mahirap gamitin ang sistema at hindi kailangan ng masyadong teknikal na kaalaman. *Using the system requires minimal effort or technical knowledge.*				
4. Maginhawa gamitin ang online platform dahil puwedeng ma-access ang mga serbisyo kahit saan at kahit kailan. *The online platform and dashboard make it convenient to access barangay services anytime and anywhere.*				
5. Komportable ako sa kabuuang ayos at takbo ng sisteme. *I am comfortable with the overall setup and functionality of the system.*				
Attitude (Towards Using the System)				
1. Masaya at kampante akong gamitin ang sistema para sa mga transaksyon at serbisyo ng barangay. *I feel positive about using the system for barangay transactions and services.*				
2. Bagay ang sistema sa mga ginagawa at proseso ng barangay ngayon. *The system fits well with the current processes and practices of barangay management.*				
3. Naniniwala ako na malaking tulong ang teknolohiyang ito sa mga gawain at serbisyo ng barangay. *I believe this technology can greatly benefit barangay operations and public service delivery.*				
4. Magandang karanasan ang paggamit ng sistemang ito dahil kapaki-pakinabang at madali itong gamitin. *Using this system is an enjoyable and beneficial experience.*				

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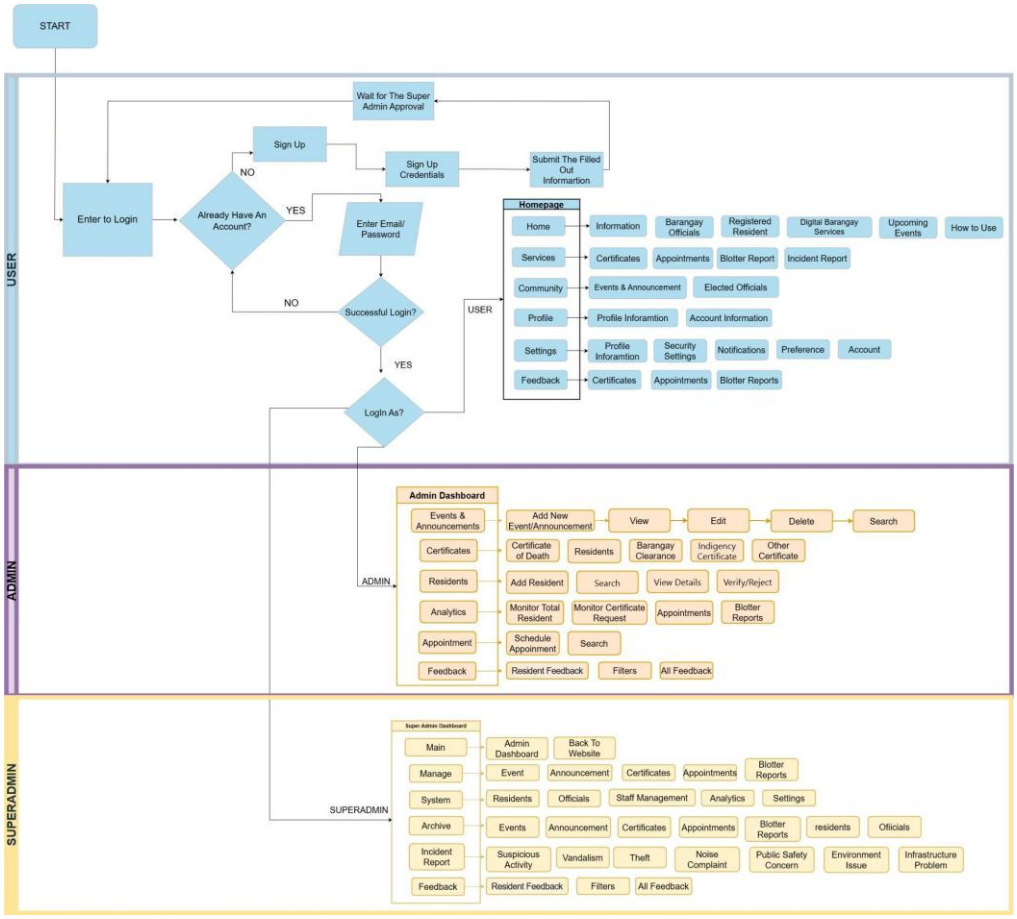
5. Irekomenda ko ang sisteme sa ibang barangay o lokal na pamahalaan. *I would recommend this system to other barangays or local government units.*				
Behavior (Intention to Use the System)				
1. Plano kong gamitin ang sistema palagi para sa mga gawain sa barangay. *I intend to use this system regularly for barangay-related activities.*				
2. Gagamitin ko ang sistemang ito bilang tulong sa pamamahala ng mga serbisyo ng barangay sa hinaharap. *I am likely to rely on this system to help manage barangay services in the future.*				
3. Handa akong gumamit pa ng ibang digital na solusyon sa barangay dahil sa karanasan ko sa sistemang ito. *I am willing to adopt other digital solutions in barangay operations based on my experience with this system.*				
4. Malamang na irekomenda ko ang sistema sa ibang residente, empleyado, at opisyal ng barangay. *I am likely to recommend this system to residents, staff, and officials in other barangays.*				
5. Balak kong gamitin ang sistemang ito bilang bahagi ng pang-araw-araw na operasyon ng barangay. *I am committed to using this system as part of daily barangay operations.*				

_____
Signature of the Respondent

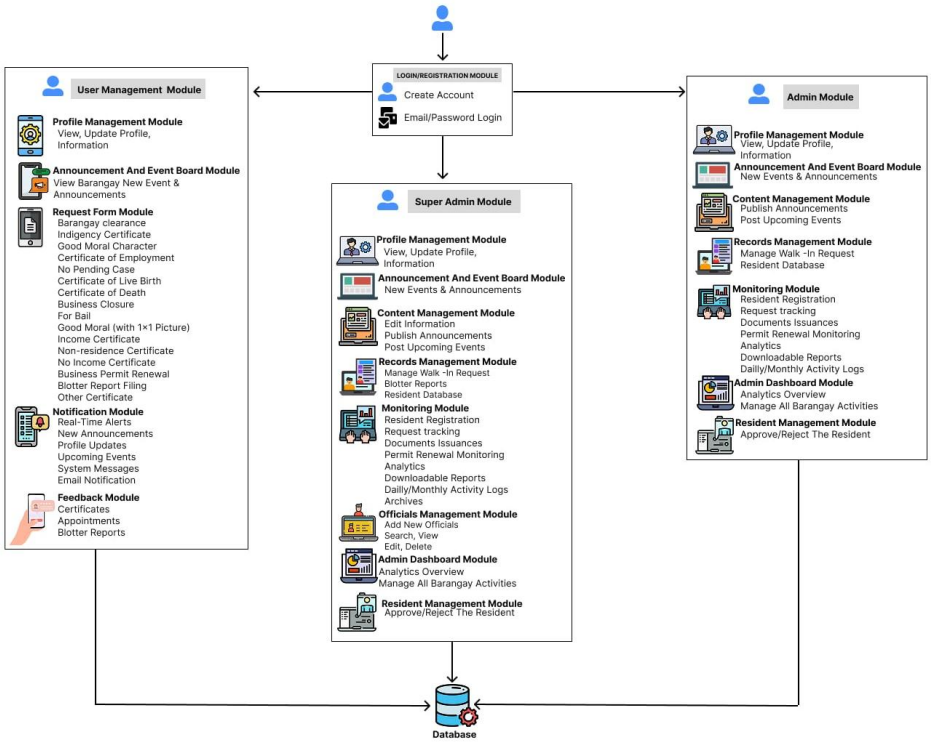
Results of Post-Evaluation for TAM Survey

			PERCEIVED USEFULNESS					Perceived Ease of Use						Attitude (Towards Using the System)					Behavior (Intention to Use the System)				
RESPON	The syste	The syste	Using this	The syste	I find the	system va	The syste	I find it ea	Using the	The onlin	I am comfortable w		I feel pos	The syste	I believe t	Using this	I would recommend		intend to	I am likely	I am willin	I am likely	I am committed to
1	4	3	4	3	4	**3.17**	4	3	4	4	3	**3.60**	3	4	4	4	4	**3.80**	4	3	4	3	4	**3.60**
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3	3	4	3	4	3	**3.40**	3	4	3	3	4	**3.40**	3	3	2	4	3	**3.00**	4	3	4	3	4	**3.60**
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50	3	4	3	4	3	**3.40**	4	3	3	3	4	**3.40**	3	4	3	3	4	**3.40**	3	4	3	4	4	**3.60**
MEAN	**3.50**	**3.60**	**3.52**	**3.56**	**3.42**	**3.51**	**3.48**	**3.58**	**3.50**	**3.44**	**3.54**	**3.51**	**3.44**	**3.46**	**3.38**	**3.56**	**3.60**	**3.49**	**3.28**	**3.44**	**3.42**	**3.40**	**3.24**	**3.36**

System Flowchart



System Architecture



APPENDIX B

USER MANUAL

https://malinta-connect.vercel.app

Smart Dashboard: A Web-Based Platform for Citizen Engagement and Data-Driven Digital Governance Using Automated Data Processing

Brgy. Malinta, Los Baños Laguna

Your one-stop platform for all barangay services. Request certificates, schedule appointments, and submit reports with ease.

TABLE OF CONTENTS

01 Registration

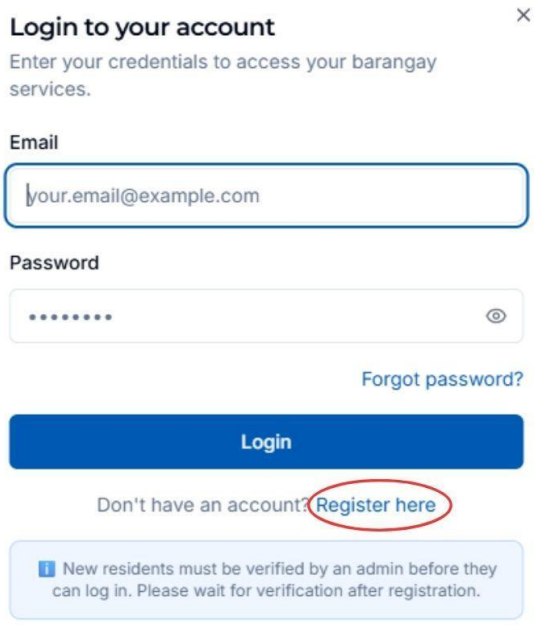
03 Log In

05 User Modules

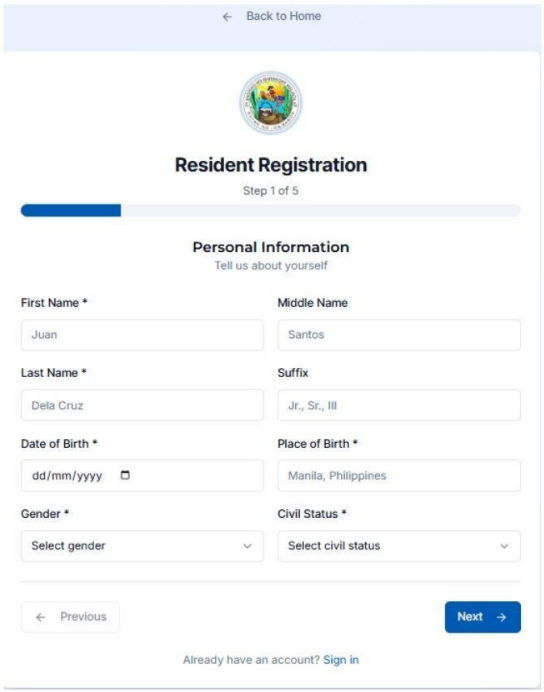
07 Super Admin Modules

03 Admin Modules

REGISTRATION



This is the login page for residents, where you can log in or register. If you don't have an account, click Register here. From there, fill out the required information and wait for admin verification. Once approved, return to this page, enter your email and password, and log in to access barangay services.



Resident Register Form

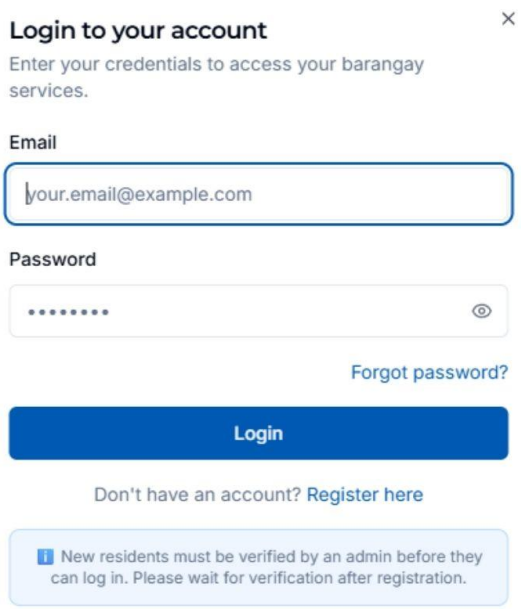
This screen shows the Resident Registration Step 1 of 5. It is the first part of a multi-step Form used by the barangay malinta system to collect a resident's personal information.

Purpose of this Registration

- To gather basic personal details of a resident for official records.
- To verify identity and create a user account inside the system.
- To ensure accurate resident profiling for services such as certificates, announcements, and other barangay transactions.

To register, fill in your First Name, Middle Name, Last Name, and suffix if you have one. Enter your Date of Birth and Place of Birth, then select your Gender and Civil Status. Once everything is complete, click Next to continue the registration process.

LOGIN



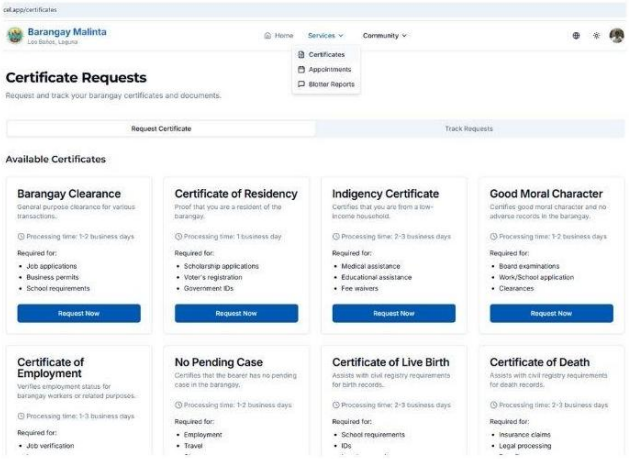
Resident Login

Residents use this section to access barangay services. They need to enter their email and password to log in. There's also a reminder that new residents must wait for an admin to verify their account before they can access the system. This helps make sure that only real residents can use the services.

Barangay Officials Login

The login page for Barangay Officials ensures that only authorized personnel can access the system. Before an official can log in, the Super Admin must first set up their account. The Super Admin is responsible for creating each official's login credentials and assigning the specific modules they are allowed to access, such as certificates, appointments, or blotter reports. This means officials cannot enter the system on their own unless their account has already been created and configured. Once the Super Admin provides their official email and password, the barangay official can log in using this form and access only the features that match their assigned role. This setup helps maintain security, proper user control, and organized management within the barangay system.

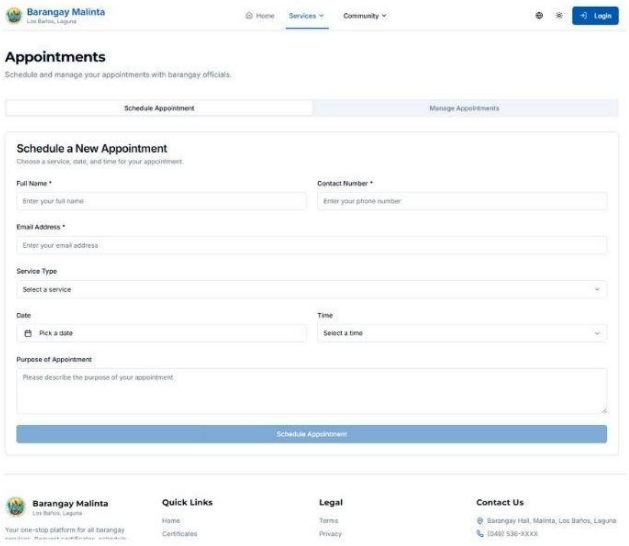
USER MODULES



User Certificate Requests Module

Certificate Request

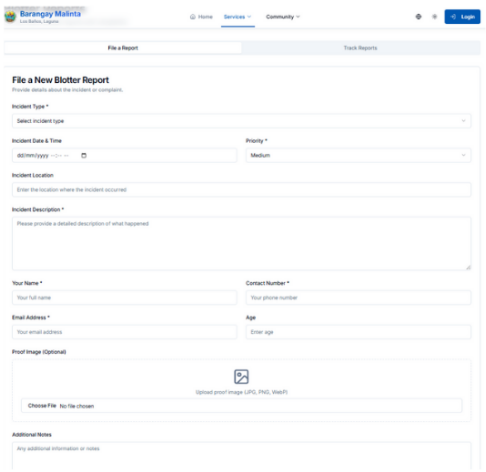
In this section, users can request certificates. Click the request button and select any certificate they want. After submitting the request. They will need to wait for the admin approval to find out if their request has been approved or rejected, they can also track the status of their request by clicking the Track Request button.



User Appointments Module

Appointments

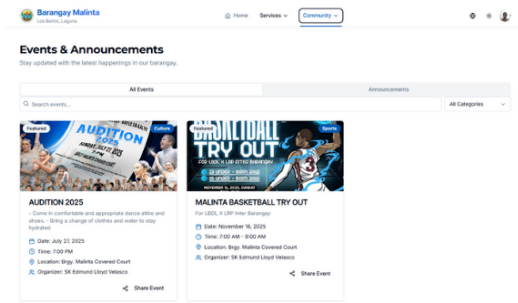
In this section, users can schedule an appointment. First, they need to fill up the form. Once submitted they need to wait for the admin approval. They can view the status of their appointment by clicking the manage appointments button.



User Blotter Reports Module

Blotter Report

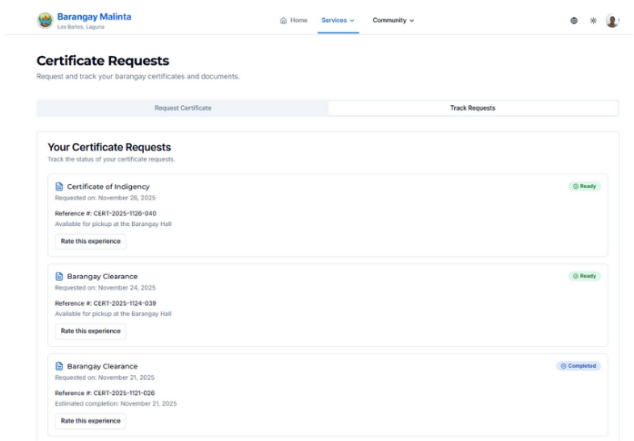
Users can file a blotter report, but first they need to fill up the form , they also have option to upload an image as proof. After submitting the report, they must wait to for an admin approval before proceeding to the barangay. Users can also check the status of their report.



User Events & Announcements Module

Events & Announcements

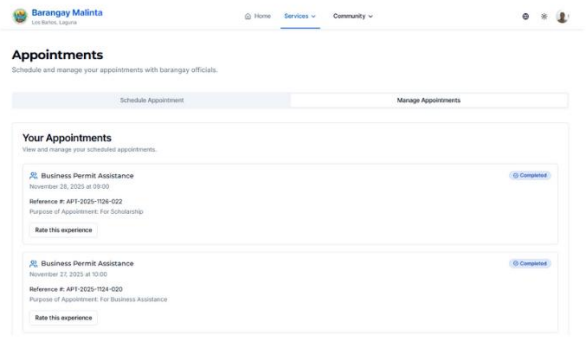
Users can view events & announcements by clicking the all events button. Once clicked they can see the list of events that admin created. The announcements page helps residents informed about what is happening in the barangay.



User Certificate Track Requests Module

User Certificate Track Requests

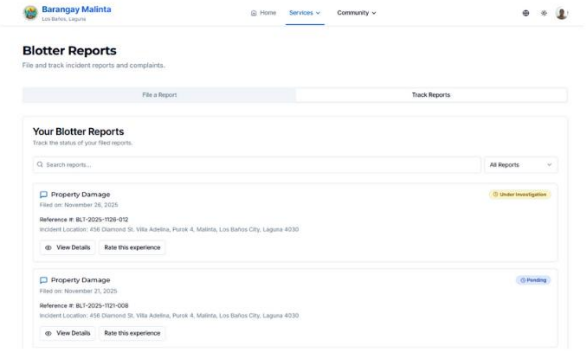
Users can track the status of their certificate requests in real time. It also includes a feedback feature where users can rate their experience, helping improve the quality of barangay services.



User Manage Appointments Module

User Manage Appointments

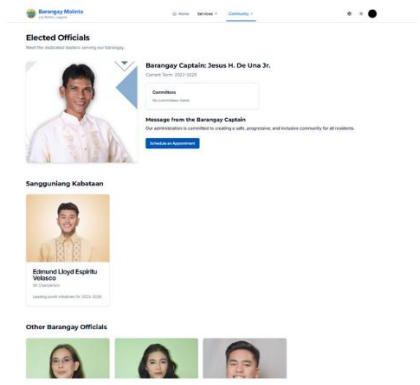
Users can view, manage, and monitor their scheduled appointments. Users can also rate their appointment experience, providing feedback to enhance service efficiency and user satisfaction.



User Blotter Track Reports Module

User Blotter Track Reports

Users can track the progress of their blotter reports through timely updates. It includes a rating feature that lets users share feedback on their experience, supporting better transparency and service improvement.

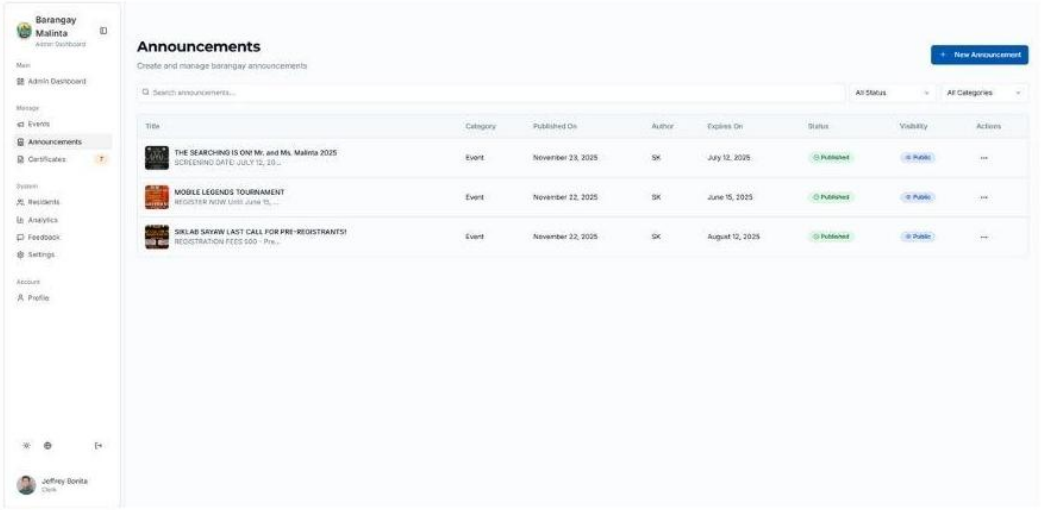


User Elected Officials Module

Elected Officials

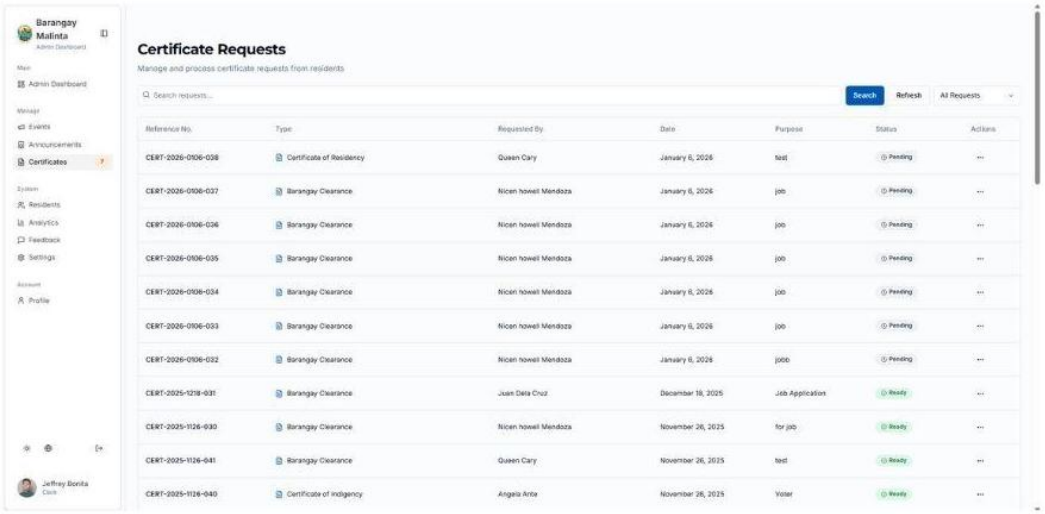
This module is designed to help the residents familiarize themselves with their elected barangay officials. Users can view information about the officials and can easily schedule an appointment by clicking the “Schedule an Appointment” button. One clicked they will redirected to the appointment page.

ADMIN MODULES



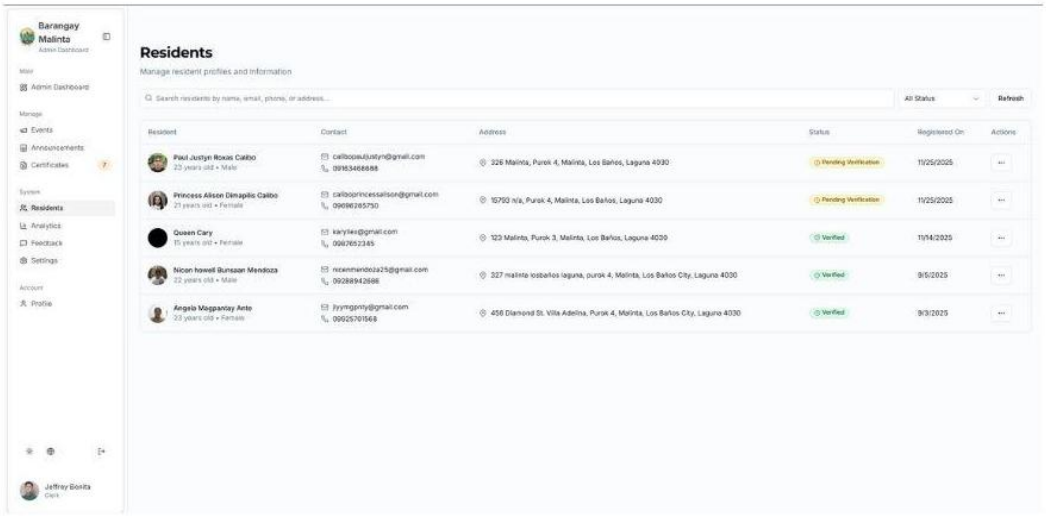
Admin Announcements Module

The Announcements page allows admins to create, edit, publish, and manage barangay announcements. It includes search and filter options for easy organization, and displays details such as title, category, status, and visibility. This module helps keep announcements updated, organized, and accessible to the community.



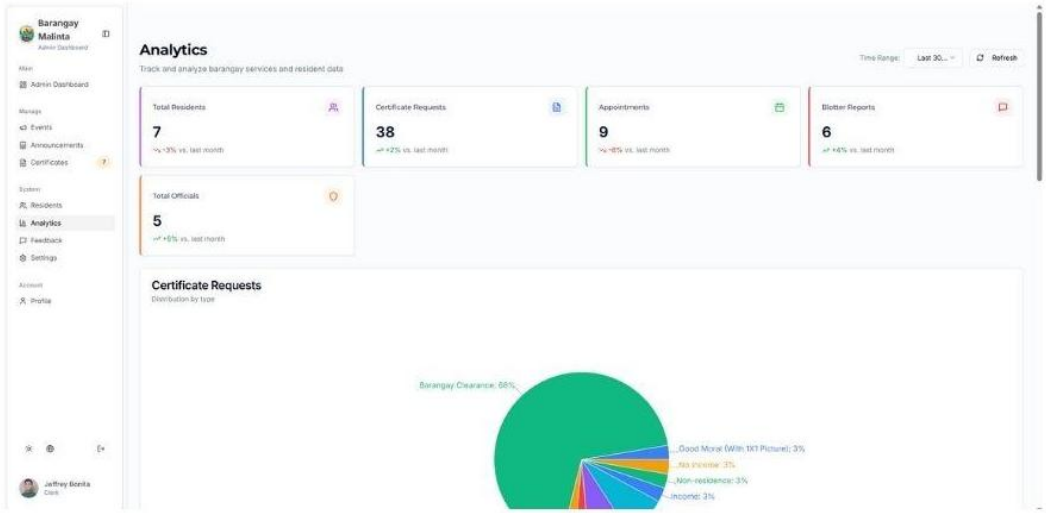
Admin Certificate Requests Module

The Certificate Request module allows admins to manage and process all certificate requests from residents. It displays key details such as request type, requester, date, purpose, and status, with search and filter options for easy organization. Admins can review requests and update their status, making certificate processing faster and more organized.



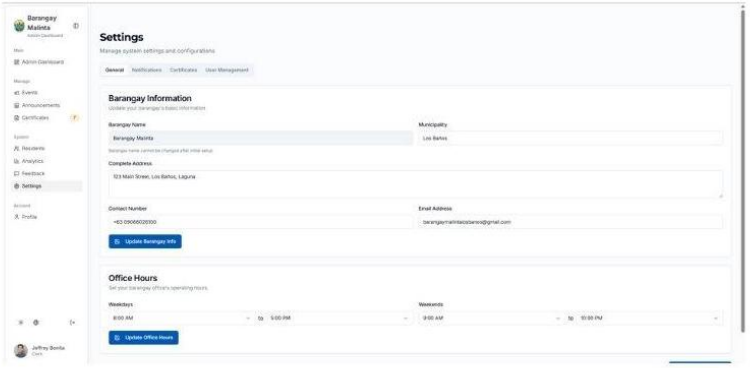
Admin Resident Module

The Residents page allows administrators to view and manage resident profiles, including personal details, contact information, addresses, and verification statuses. It includes search and filter options to easily find residents and manage pending verifications, helping to maintain accurate barangay records.



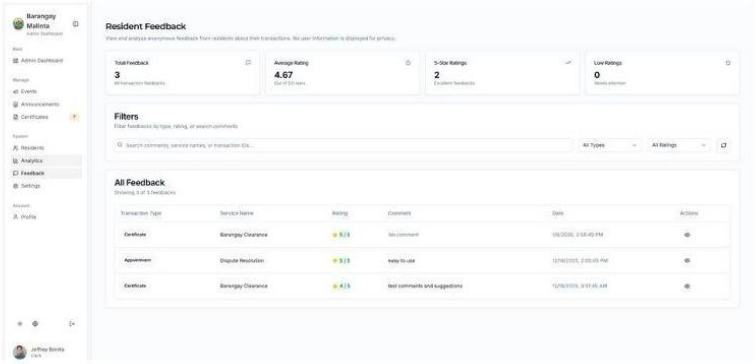
Admin Analytics Module

The Analytics section of the admin dashboard provides a clear overview of barangay data, including totals for residents, officials, certificate requests, appointments, and blotter reports, with trends compared to the previous month. Visual charts such as pie charts for request types for appointment statuses, and bar graphs for blotter categories help the admin easily track service usage and common issues, supporting data-driven decisions and effective monitoring of barangay services.



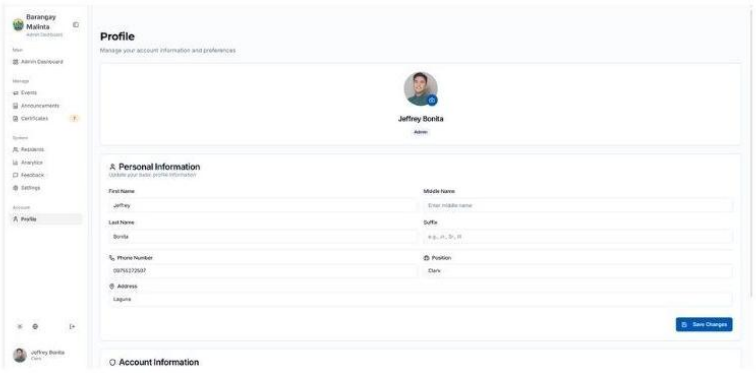
Admin Settings Module

The Settings module allows the admin to manage the Barangay Smart Dashboard's information and preferences. Here, the admin can update official details, set office hours, customize notifications, manage certificate types, and assign user roles. It serves as the system's control center, ensuring accurate information and smooth platform management.



Admin Resident Feedback Module

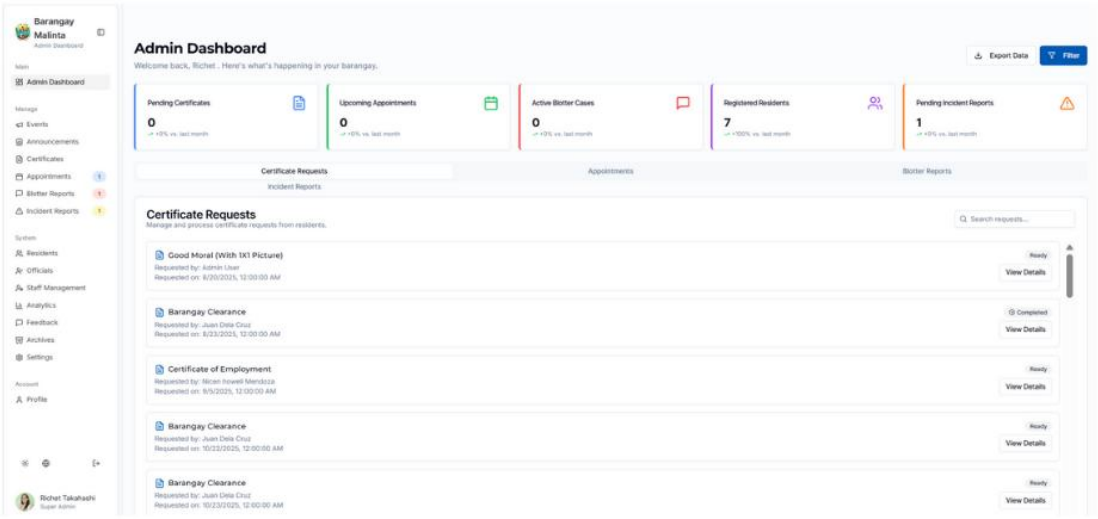
The resident feedback page lets administrators review anonymous feedback on barangay services. It provides summaries of ratings, filtering options, and a detailed of transactions, comments and dates - helping officials evaluate service quality and identify improvements while protecting user privacy.



Admin Profile Module

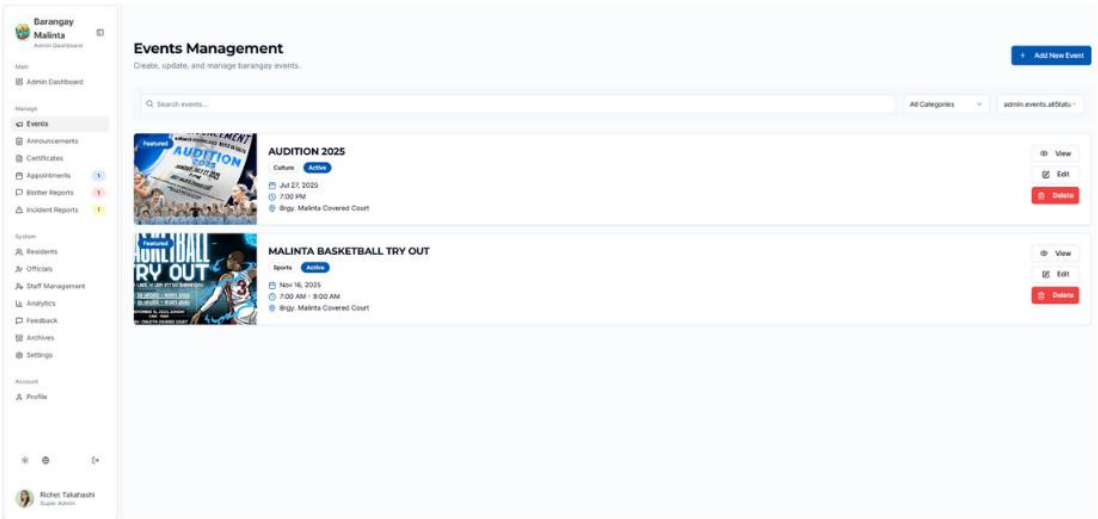
The Admin Profile page, where the administrator can view and update their personal details. The editable fields include first name, last name, middle name, suffix, phone number, address, and their assigned position. The admin can also change their profile picture and save any updates using the Save Changes button.

SUPER ADMIN MODULES



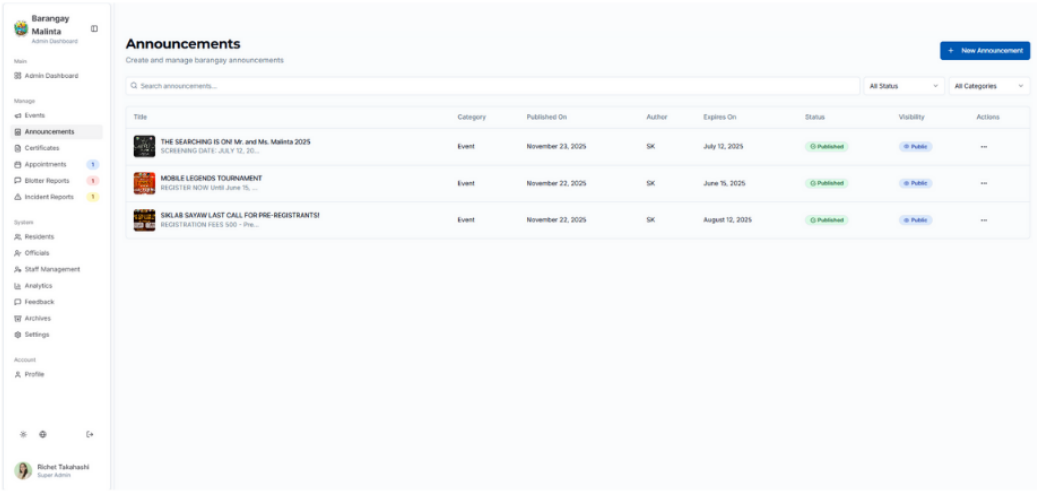
Super Admin Dashboad Module

The Admin Dashboard gives the Super Admin an overview of the barangay system, with summary cards showing pending certificates, upcoming appointments, active blotter cases, and registered residents, including monthly trends. Below, tabs for Certificate Requests, Appointments, and Blotter Reports let the admin view details, track statuses, and process requests efficiently.



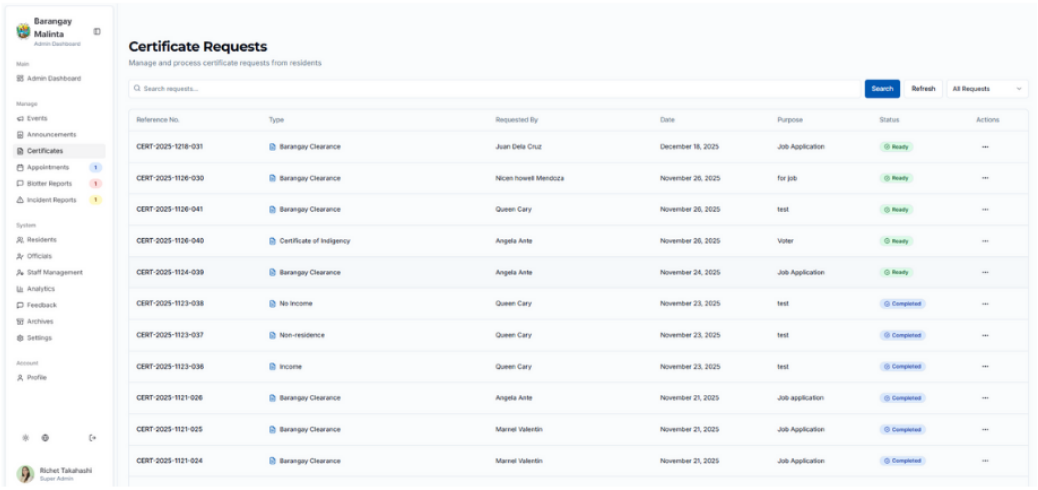
Super Admin Event Mangement Module

The Events page lets the admin manage barangay events, with a search bar and filters to find upcoming, ongoing, or completed events. Each event shows its title, date, location, organizer, and status, with options to edit, publish, or archive, keeping announcements accurate and up to date.



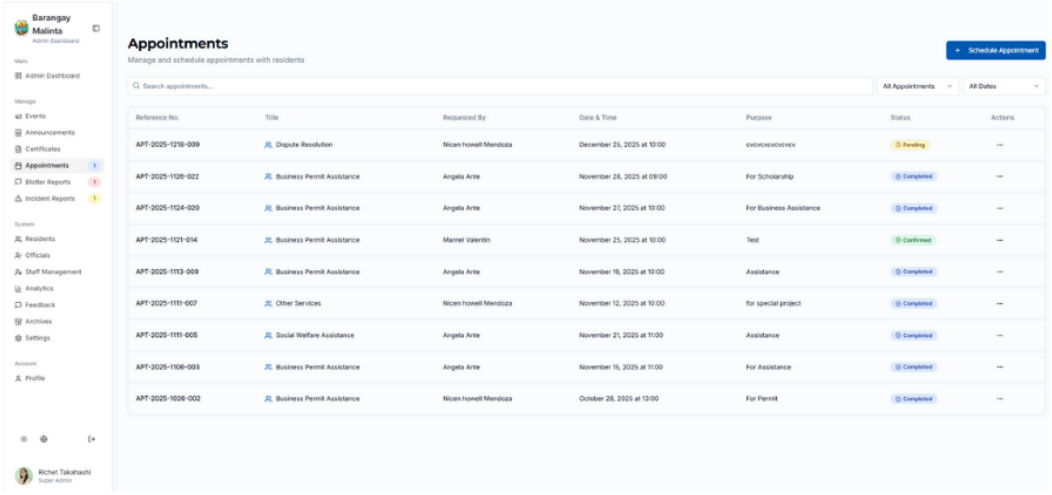
Super Admin Event Announcement Module

The Announcements page allows the Super Admin to manage all barangay posts, with search and filters to sort by category or status. Each announcement shows key details and visibility, with options to update, publish, unpublish, or archive, keeping announcements organized and current.



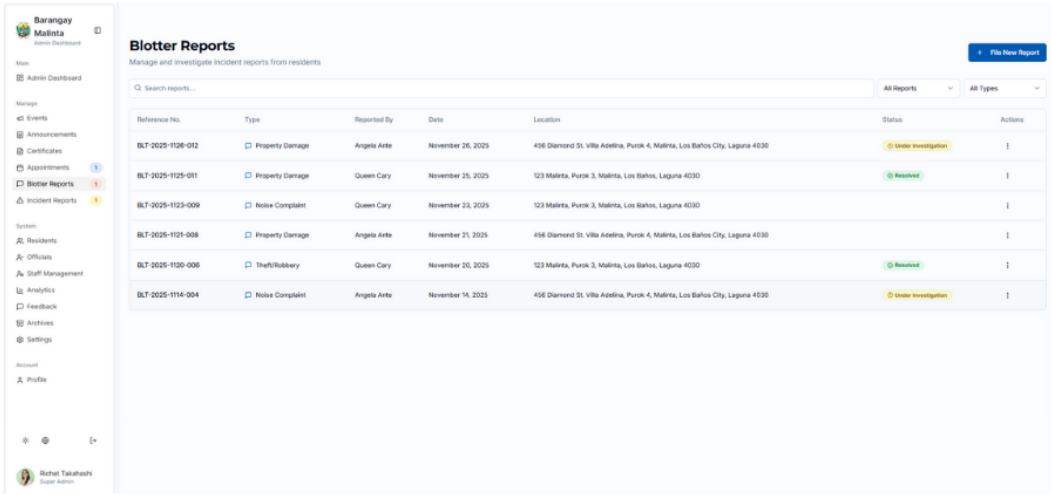
Super Admin Event Certificate Request Module

The Certificates page allows the Super Admin to manage certificate types and resident requests. The Super Admin can view, process, and update requests, as well as create or edit certificate templates, ensuring certificates are handled efficiently.



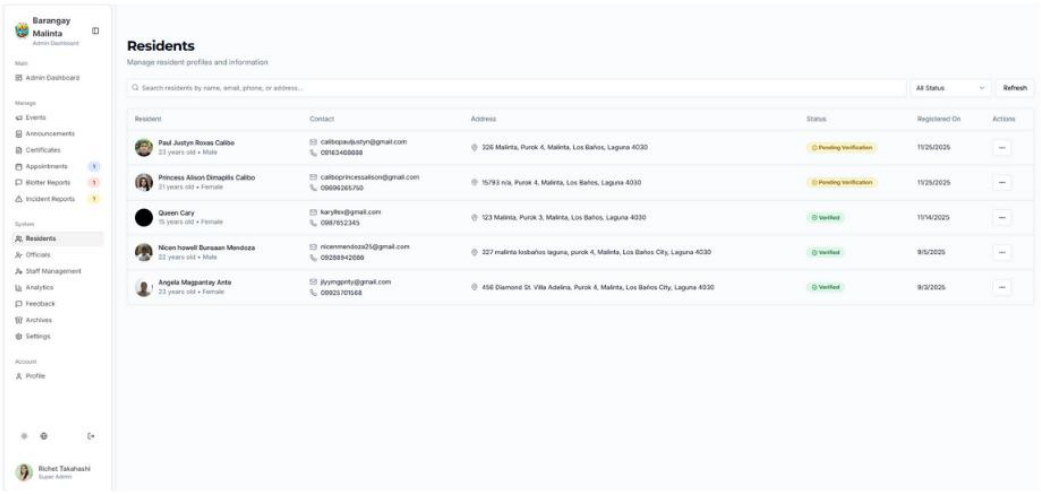
Super Admin Appoinment Module

The Appointments page allows the Super Admin to manage all resident schedules using search and filter options. Each appointment shows the resident, purpose, date, time, and status, with actions to approve, reschedule, or cancel, ensuring organized and conflict-free scheduling.



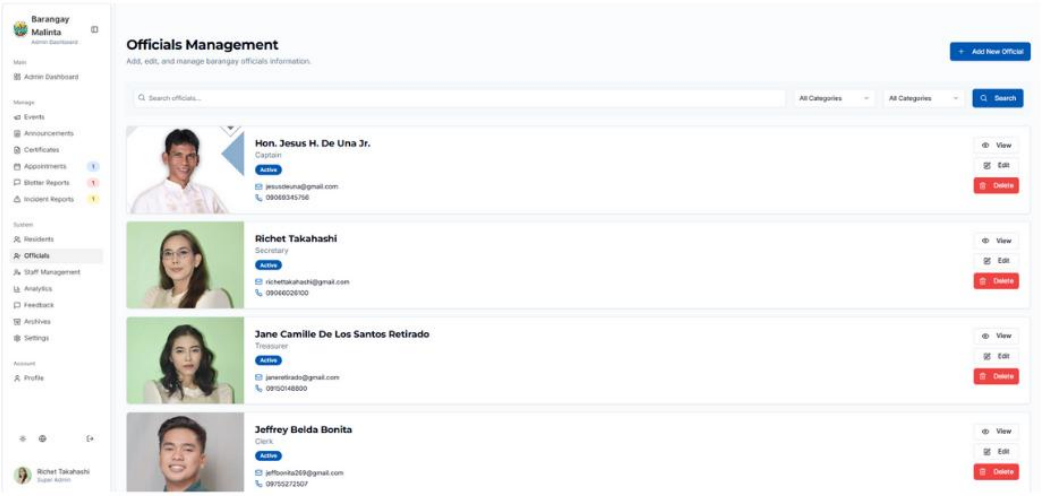
Super Admin Blotter Report Module

The Blotter Reports page allows the Super Admin to manage all reported barangay incidents using search and filter options. Each report shows key details and status, with actions to view, update, or delete cases, ensuring accurate and organized incident records.



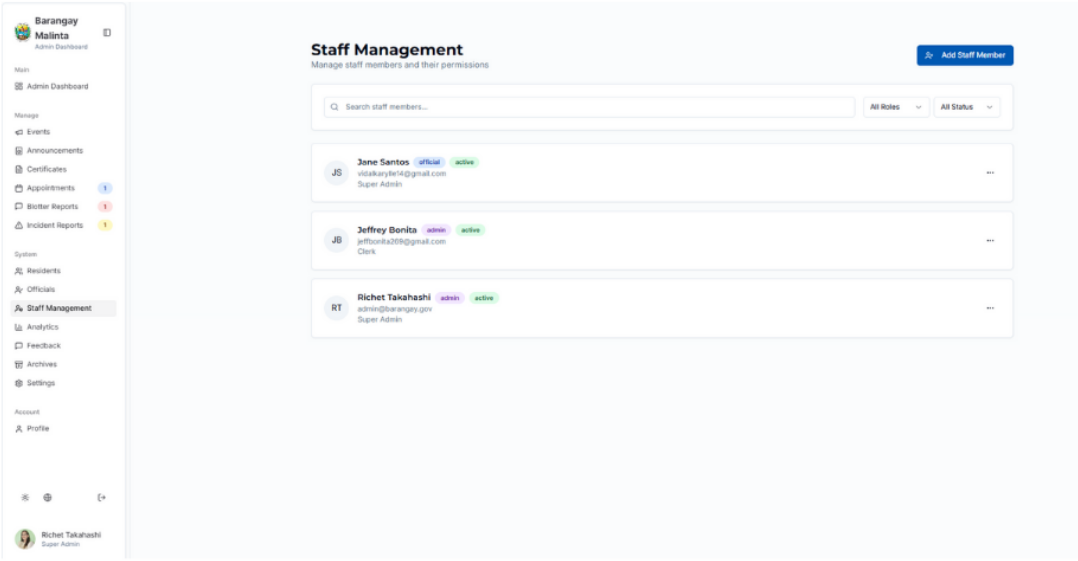
Super Admin Resident Module

The Residents page allows the Super Admin to manage all registered residents using search and filter options. Each record shows key resident details, verification status, and registration date, with actions to view and manage profiles, ensuring accurate and verified resident information.



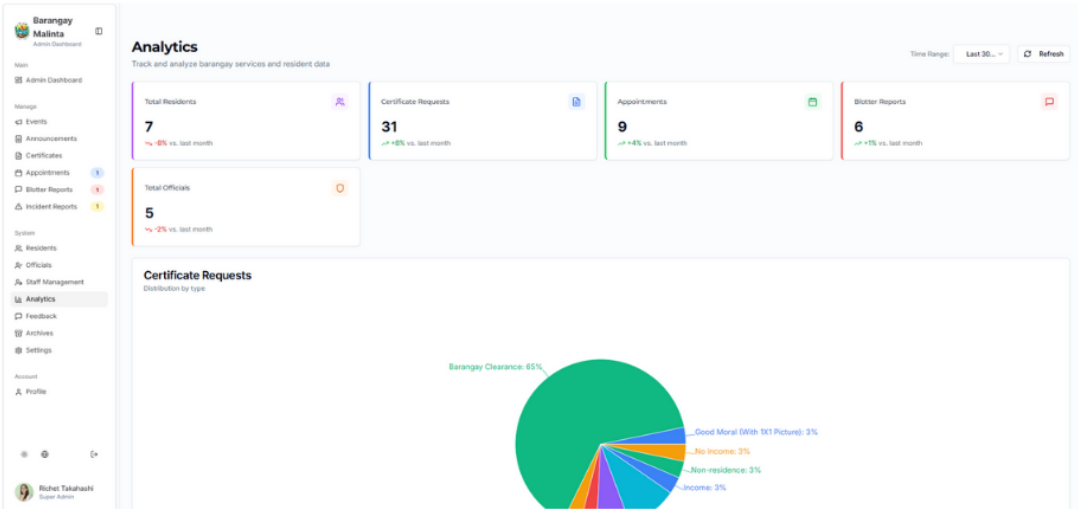
Super Admin Official Management Module

The Officials page allows the Super Admin to manage all barangay officials using search options. Each record shows the official's details and term information, with actions to update, replace, or manage profiles, ensuring accurate and updated leadership records.



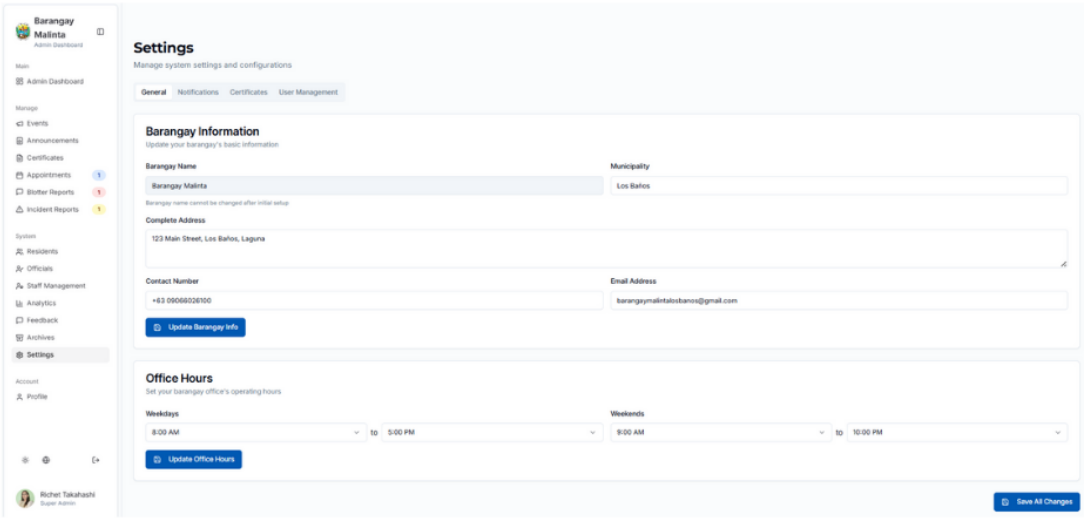
Super Admin Staff Management Module

The Staff Management page allows the Super Admin to manage staff accounts and permissions using search and filter options. The Super Admin can add staff members, edit roles, activate or deactivate accounts, and remove users, ensuring secure and controlled system access.



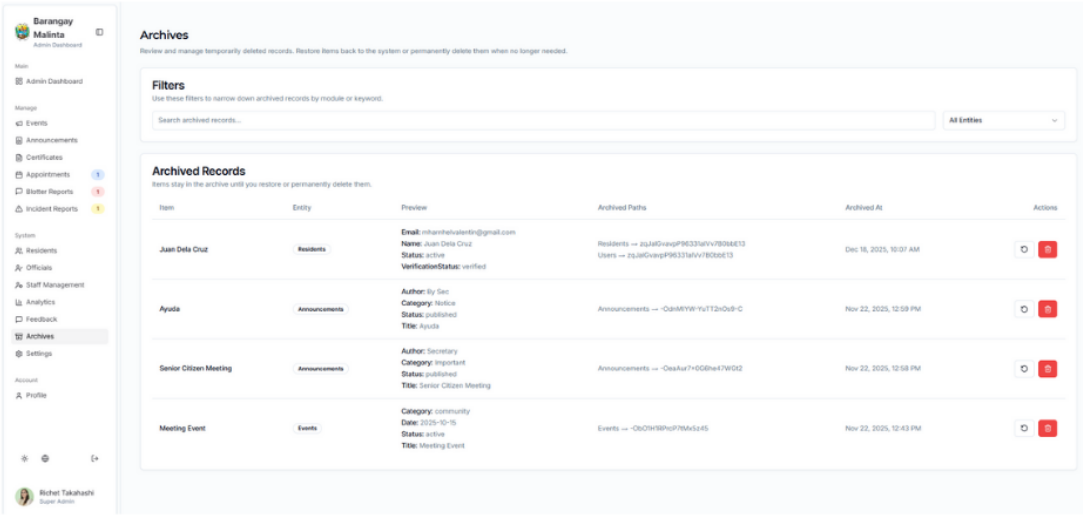
Super Admin Analytics Module

The Analytics page displays visual reports and data summaries about system usage. The Super Admin can view charts showing trends in resident registrations, certificate requests, blotter cases, and appointments. Each summary card highlights important statistics and key changes over time. This page helps the admin make data-driven decisions and understand the overall activity and performance of the barangay.



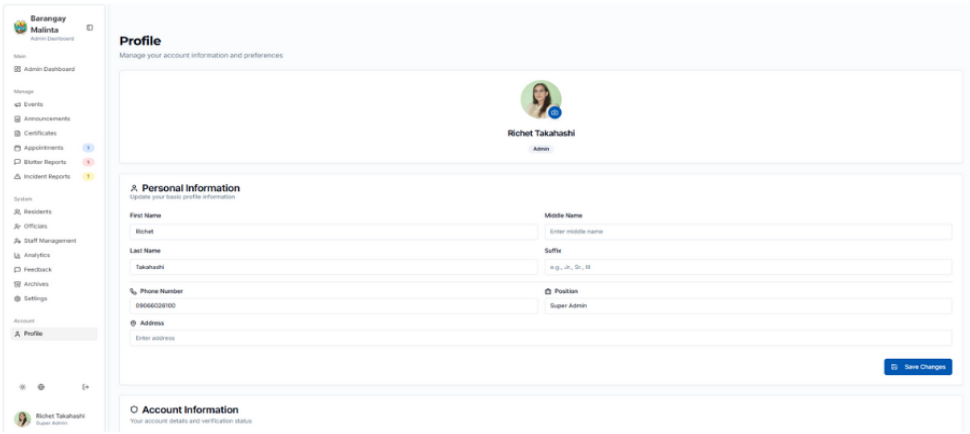
Super Admin Settings Module

The Settings page allows the admin to configure system preferences and barangay information. Here, the admin can update barangay profile details, change system settings, manage authentication preferences, and customize certain features.This page ensures the system works according to barangay standards and admin preferences.



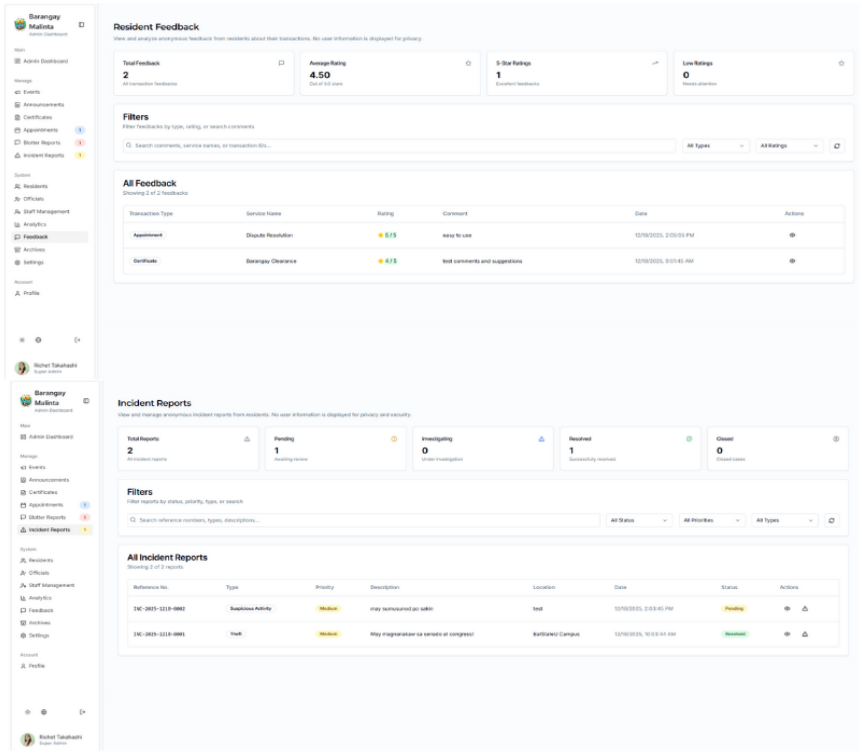
Super Admin Archives Module

The Archives page stores all previously removed or completed records from different modules. The search bar and filters help the Super Admin find archived items by name, date, or category. Each archived item displays its title, type, date archived, and the module it originally belonged to. The action menu allows restoration back to the original section or permanent deletion. This keeps the system organized while preserving past records when needed.



Super Admin Profile Module

The Profile page of the Super Admin dashboard. It allows the user to update their basic information, such as first name, last name, middle name, suffix, phone number, address, and their position in the barangay. There is also a section to upload or change the profile picture. A Save Changes button is provided at the bottom to apply updates.



Super Admin Feedback, Incident Reports Module

The Super Admin Feedback and Incident Reports allow the super administrator to review resident feedback and manage incident reports in an organized manner. These modules provide clear ratings, comments, and report statuses, helping evaluate service quality, ensure faster responses, and improve monitoring and accountability within the barangay.

APPENDIX C

SOURCE CODE

```tsx
"use client";

import Link from "next/link";
import Image from "next/image";
import { Button } from "@/components/ui/button";
import {
  Card,
  CardContent,
  CardDescription,
  CardFooter,
  CardHeader,
  CardTitle,
} from "@/components/ui/card";
import { ChevronRight, Users } from "lucide-react";
import { ServicesSection } from "@/components/services-section";
import { EventsSection } from "@/components/events-section";
import { useLanguage } from "@/contexts/language-context";
import { useAuth } from "@/contexts/auth-context";
import { useRouter } from "next/navigation";
import { useEffect, useState } from "react";
import { requestForToken } from "../firebase/firebase";
import { useFCMToken } from "@/hooks/use-fcm-token";
import { getAllOfficialsAction, type Official } from "@/app/actions/officials";

export default function Home() {
  const { t } = useLanguage();
  const { user, userProfile, loading } = useAuth();
  const router = useRouter();
  const { updateToken, hasToken } = useFCMToken();
  const [officials, setOfficials] = useState<Official[]>([]);
  const [officialsLoading, setOfficialsLoading] = useState(true);

  // Helper function to categorize officials by their roles
  const categorizeOfficials = (officials: Official[]) => {
    const captain = officials.find(
      (official) => official.position === "captain"
    );
    const councilors = officials.filter(
      (official) => official.position === "councilor"
    );
    const skChairperson = officials.find(
      (official) => official.position === "skChairperson"
    );
    const barangayStaff = officials.filter((official) =>
      ["secretary", "treasurer", "clerk"].includes(official.position)
    );

    return { captain, councilors, skChairperson, barangayStaff };
  };

  // Fetch officials data
  const fetchOfficials = async () => {
    try {
      setOfficialsLoading(true);
      const result = await getAllOfficialsAction();
      if (result.success && result.officials) {
        // Filter only active officials
        const activeOfficials = result.officials.filter(
          (official) => official.status === "active"
        );
        setOfficials(activeOfficials);
      }
    } catch (error) {
      console.error("Error fetching officials:", error);
    } finally {
      setOfficialsLoading(false);
    }
  };

  // Load officials on component mount
  useEffect(() => {
    fetchOfficials();
  }, []);

  // Auto-redirect based on user role
  useEffect(() => {
    if (!loading && user && userProfile) {
      console.log(userProfile.role);
      // Redirect based on role
      switch (userProfile.role) {
        case "admin":
          router.push("/admin");
          break;
        case "official":
          router.push("/admin");
          break;
        case "resident":
          // Residents stay on home page
          break;
        default:
          // Default case - stay on home page
          break;
      }
    }
  }, [user, userProfile, loading, router]);
```

```
96
97  // Request FCM token for push notifications
98  useEffect(() => {
99    const getFCMToken = async () => {
100     // Only request token if user is logged in
101     if (!user || !userProfile) return;
102
103     const vapidKey = process.env.NEXT_PUBLIC_FIREBASE_VAPID_KEY;
104
105     if (!vapidKey) {
106       console.error(
107         "VAPID key not configured. Please add NEXT_PUBLIC_FIREBASE_VAPID_KEY to your environment variables."
108       );
109       return;
110     }
111
112     const token = await requestForToken(vapidKey, user.uid, userProfile.role);
113     if (token) {
114       console.log("FCM Token received and stored successfully");
115       // Update the token in the hook
116       if (user?.uid && userProfile?.role) {
117         updateToken(token, user.uid, userProfile.role);
118       }
119     }
120   };
121
122   getFCMToken();
123 }, [user, userProfile, updateToken]);
124
125 // Show loading while checking auth state
126 if (loading) {
127   return (
128     <div className="flex items-center justify-center min-h-screen">
129       <div className="animate-spin rounded-full h-32 w-32 border-b-2 border-primary"></div>
130     </div>
131   );
132 }
133
134 // If user is admin or official, don't render the home page content
135 // (they will be redirected)
136 if (
137   user &&
138   userProfile &&
139   (userProfile.role === "admin" || userProfile.role === "official")
140 ) {
141   return null;
142 }
143
144 return (
145   <div className="flex flex-col">
146     {/* Hero Section with Image Background and White Overlay */}
147     <section className="relative w-full min-h-screen flex items-center">
148       {/* Background Image */}
149       <div className="absolute inset-0 z-0">
150         <Image
151           src="/images/front.jpg"
152           alt="Barangay Malinta Background"
153           fill
154           className="object-cover"
155           priority
156         />
157       </div>
158
159       {/* White Overlay with gradient - dark mode compatible */}
160       <div className="absolute inset-0 z-10 bg-gradient-to-r from-white/65 via-white/90 to-white/80 dark:from-black/95 dark:via-black/90 dark:to-black/80"></div>
161
162       {/* Content */}
163       <div className="container relative z-20 px-4 md:px-6 py-16 md:py-24">
164         <div className="grid grid-cols-1 md:grid-cols-14 lg:grid-cols-12 gap-8 items-center">
165           <div className="md:col-span-5">
166             <div className="flex items-center gap-4 mb-6">
167               <div className="relative h-20 w-20 overflow-hidden rounded-full border-4 border-white dark:border-gray-800 shadow-md">
168                 <Image
169                   src="/images/malinta_logo.jpg"
170                   alt="Barangay Malinta Logo"
171                   fill
172                   className="object-cover"
173                 />
174               </div>
175               <div className="bg-primary/10 rounded-full py-1.5 px-4">
176                 <span className="text-sm font-medium text-primary">
177                   Bayan ng Los Baños, Laguna
178                 </span>
179               </div>
180             </div>
181
182             <p className="text-xl md:text-2xl text-foreground/80 font-medium mb-4 max-w-2xl">
183               {t("home.hero.title")}
184             </p>
185             <h1 className="text-4xl md:text-5xl lg:text-6xl font-bold tracking-tight mb-4">
186               <span className="text-gradient dark:text-white">
187                 Barangay Malinta
188               </span>
189             </h1>
190
191             <p className="text-muted-foreground text-lg max-w-2xl mb-8">
192               {t("home.hero.subtitle")}
193             </p>
194
195             <div className="flex flex-col sm:flex-row gap-4">
196               <Button asChild size="lg" className="rounded-md shadow-sm px-8">
197                 <Link href="/certificates">
198                   {t("home.hero.getStarted")}
```

APPENDIX D

COMMUNICATION LETTER AND APPROVALS

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

March 31, 2025

Hon. Jesus H. De Una Jr.
Barangay Malinta, Los Baños, Laguna

REQUEST FOR AUTHORIZATION TO CONDUCT AN IN-DEPTH STUDY FOR THE BARANGAY MALINTA DASHBOARD SYSTEM PROJECT

Dear Mr. De Una Jr.,

Good day!

We are writing to formally request your approval and support to conduct an in-depth study for our website project titled *"Dashboard System Project."* Our team is dedicated to creating a system that enhances Barangay Malinta's management, aiming to improve its operational efficiency significantly. We believe this study will provide valuable insights that will contribute to the barangay's resource optimization and management.

This system aims to streamline operations, enhance transparency, and improve service delivery to our community. We assure you that all data collected throughout the study will be handled with the utmost confidentiality and will not be disclosed to any external parties. The insights obtained from this study will assist in improving overall barangay management.

We welcome any questions, suggestions, or recommendations you may have regarding this project. We are open to discussions to ensure that the study aligns with the company's objectives and expectations.

Thank you for your time and consideration. We appreciate your support in advancing this initiative and look forward to your positive response.

Sincerely,

ANGELA M. ANTE

NICEN HOWELL B. MENDOZA

KARYLLE R. VIDAL

HON. JESUS H. DE UNA JR.
Barangay Captain

RECEIVED BY:
RICHET E. TAKAHASHI
BRGY. SEC
03-31-2025

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

"SMART DASHBOARD: A WEB-BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED DATA PROCESSING"

ANGELA ANTE, NICEN HOWELL MENDOZA, and KARYLLE VIDAL have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

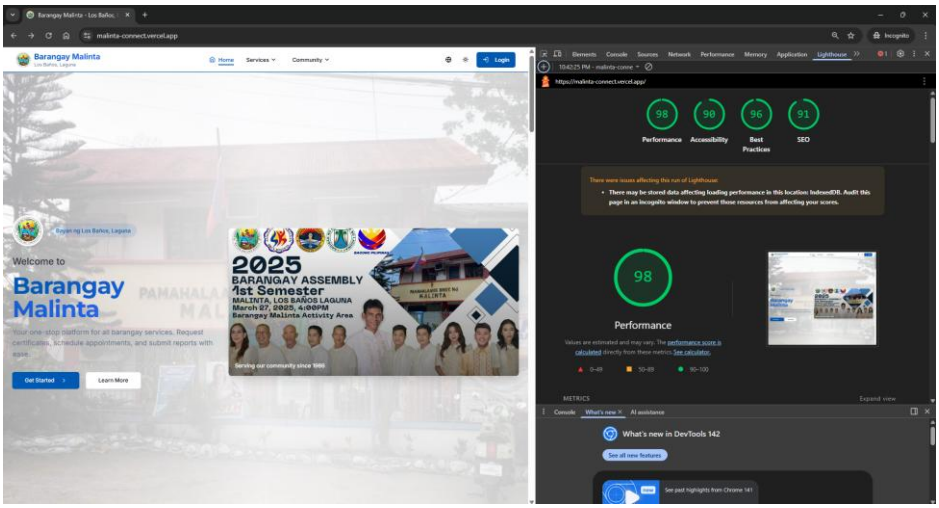
JEFFERSON LERIOS
Capstone Project Adviser

APPENDIX E

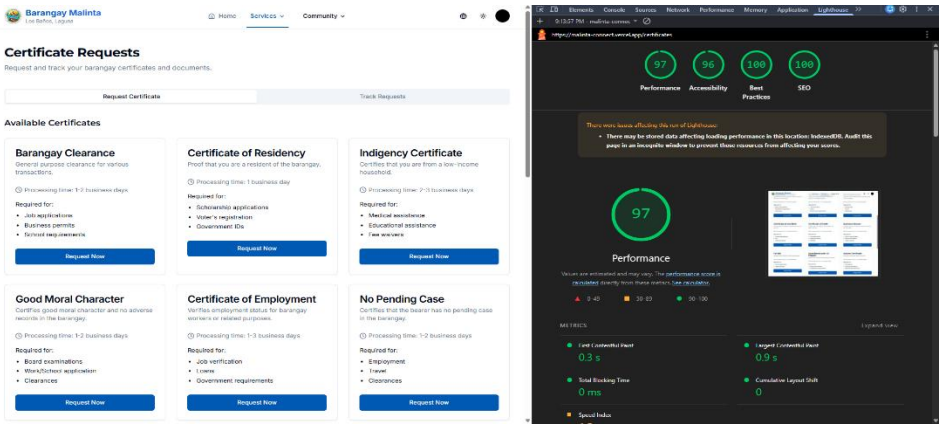
TESTING RESULTS

COMPATIBILITY TESTING USING LIGHTHOUSE FOR GOOGLE CHROME

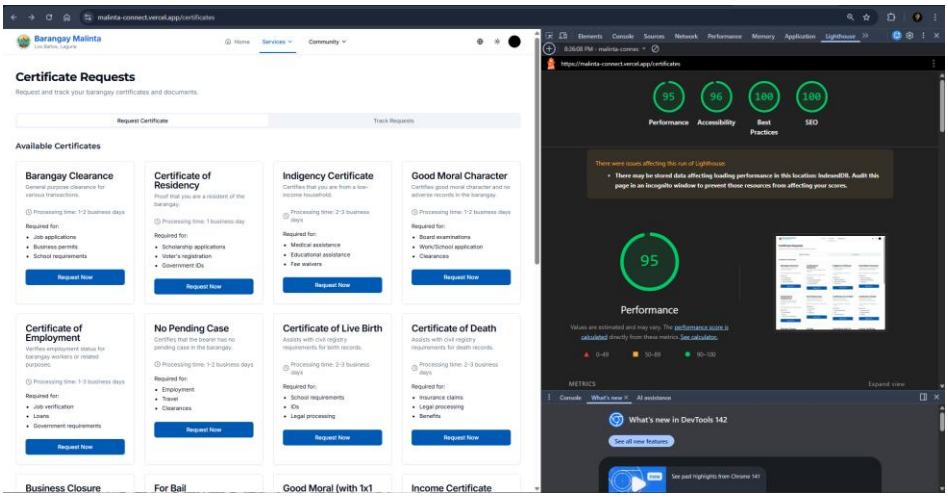
Log In Page



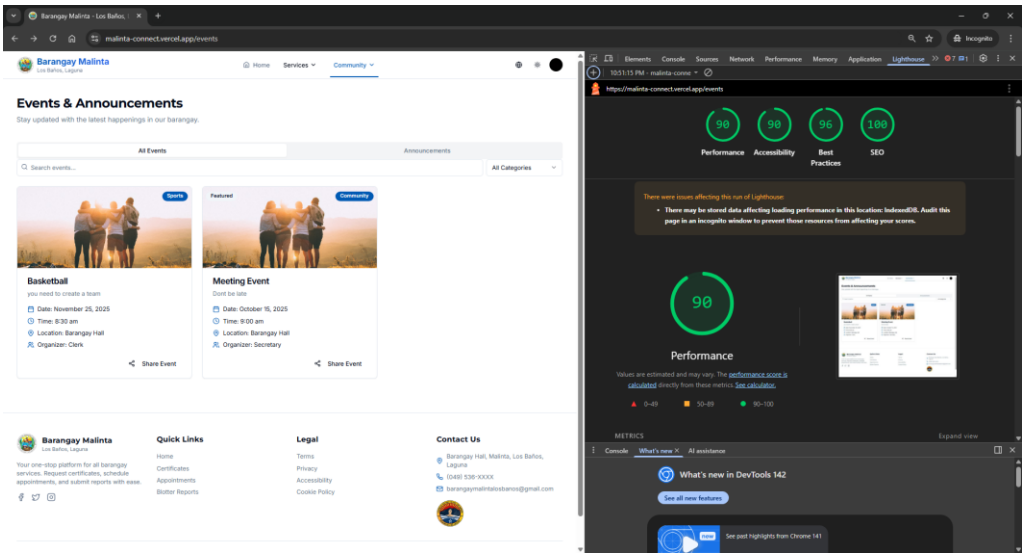
Certificate Request



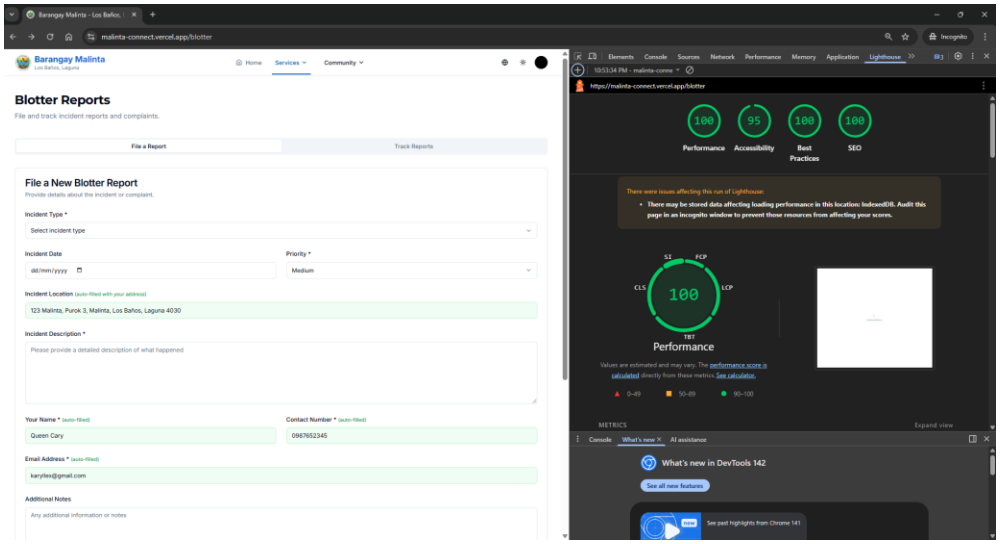
Users Track Request



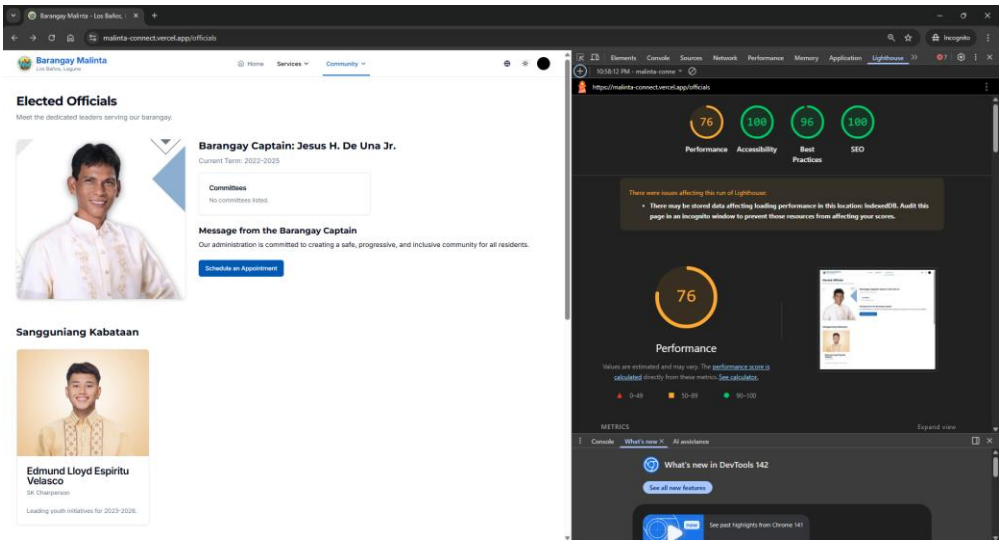
Events & Announcements



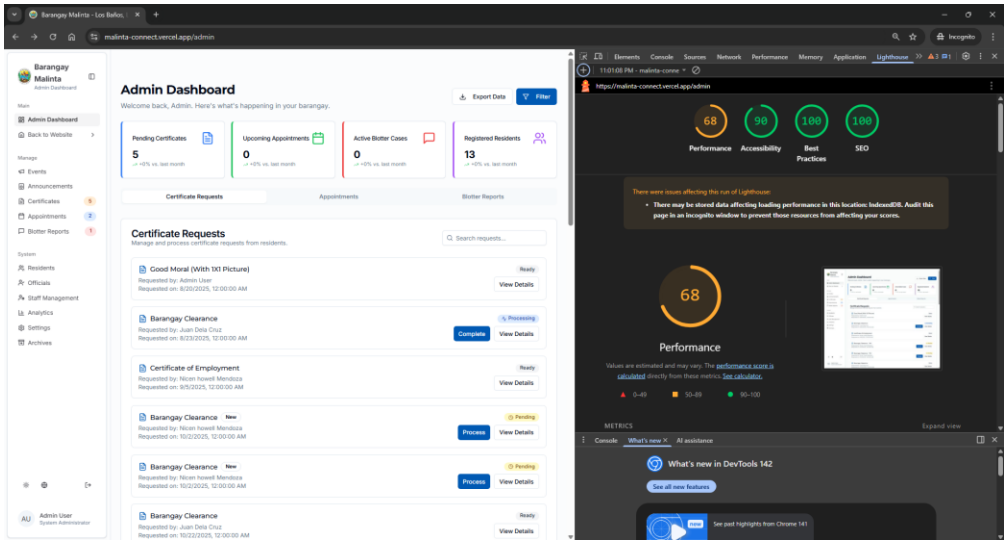
Blotter Reports



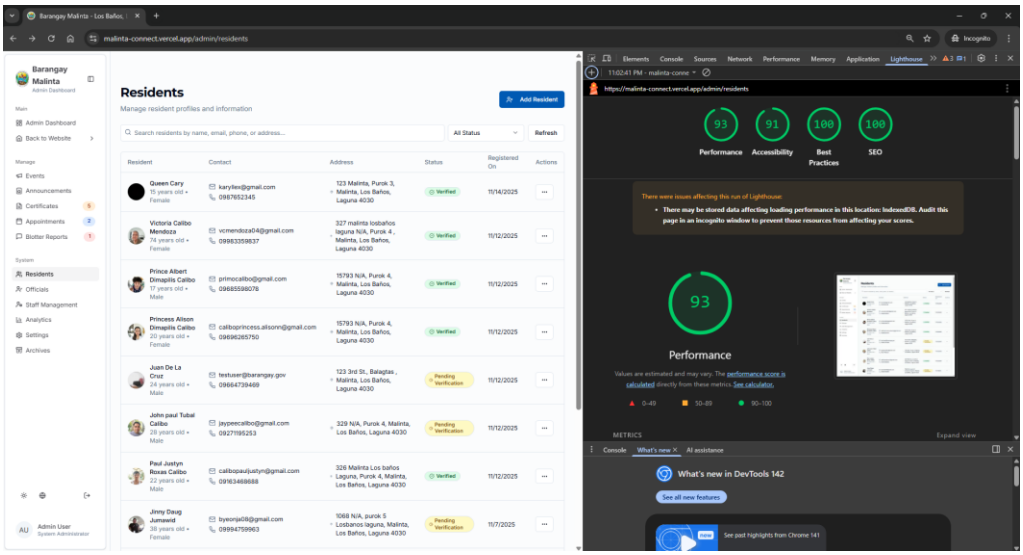
Elected Officials



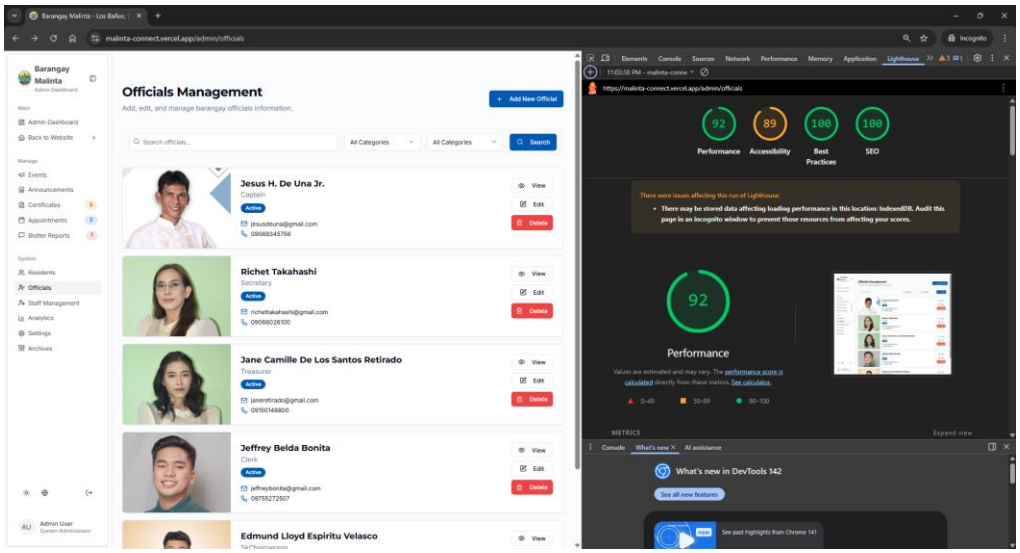
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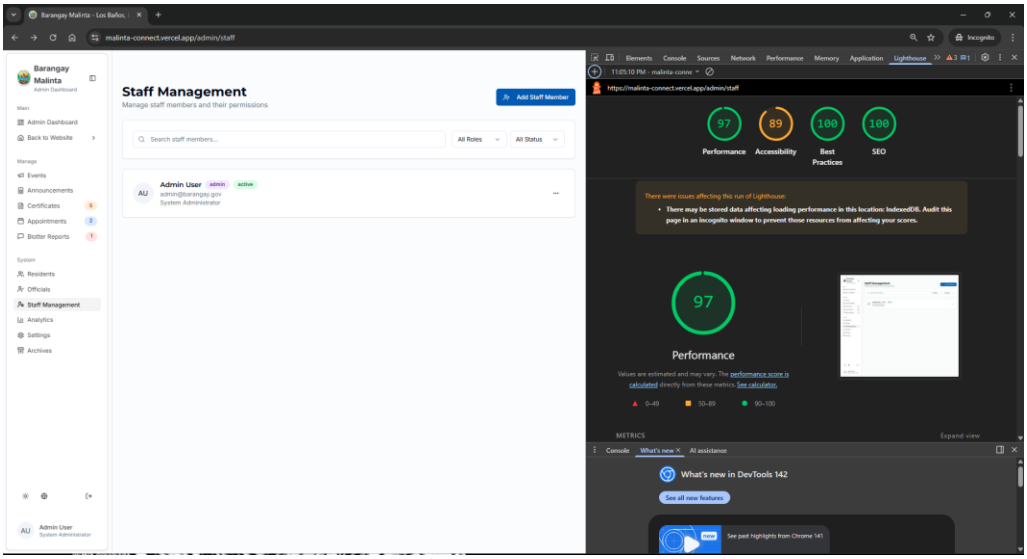
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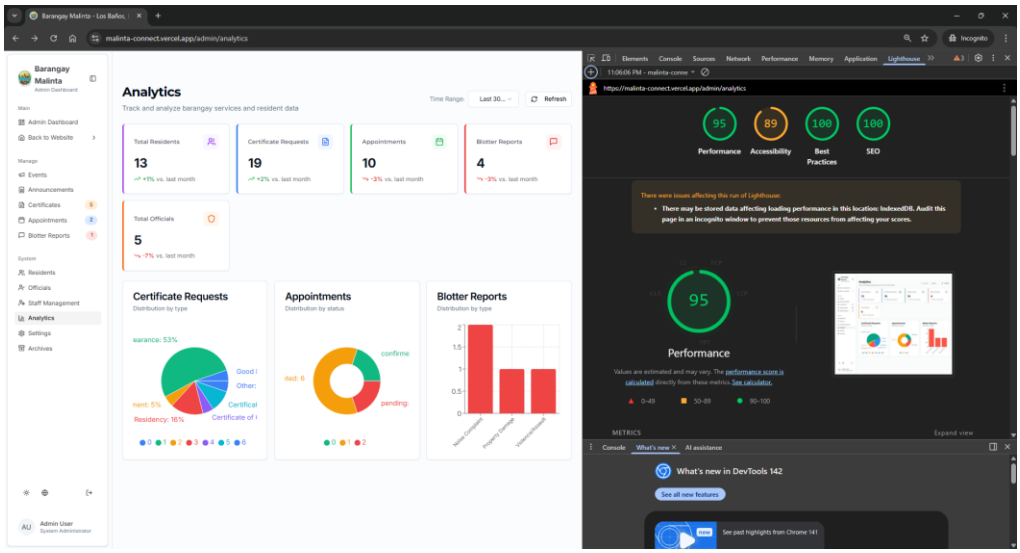
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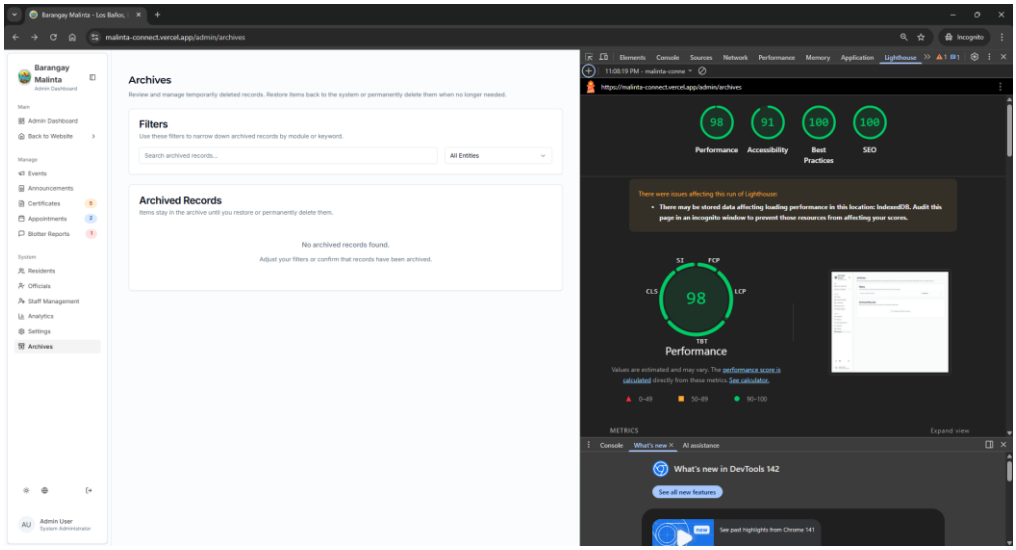
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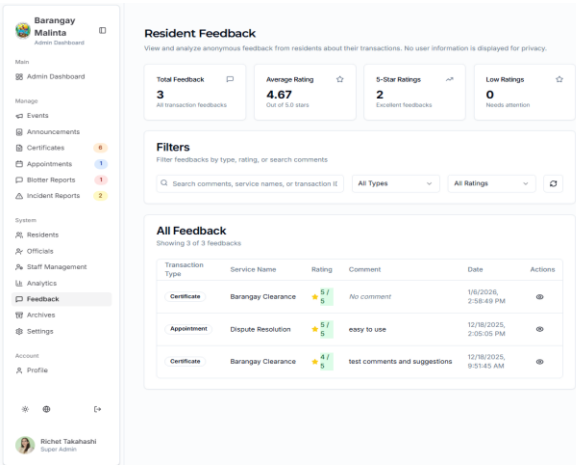
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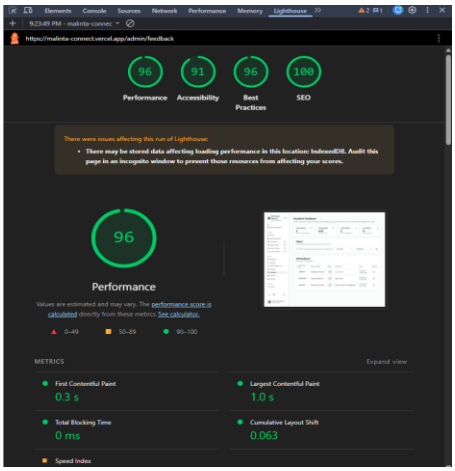


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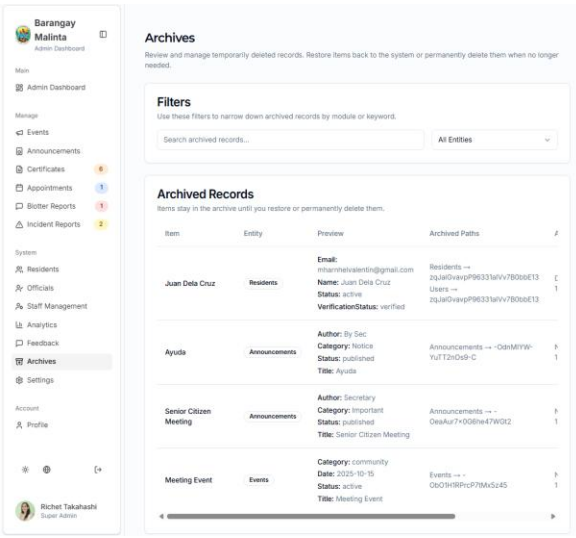


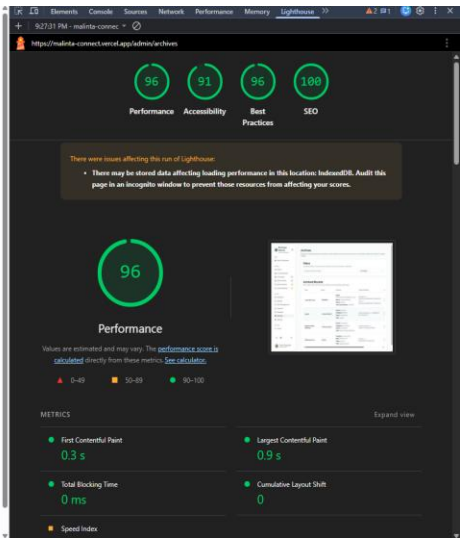
Feedback





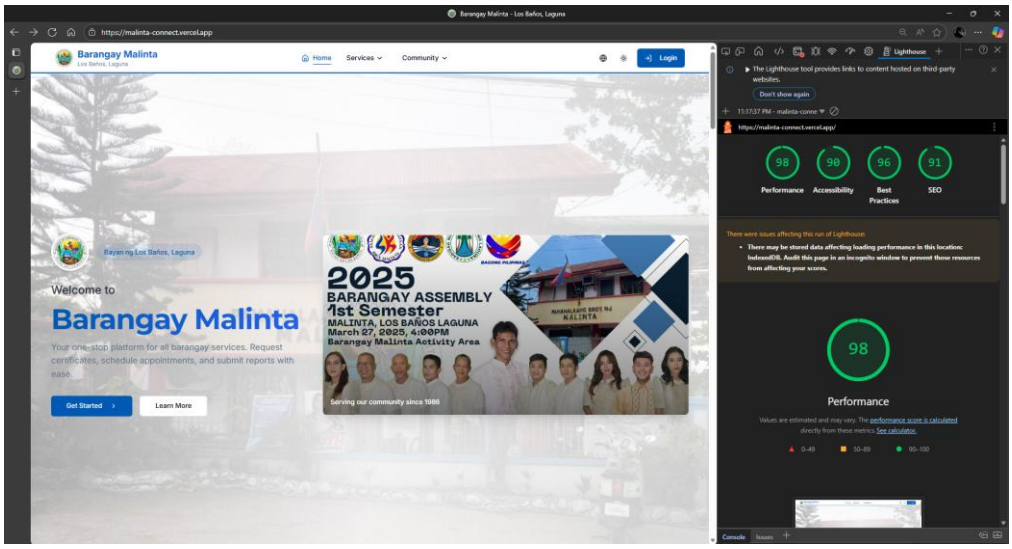
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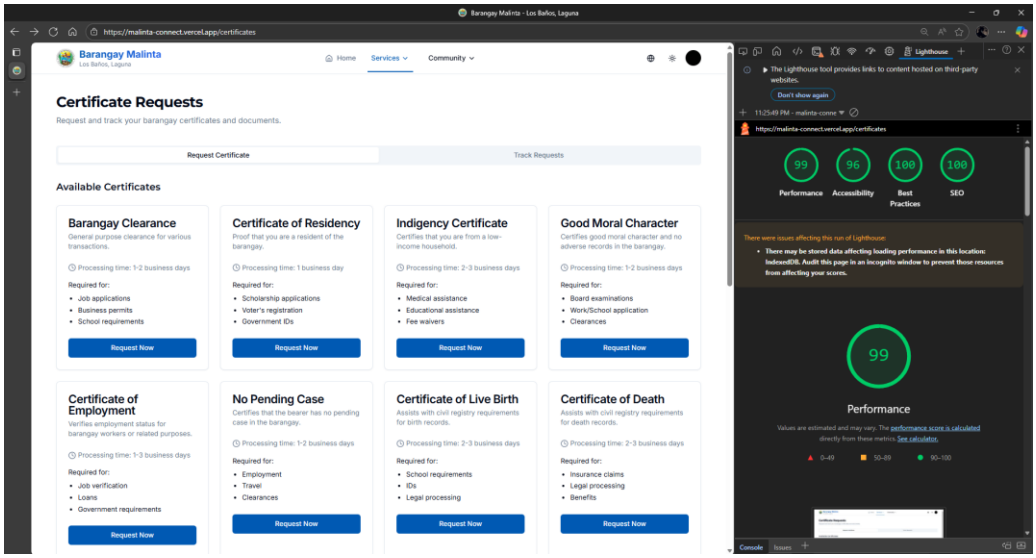


COMPATIBILITY TESTING USING LIGHTHOUSE FOR MICROSOFT EDGE

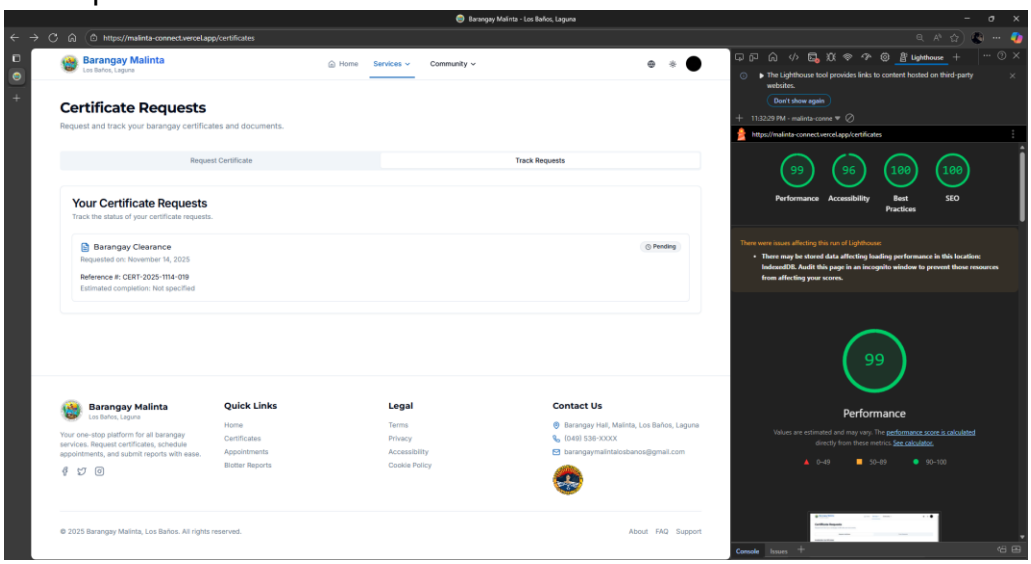
Log In Page



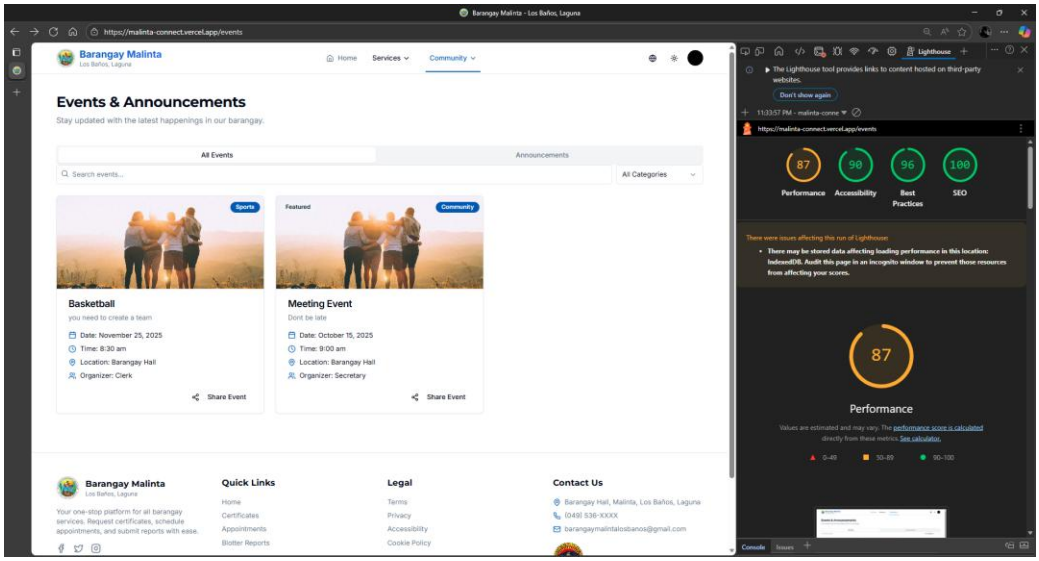
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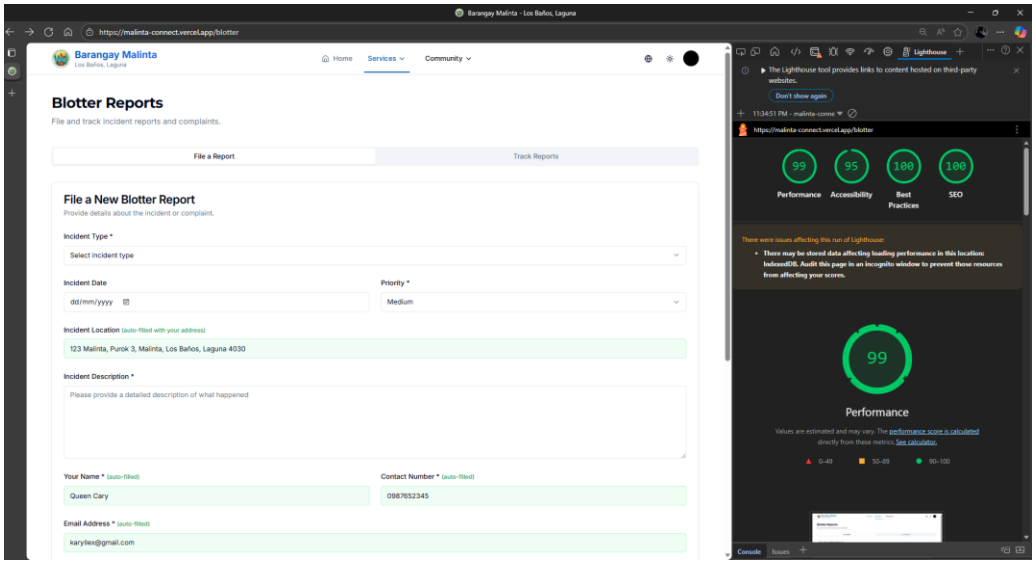
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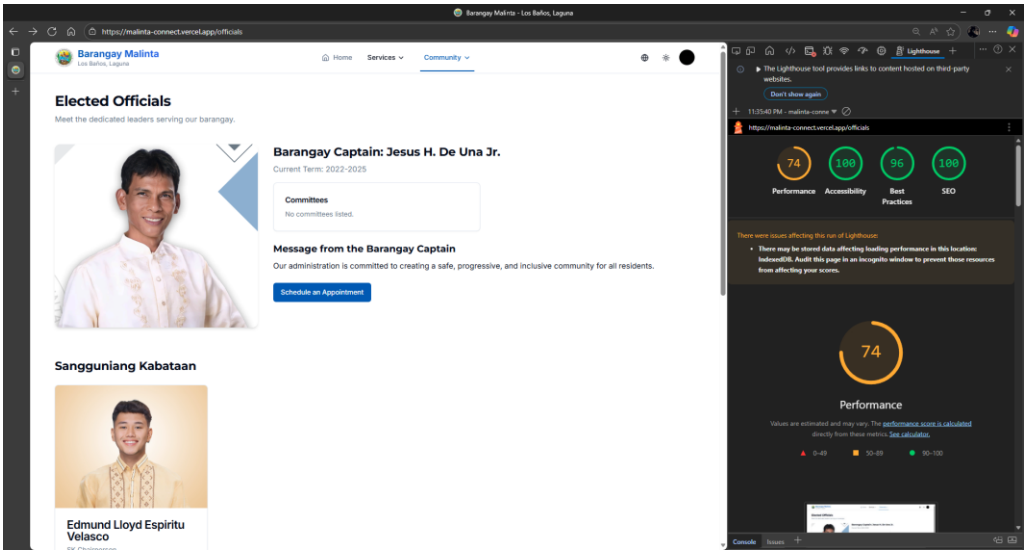
Events & Announcements



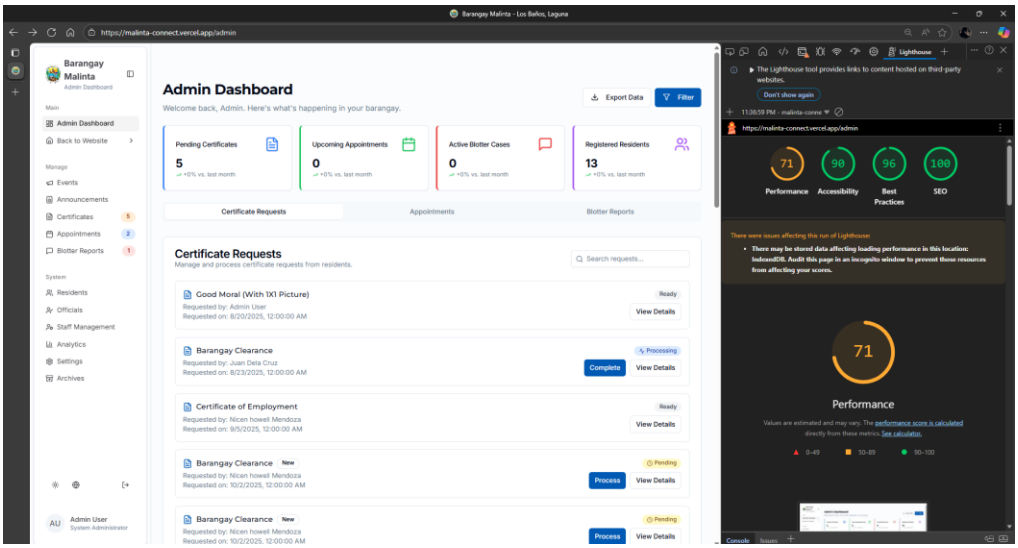
Blotter Reports



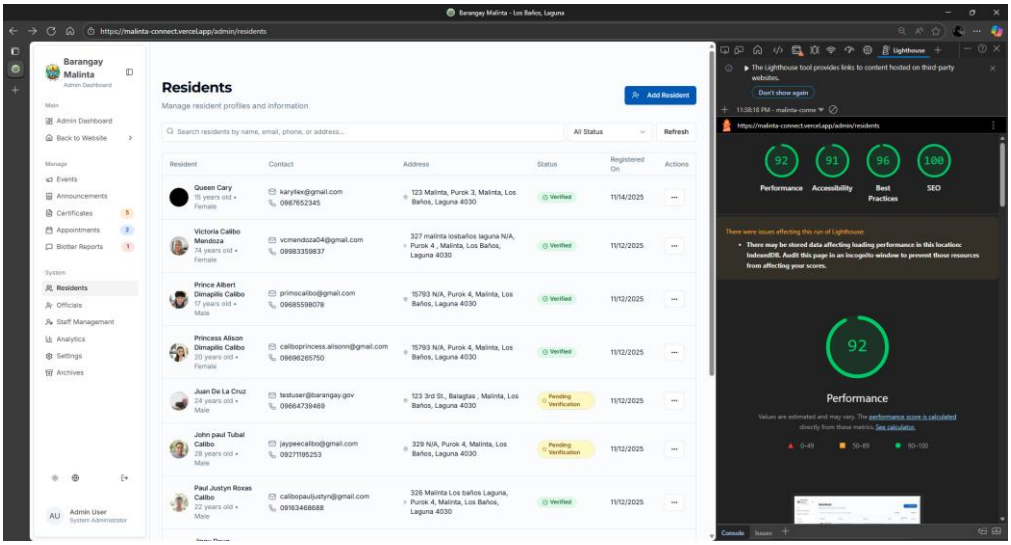
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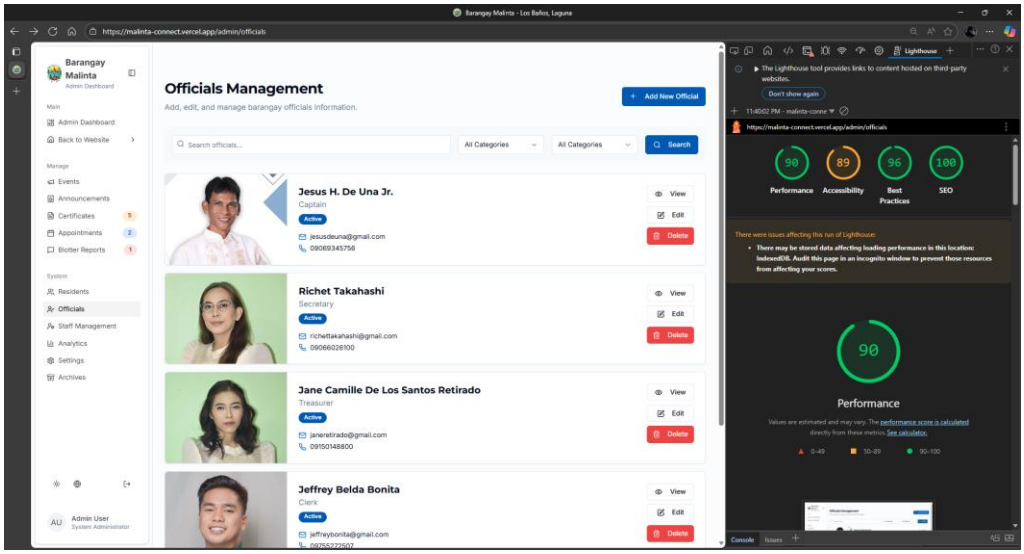
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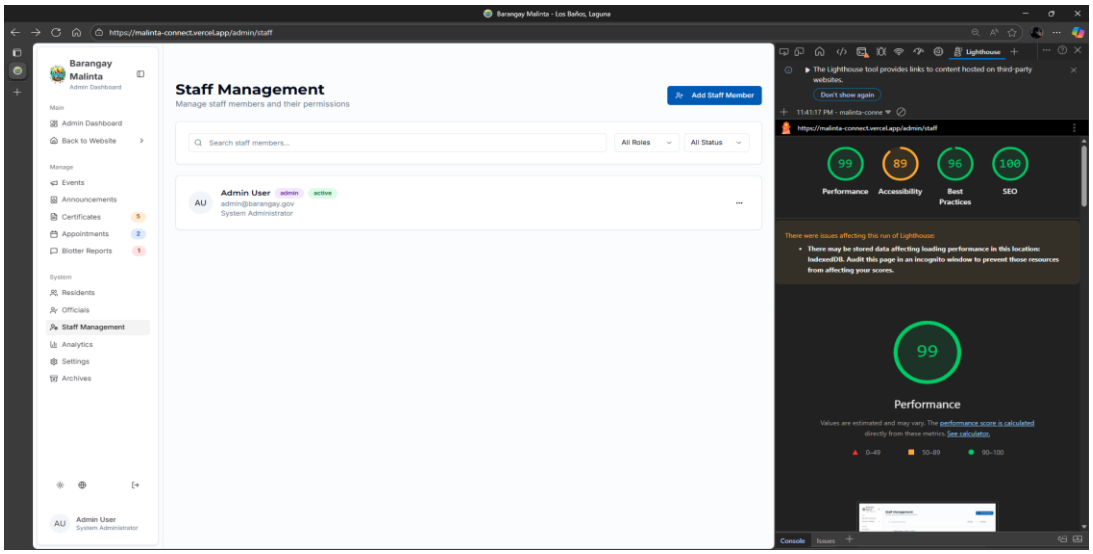
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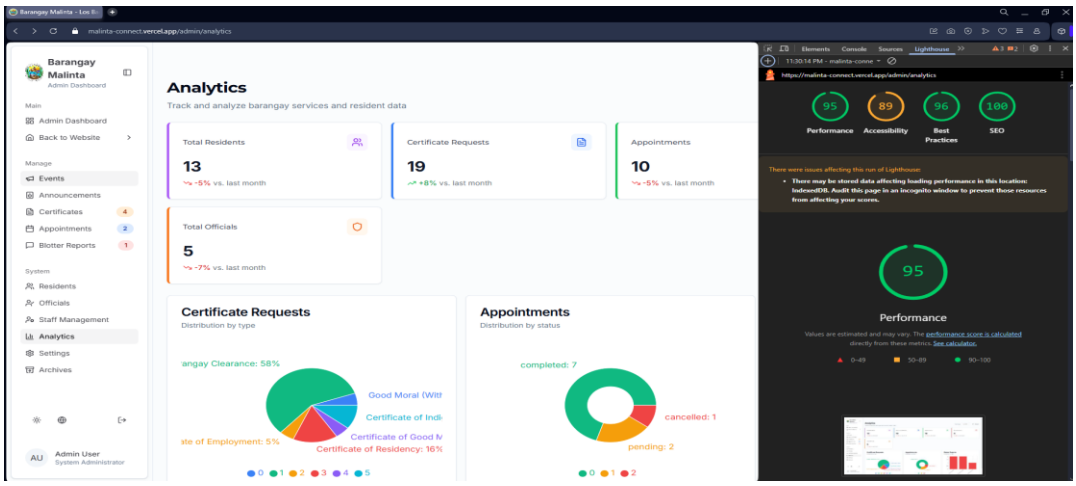
Officials Management



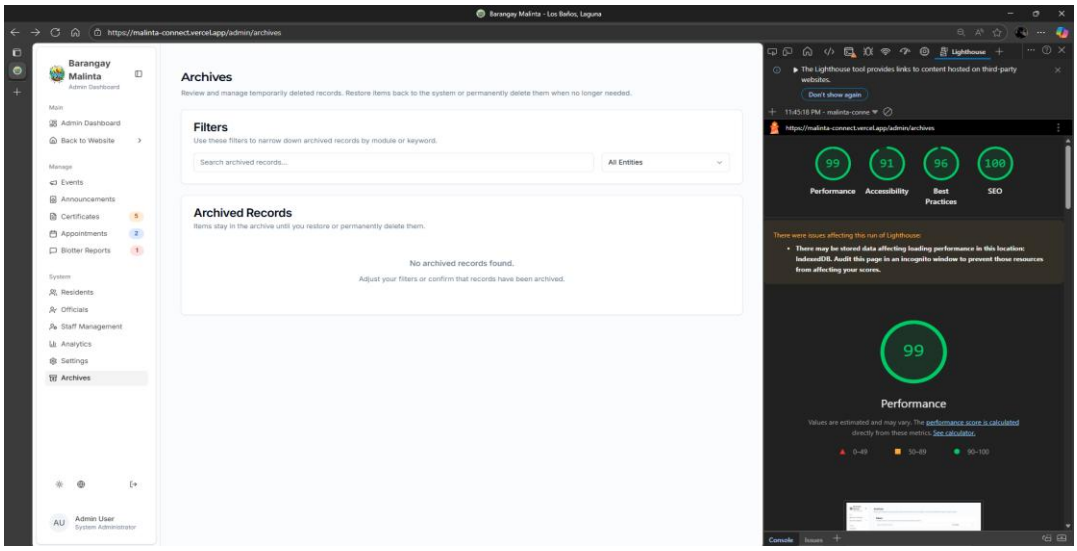
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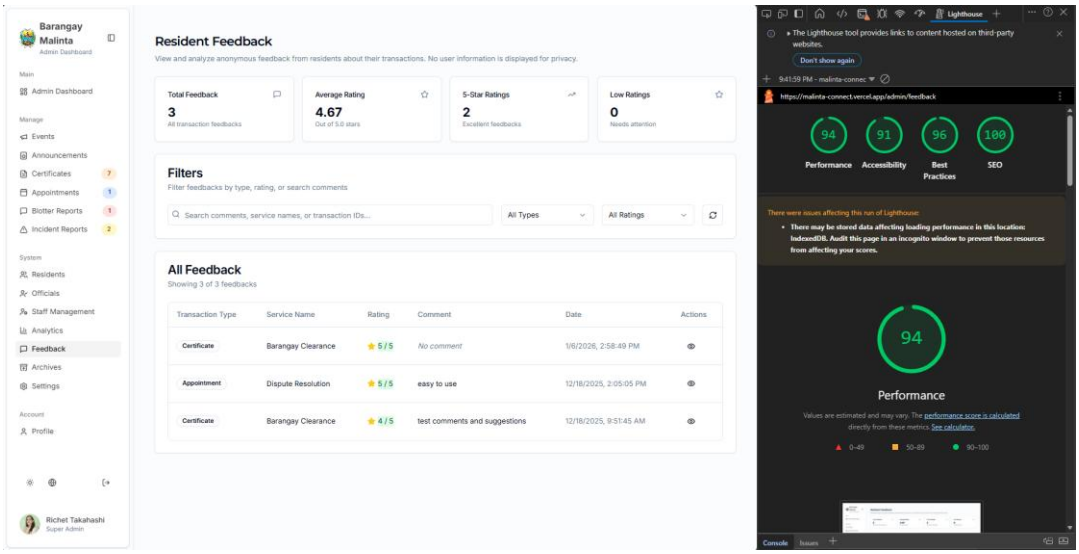
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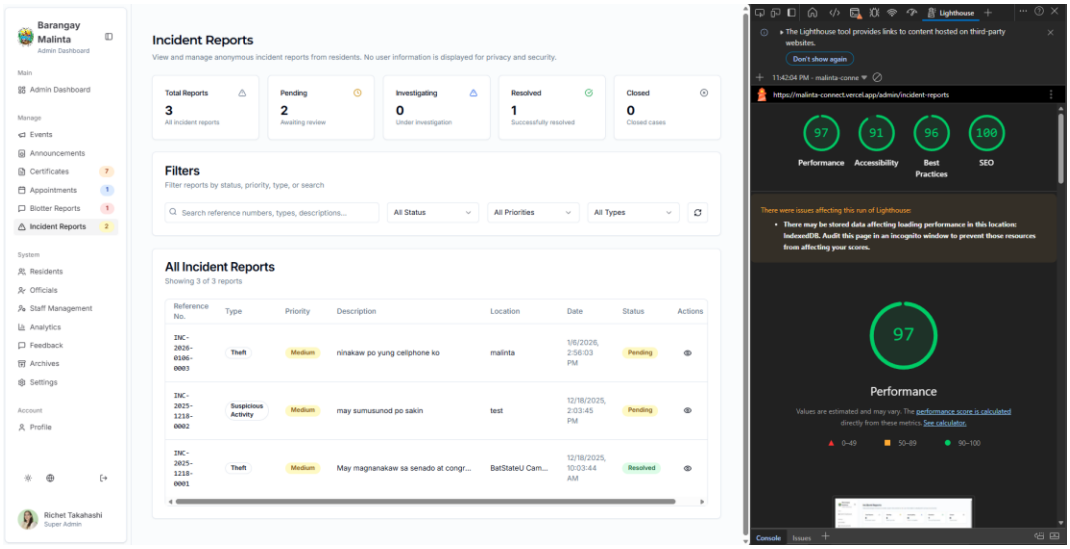
Archives



Feedback

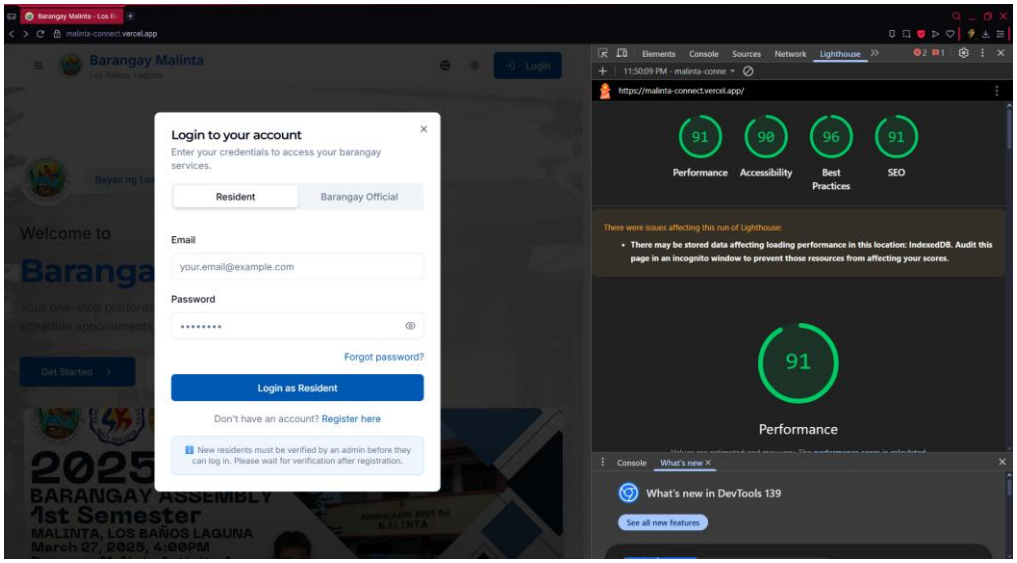


Incident Reports

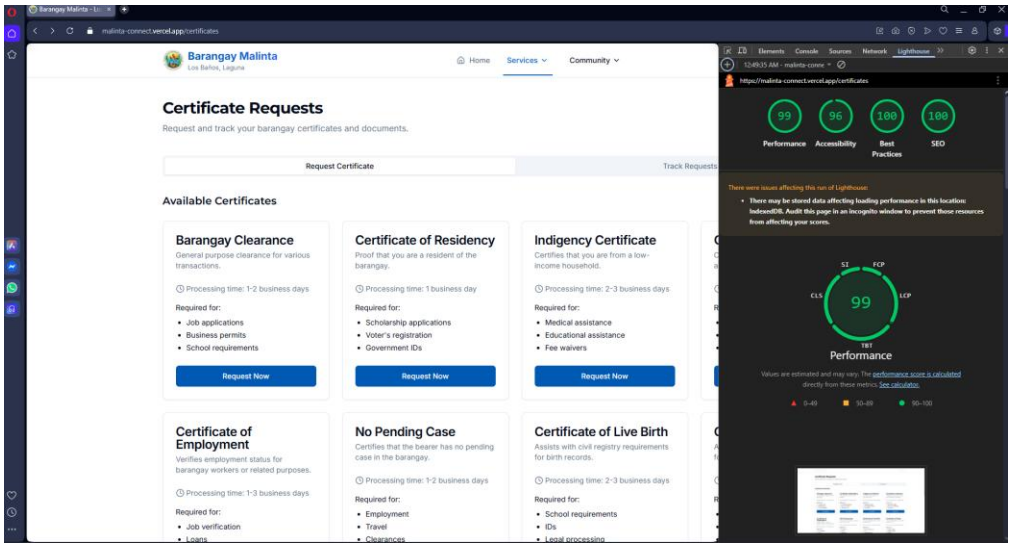


COMPATIBILITY TESTING USING LIGHTHOUSE FOR OPERA BROWSER

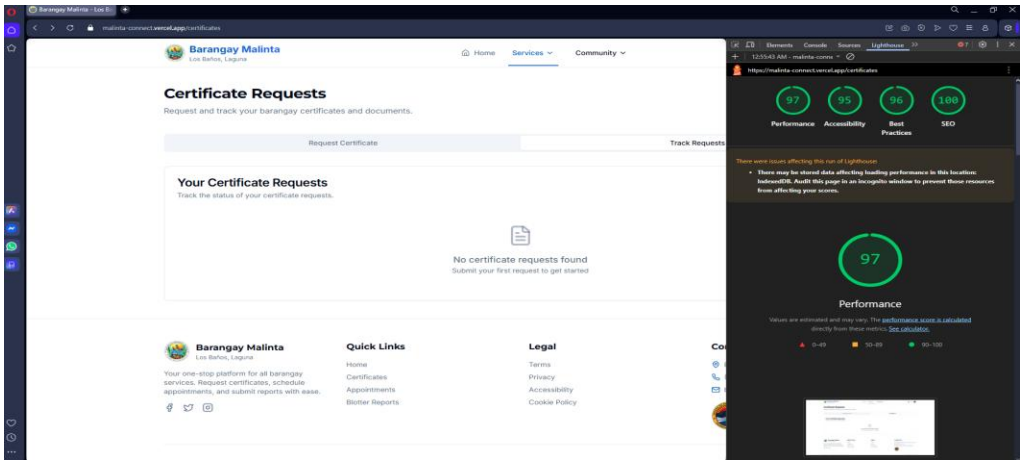
Log In Page



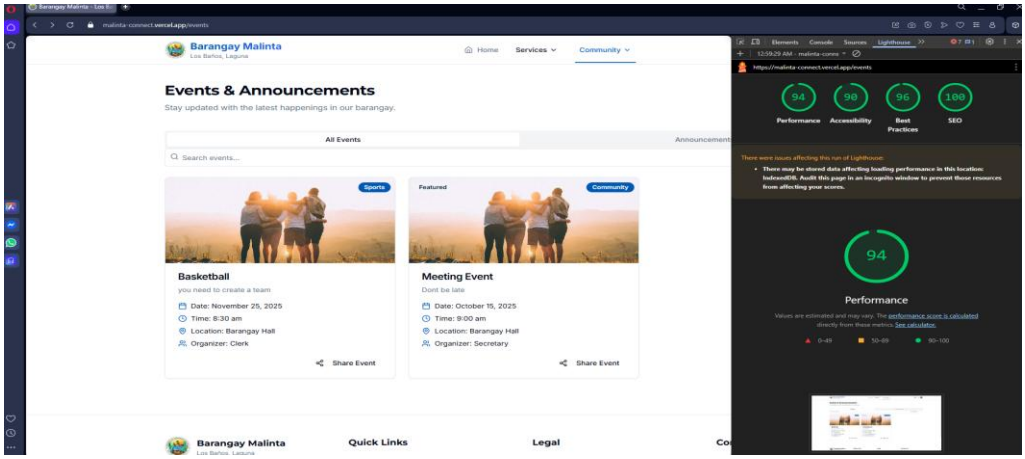
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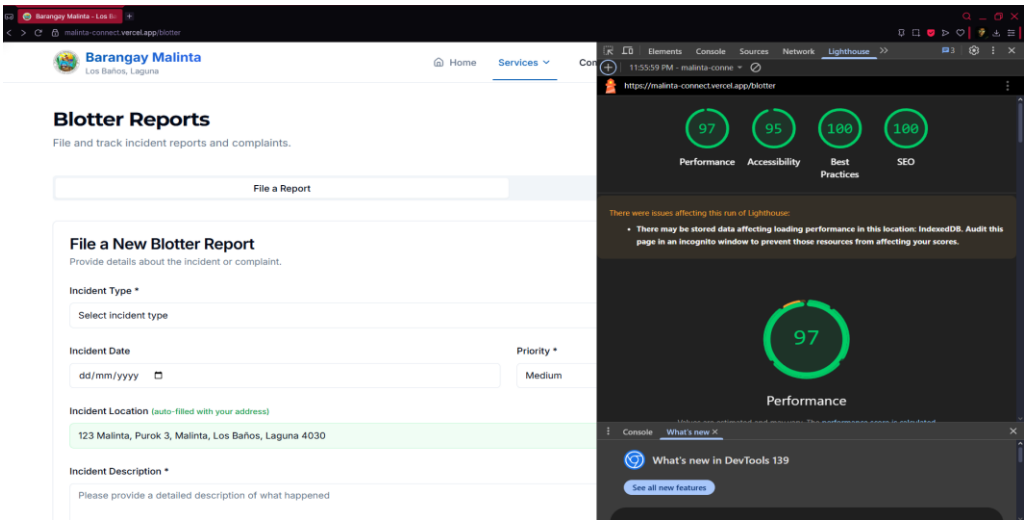
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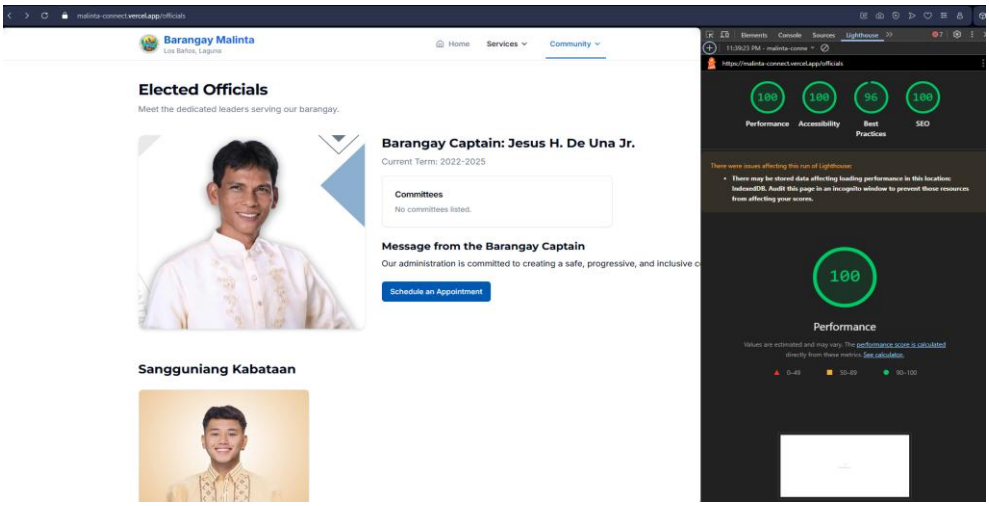
Events & Announcements



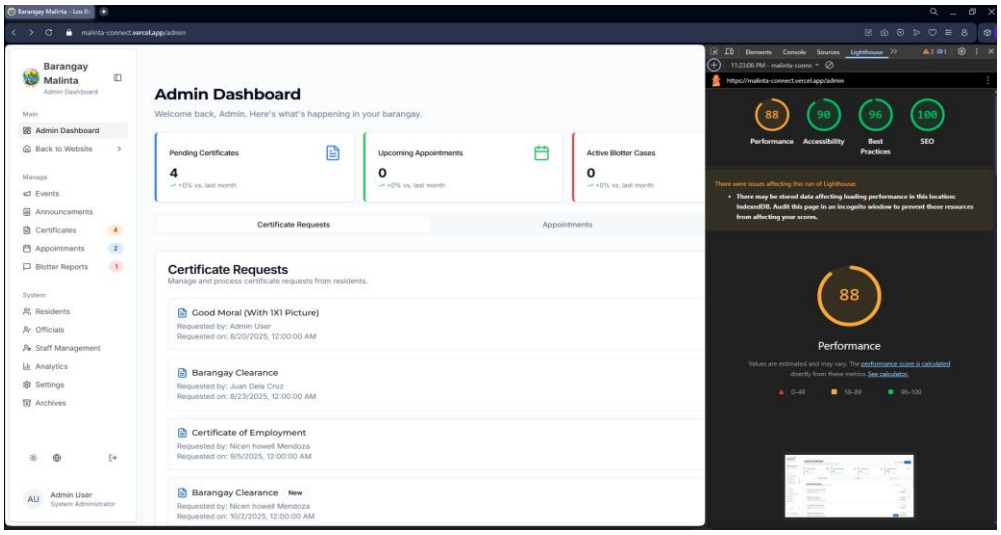
Blotter Reports



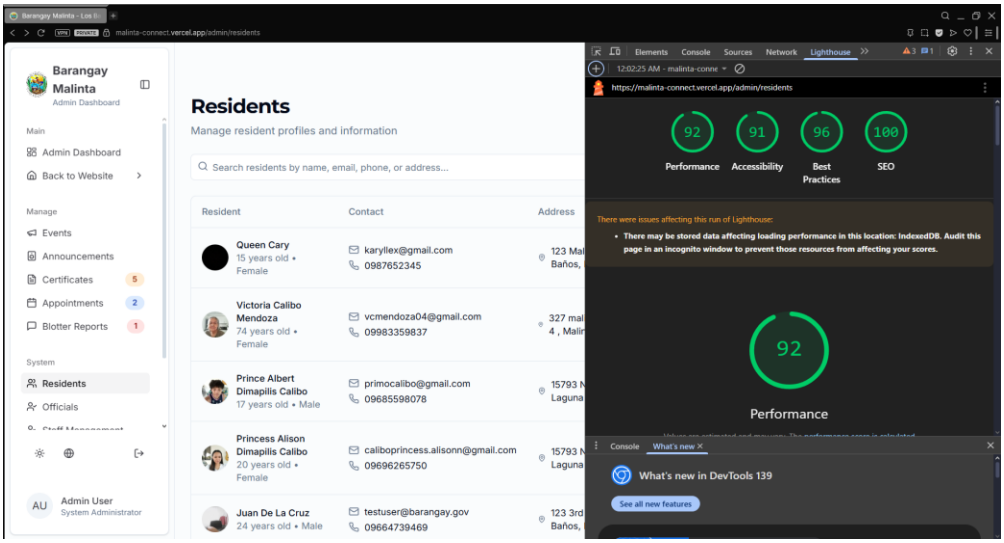
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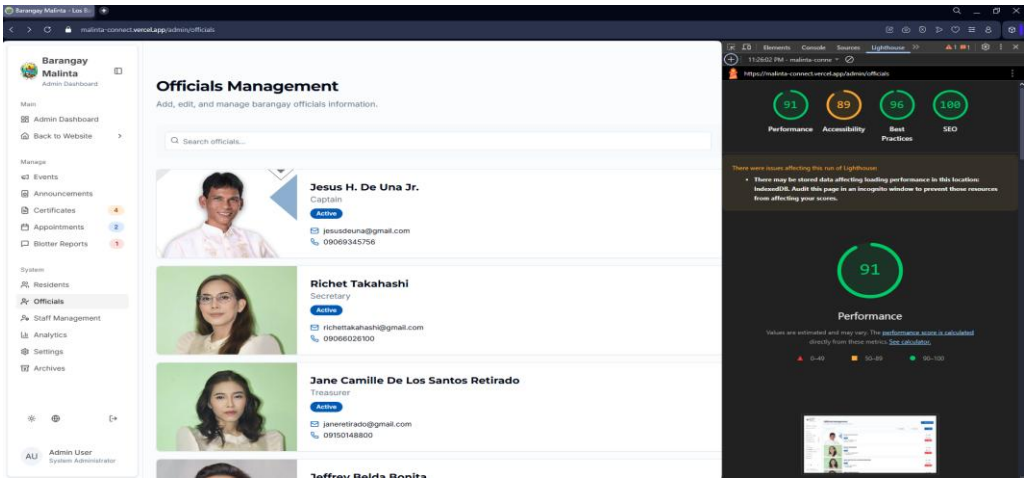
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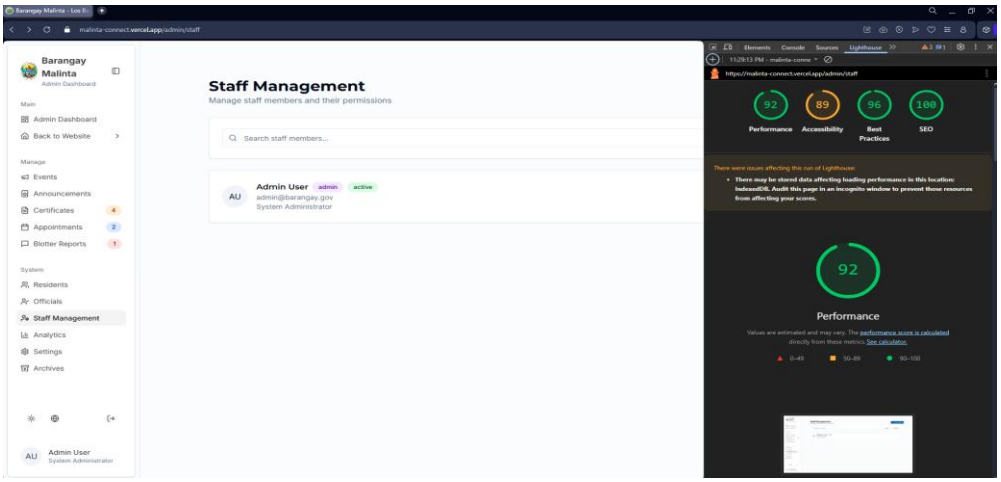
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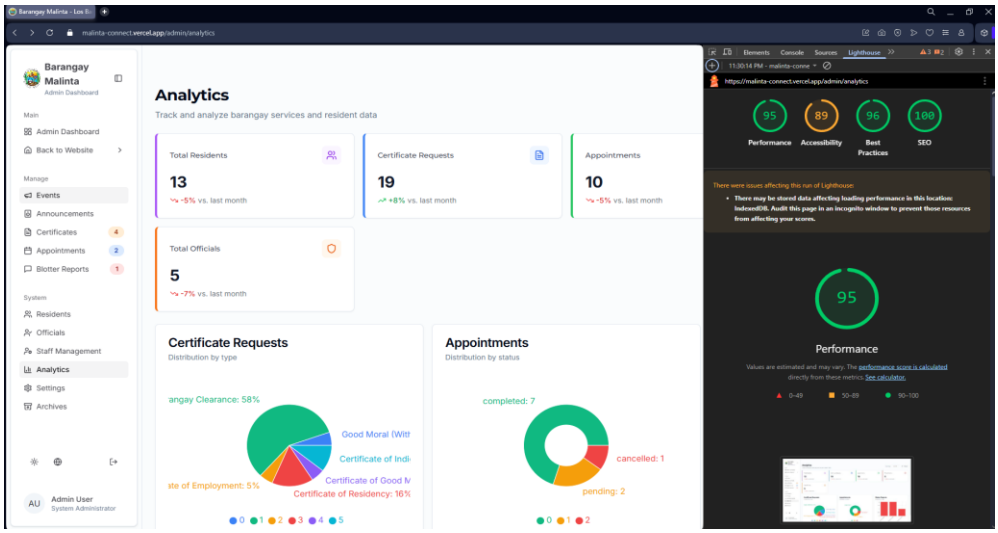
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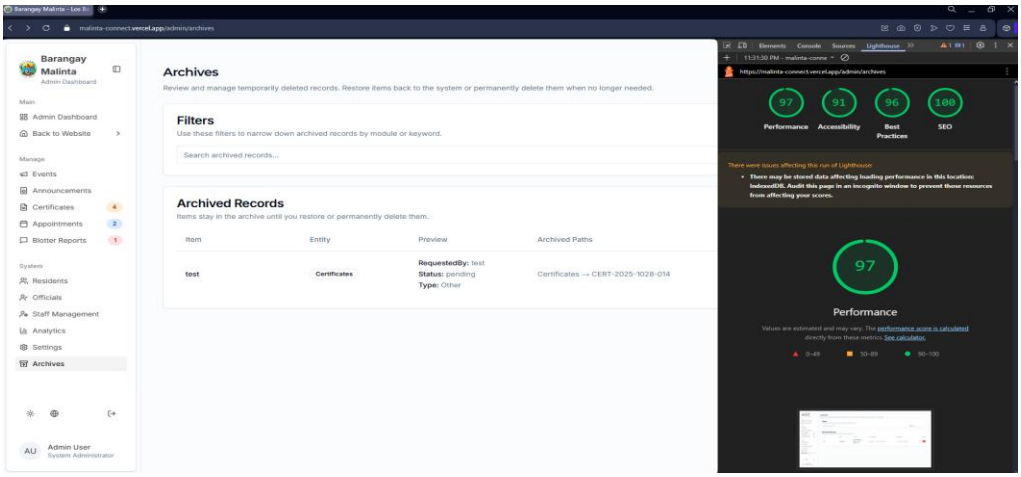
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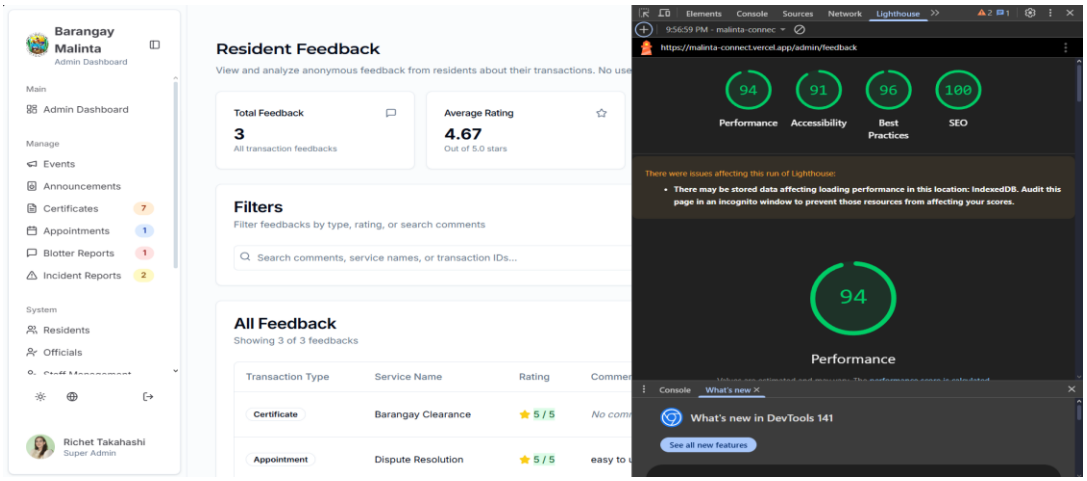
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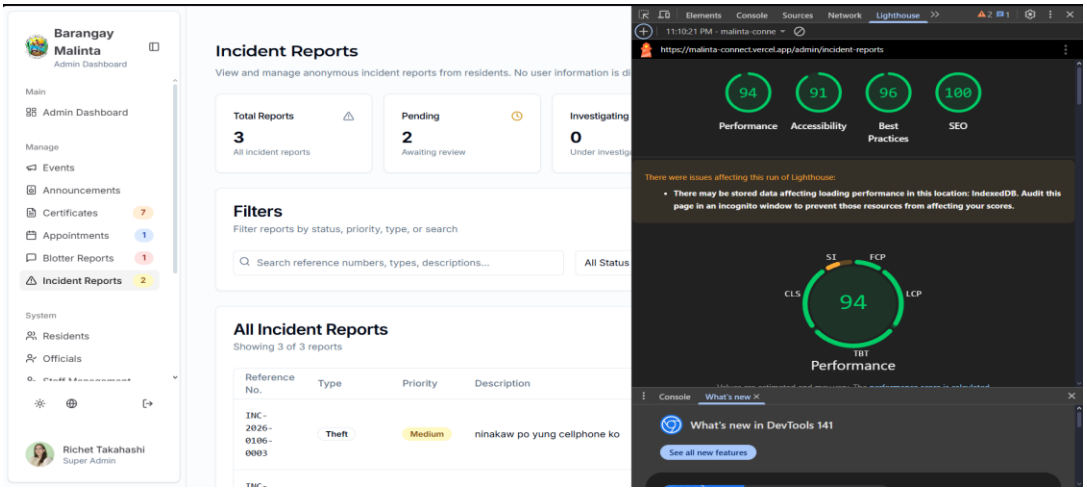
Archives



Feedback



Incident Reports



APPENDIX F

PUBLISHABLE PAPER

SMART DASHBOARD: A WEB-BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED PROCESSING

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ARTICLE INFORMATION

Keywords:
Web-based Platform
Centralized Digital Platform
Realtime Analytics

ABSTRACT

The Barangay Malinta Smart Dashboard is a web-based management platform developed to address the recurring problems in manual barangay processes such as slow certificate requests, disorganized records, and limited access to information. The project was created to modernize barangay operations and improve service delivery for both residents and officials by providing a centralized and accessible digital system. The platform integrates essential modules including resident registration, certificate request processing, appointment scheduling, event and announcement posting, blotter reporting, archives management, and real-time analytics for data-driven decision-making. The system is built with modern tools like Next.js, React, TypeScript, and Firebase, while Cloudinary handles image storage, Nodemailer manages email notifications, and PDF utilities generate official documents, ensuring a fast, reliable, and easily maintainable platform. A user-centered design approach guided development to ensure a simple and functional interface suited for community use. The project underwent system testing, compatibility testing, and user acceptance evaluation using the Technology Acceptance Model (TAM), all of which achieved positive and functional results. The ISO/IEC 25010 evaluation conducted by 2 IT experts yielded an overall compatibility rating of 5.0, indicating that the system strongly meets quality standards in reliability, usability, and performance. Overall, findings show that the Barangay Malinta Smart Dashboard effectively streamlines transactions, enhances transparency, improves communication within the community, and stands as a reliable and efficient tool for digital governance

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1. INTRODUCTION

Across the Philippines, thousands of barangays serve as the foundation of local governance, directly assisting millions of citizens with everyday services and community concerns. From handling documents to managing public requests, these local units are expected to operate efficiently. Yet many still depend on manual, paper-based systems. This reliance on traditional methods often causes slow processing, disorganized records, and repeated on-site visits, creating challenges for both barangay staff and the people they serve. In light of this, the Philippine

government is now pushing for digital transformation at the barangay level. There has been strong collaboration between public institutions and private organizations, resulting in systems like BARS Philippines, which helps automate resident information management and TECHSTATIC's Barangay eGovernance guided the development process to ensure functionality, ease of use, and accessibility. Modern web technologies were utilized to automate tasks, reduce paperwork, and minimize human error, while the analytics module provided tools for generating visual insights and supporting data-driven decision-making. System testing showed that the Smart Dashboard effectively streamlines transactions, enhances transparency, and improves communication between the barangay and its residents. In conclusion, the Barangay Malinta Smart Dashboard demonstrates a significant improvement in local governance by offering a reliable, efficient, and data-driven platform for managing essential barangay services.

OBJECTIVES OF THE STUDY

The specific objectives of this study are

1. To develop a web-based Smart Dashboard platform that promotes citizen engagement and facilitates data-driven digital governance through the use of automated data processing Specifically, it aims to:

2. To design and develop an interactive web-based Smart dashboard system with key features such as an announcements/events board, request submission forms, and automated notifications, tailored to enhance community engagement and governance transparency.

3. To provide a platform that allows citizens to efficiently, reliably request and track various barangay certifications online. It provides a range of certificate services, including Barangay Clearance, Certificate of Residency, Indigency Certificate, Good Moral Character, Certificate of Death, Business Closure, For Bail, Good Moral (with 1x1 Picture), and Income Certificate. Non-Residence Certificate, No Income Certificate, and other certificates. The system enables residents to submit document requests for job, education, legal, and other official purposes without the need for manual transactions, resulting in a faster, more transparent, and user-friendly experience.

CONCEPTUAL FRAMEWORK

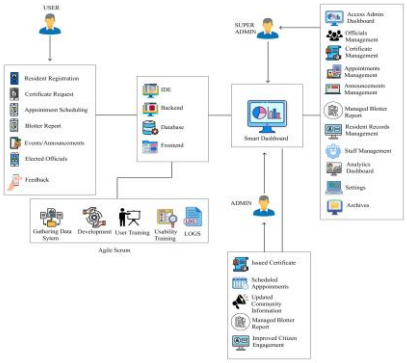


Figure 1. Conceptual Framework

Figure 1 presents the conceptual framework of the barangay information system, showing the interaction between users, administrators, and system processes. Users can register, request certificates, schedule appointments, submit blotter reports, and view events or officials. At the center, the framework highlights the Agile Scrum development process, including data gathering, system development, user training, testing, and continuous improvement.

METHODOLOGY

Project Design

Figure 2 below shows that the system architecture is composed of various modules working together to ensure functionality and efficiency. These include the User Authentication Module, which allows residents and barangay officials to securely access the system, and the Request Management Module, where residents can submit and track requests. The Business Permit Renewal Tracking Module helps barangay officials monitor business permit applications, while the Announcements Module enables officials to post updates for residents. Additionally, the Barangay Calendar Module helps schedule and manage events and activities.

The system architecture was designed to allow seamless navigation and interaction among

the different components, ensuring ease of use for all stakeholders.

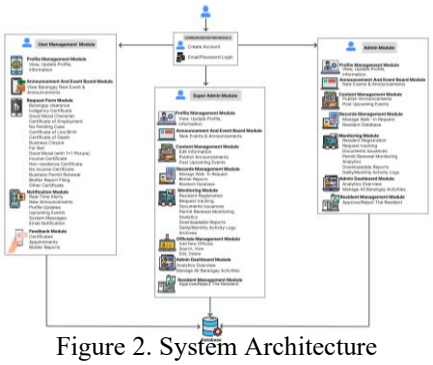


Figure 2. System Architecture

Project Development

Employing the Agile Software Development Cycle led to effective design and development of the smart dashboard web-based system built as a Progressive Web App (PWA) to enhance user experience. The Agile Scrum Methodology improved the smart dashboard by encouraging continuous improvement and adaptability to changing requirements. It emphasized active user interaction, regular feedback, simplified functionality, and ongoing interface refinement. The system ensured compliance with modern regulations through the use of the most recent valid instructions via repeated sprint cycles. Agile Scrum helped a responsive system by guaranteeing consistent value delivery and breaking down complex legal requirements into manageable tasks.

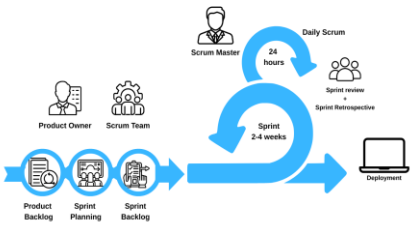


Figure 3. Agile Scrum Diagram

Sampling Design

Purposive sampling was employed in this study, wherein participants were selected based on their relevance to the objectives of the research.

A total of 50 respondents were involved in the study, including barangay personnel and residents, as well as two (2) IT experts with knowledge and experience in web-based systems. The barangay participants were selected based on their familiarity with the community's processes and services, while the IT experts were chosen for their expertise in evaluating system usability and technical functionality.

Population of the Study

The population of this study consisted of barangay staff, residents, and IT professionals who participated in the evaluation and testing of the Smart Dashboard system. These individuals represented the primary users of the system.

Barangay staff were chosen to assess the system using the Technology Acceptance Model (TAM) questionnaire to evaluate user perception regarding ease of use and usefulness. Their experiences were crucial in determining the system's effectiveness.

Data Collection Instrument

To collect data, the study used structured questionnaires and semi-structured interviews. In-depth interviews were conducted to gather detailed opinions on the system's design, accessibility, security, and data accuracy. A Likert Scale ranging from 1 ("Strongly Disagree") to 4 ("Strongly Agree") was used to measure user satisfaction. This dual-feedback strategy captured overall user experience and insights for system improvement.

Table 1. Likert Scale for User Feedback

RATING	RANGE	INTERPRETATION
4	3.40 – 4.19	Strongly Agree
3	2.60 – 3.39	Agree
2	1.80 – 2.59	Disagree
1	1.00 – 1.79	Strongly Disagree

Statistical Treatment

The IT Expert Evaluation was conducted to assess the technical quality of the Smart Dashboard system based on ISO 25010

software quality standards. A total of IT experts participated in the study and provided their feedback through structured questionnaires.

The IT experts rated each system feature using a 5-point scale, as shown below:

Table 2. Interpretation of Rating Scale for Evaluating System

Rating	Interpretation
5	Excellent
4	Good
3	Fair
2	Needs Improvement
1	Poor

RESULTS AND DISCUSSION

This chapter presents the results and discussions of the Smart Dashboard Web-Based platform, highlighting its key features, interfaces, and functionalities. The results emphasize the performance of various modules such as certificates management, appointment scheduling, feedback handling, analytics, and administrative tools showing how the platform enhances efficiency, transparency, and accessibility in barangay operations.

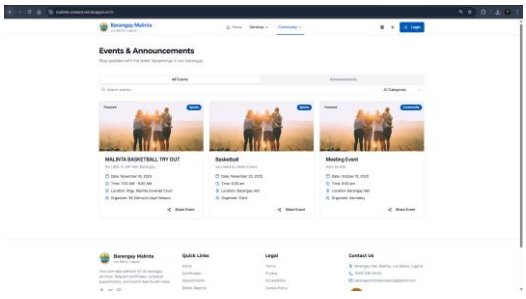


Figure 3. Events & Announcements

In this figure the events & announcements serve as the interface to keep the residents updated on the barangay activities. Including community meetings, health programs, and other local events. This platform not only spreads important information but also encourages resident participation and strengthens a sense of community engagement within the barangay.

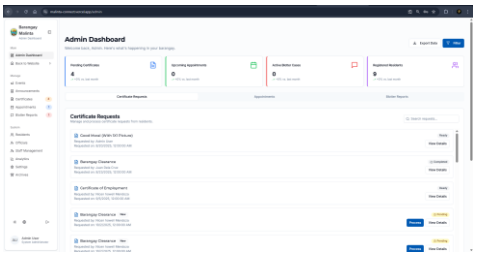


Figure 4. Administrator Dashboard

Figure 4 presents the Administrator Dashboard, which supports the Smart Dashboard's goal of improving data management, monitoring, and decision-making in the barangay. It provides administrators with a centralized view of real-time analytics, service summaries, certificate requests, user activities, and appointment updates. This organized interface enables efficient monitoring of system operations, quicker response to requests, and data-driven decision-making, ultimately enhancing operational efficiency and service delivery.



Figure 5. Analytics Overview Administrator

Figure 5 presents the Analytics Overview for administrators, showing visual summaries of key barangay operations such as certificate requests, appointments, and system activities. It provides real-time insights that help identify trends, monitor community needs, and evaluate service effectiveness. This feature supports informed decision-making, efficient resource allocation, and improved transparency and performance.

Table 13. TAM User Acceptance Evaluation

ITEMS	MEAN	SD	INTERPRETATION
Perceived Usefulness	3.49	0.26	Strongly Agree
Perceived ease of use	3.56	0.25	Strongly Agree
Attitude Toward Using the System	3.54	0.27	Strongly Agree
Behavior Intention to use the system	3.59	0.32	Strongly Agree

Table 14. ISO 2510 User Acceptance Evaluation

ITEMS	MEAN	INTERPRETATION
Functional Suitability	3.33	Strongly Agree
Performance Efficiency	3.33	Strongly Agree
Compatibility	3.50	Strongly Agree
Usability	3.50	Strongly Agree
Reliability	3.50	Strongly Agree
Security	3.33	Strongly Agree
Maintainability	3.00	Agree
Portability	3.50	Strongly Agree

Compatibility Testing Results

The researchers evaluated the system using Google Lighthouse. Chrome outperformed Microsoft Edge in performance, accessibility, best practices, and SEO, while Edge scored lower. Opera delivered the most balanced results, scoring highest in accessibility and SEO, making it the most optimized and reliable browser for the system.

Browser	Web Page	Performance	Accessibility	Best Practice	SEO
Google Chrome	Log in Page	96	88	96	90
	Certificate Requests	99	96	100	100
	Track Requests	100	96	100	100
	Events & Announcements	86	90	96	100
	Blotter Reports	100	95	100	100
	Elected Officials	80	100	96	100
	Dashboard	89	90	100	100
	Residents	94	91	100	100
	Officials Management	78	89	100	100
	Staff Management	98	89	100	100
	Analytics	93	89	100	100
	Archives	89	91	100	100
	Feedback	97	91	100	100
	Incident Reports	98	91	96	100
Microsoft Edge	Log In Page	97	90	96	91
	Certificate Requests	100	96	100	100
	Track Requests	99	96	100	100
	Events & Announcements	89	90	96	100
	Blotter Reports	99	95	100	100
	Elected Officials	73	100	96	100
	Dashboard	72	90	96	100
	Residents	91	91	96	100
	Officials Management	90	89	96	100
	Staff Management	98	89	96	100
	Analytics	95	89	96	100
	Archives	99	91	96	100
	Feedback	94	91	96	100
	Incident	98	91	96	100
Opera	Log In Page	91	90	96	91
	Certificate Requests	99	96	100	100
	Track Requests	97	95	96	100
	Events & Announcements	94	90	96	100
	Blotter Reports	97	95	100	100
	Elected Officials	100	100	96	100
	Dashboard	88	90	96	100
	Residents	92	91	96	100
	Officials Management	91	89	96	100
	Staff Management	92	89	96	100
	Analytics	95	89	96	100
	Archives	97	91	96	100
	Feedback	94	91	96	100
	Incident	94	91	96	100

Table 27. System Compatibility Testing for Browsers in Desktop

CONCLUSION

1. Development of the Smart Dashboard Web-Based Platform for enhanced community engagement and governance transparency

The Smart Dashboard is a web-based platform that helps barangay administrators manage operations efficiently. It integrates modules for analytics, resident records, certificates, appointments, blotter reports, events, and archived data, reducing manual work and improving service delivery. Through organized data and real-time insights, the system promotes accuracy, transparency, and effective governance.

2. Development of the Smart Dashboard Web-Based Platform for generating barangay certificate requests and blotter reports

The Smart Dashboard offers a user-friendly platform for managing certificates, appointments, and blotter reports. Residents can easily request and track certificates, schedule and manage appointments, and submit and monitor blotter reports in real time, improving transparency and communication. The Archives module supports secure and organized record management for administrators. Overall, the system enhances efficiency, organization, and transparency for both residents and barangay staff.

3. Evaluation of the system's user acceptance using the Technology Acceptance Model (TAM) and software quality characteristics using the ISO 25010 standard.

The study results showed that the system passed the Technology Acceptance Model (TAM), indicating it is useful and easy to use. IT experts also confirmed that it met ISO/IEC 25010 standards, proving its quality and reliability. Based on expert and instructor suggestions, improvements were identified to further enhance performance, security, and usability for long-term use.

RECOMMENDATIONS

The Smart Dashboard improved barangay processes by centralizing and automating key services, enhancing efficiency, transparency, and accessibility. TAM and ISO 25010 evaluations confirmed its effectiveness. Further improvements include developing a PWA, adding online payments, and SMS notifications to provide convenient access, faster transactions, and timely updates even without the internet.

1. Integrating offline access capabilities is suggested so the system can still work even without an internet connection. This allows users and barangay staff to view important information and continue basic tasks during network interruptions, making the system more reliable for areas with weak or unstable internet.
2. Using blockchain technology is suggested to make certificates secure and trustworthy. This ensures documents cannot be tampered with and can be verified easily.
3. A digital signature QR code verified documents as authentic, confirmed the barangay's captain approval and improved transparency.
4. Adding an API Scanner for ID Verification is also recommended. This feature will allow the system to scan and validate resident IDs automatically, making the verification process faster, more accurate, and more secure.

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APPENDIX G

GRAMMARLY RESULT

SMART DASHBOARD: A WEB BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED PROCESSING

by Ronah joy Ramirez

General metrics

5,986	839	42	3 min 21 sec	6 min 27 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 82% of all texts checked by Grammarly

Writing Issues

28		28
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

SMART DASHBOARD: A WEB BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED PROCESSING

by Ronah joy Ramirez

General metrics

97,459	13,415	721	53 min 39 sec	1 hr 43 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 86% of all texts checked by Grammarly

Writing Issues

340	2	338
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

APPENDIX H

PLAGIARISM RESULT

SMART DASHBOARD A WEB BASED PLATFORM FOR CITIZEN ENGAGEMENT AND DATA-DRIVEN DIGITAL GOVERNANCE USING AUTOMATED PROCESSING .pdf

ORIGINALITY REPORT

8%	4%	4%	5%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Laguna State Polytechnic University Student Paper	2%
2	Submitted to Asia Pacific Instutute of Information Technology Student Paper	1%
3	Submitted to Technological Institute of the Philippines Student Paper	<1%
4	www.coursehero.com Internet Source	<1%
5	Submitted to Southern Luzon State University Student Paper	<1%
6	Submitted to Jose Rizal University Student Paper	<1%
7	Matlala, Lesedi Senamele. "Social Media as an Effective Citizen-Based Monitoring Tool for Public Service Delivery in South Africa",	<1%

APPENDIX I

CURRICULUM VITAE



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Sitio. Sawmill Brgy. Balansay
Mamburao, Occidental Mindoro

EDUCATION

Elementary
Balansay Elementary School
2015 - 2016

Junior Highschool
Occidental Mindoro National High School
2019 - 2020

Senior Highschool
Occidental Mindoro National High School
2021 - 2022

College
Laguna State Polytechnic University Brgy. Malinta, Los Banos, Laguna Bachelor of Science in Information Technology
2022 - Present

Angela M. Ante

SUMMARY

Information Technology student with hands on experience in web development, server configuration, and database management. Skilled in deploying applications, integrating features, and designing user interfaces using Figma and Canva. Knowledgeable in web technologies such as HTML, CSS, JavaScript, and React, with a strong commitment to continuous learning and improving technical skills.

SKILLS

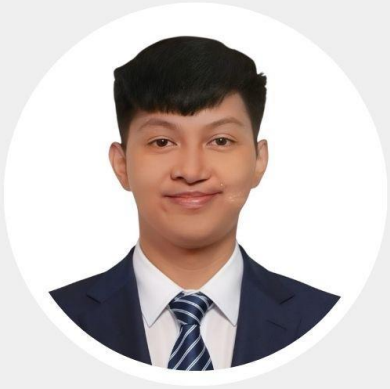
- UI/UX Design (Figma, Canva)
- Web Designer
- Web Development: HTML, CSS, JavaScript, React

EXPERIENCES

- Configured a local server for testing and deployment, managing basic server settings and database connections.
- Deployed web applications via FTP, organized server files, and ensured the live site functioned properly.
- Installed and configured web server packages on Ubuntu, set up hosting environments, and managed system permissions.

AWARDS/ACHIEVEMENTS

- With honors 2021-2022
- Dean’s Lister Award 2022



Nicen Howell B. Mendoza

0928 894 2686

nicenmendoza25@gmail.com

327 Purok 4, Brgy. Malinta, Los Baños, Laguna

SUMMARY

Information Technology student with a strong foundation in programming, database management, and technical operations. Skilled in Python, Java, and JavaScript, with hands-on experience working with MySQL databases and operating in both Windows and Linux environments. Gained practical organizational and problem-solving skills as a Student Assistant at DOST, where responsibilities included inventory management, stock organization, proper labeling of devices, and identifying issues with stored items.

EDUCATION

PRIMARY

LICEO DE LOS BAÑOS (2010-2012)
Brgy. Timugan, Los Baños, Laguna

GATEWAY LEARNING CENTER (2012-2016)
Brgy. Anos, Los Baños, Laguna

SECONDARY

BN CALARA NATIONAL HIGHSCHOOL (2016-2020)
Brgy. Anos, Los Baños, Laguna

BN CALARA NATIONAL HIGHSCHOOL (2020-2022)
Accountancy, Business, and Management Strand
Brgy. Anos, Los Baños, Laguna

TERTIARY

LAGUNA STATE POLYTECHNIC UNIVERSITY-LOS BAÑOS CAMPUS (2022-PRESENT)
BS in Information Technology
Brgy. Malinta, Los Baños, Laguna

SKILLS

- Programming Languages: Python, Java, JavaScript, React JS, C#, C++
- Database Managemen: MySQL, Firebase
- Operating Systems: Windows, Linux

EXPERIENCES

- **DOST**

Student Assistant

Inventory of stocks in the storage room, organize and labeling the items or devices, and look for any issues.

- **Upwork**

Virtual Assistant

Product listing, and launching of advertisements.



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0536 Sirang Lupa, Calamba City, Laguna

Karylle R. Vidal

SUMMARY

Information Technology student with expertise in web system design, graphic design and UI/UX interface development using Adobe Photoshop and Figma. Highly interested in engaging more in designing interfaces and software. Possesses foundational to average knowledge in various programming languages and perpetually improving technical abilities. Wants to grow intellectually, morally, and emotionally in new working environment.

SKILLS

- Web Designer
- Web Developer
- Computer Literate individual with extensive proficiency in a wide variety of software application.

EXPERIENCES

- Work Immersion from Senior High School, deployed a Web System and provided support to the company's IT operations.

AWARDS/ACHIEVEMENTS

- Graduated Senior High School with High Honors

EDUCATION

PRIMARY
Sirang Lupa Elementary School
Sirang, Lupa Calamba City
2013 - 2016

SECONDARY
Calamba - Institute Canlubang
Kapayapaan Village Canlubang, Calamba City
2016-2020

Calamba - Institute Canlubang
Kapayapaan Village Canlubang, Calamba City
2020-2022

TERTIARY
Laguna State Polytechnic University Brgy. Malinta, Los Banos, Laguna
Bachelor of Science in Information Technology
2022 - Present